



Principles of Finance Academic Year 2025/26 B-Track

Prof. Dr. habil. Dr. Stefan Prigge

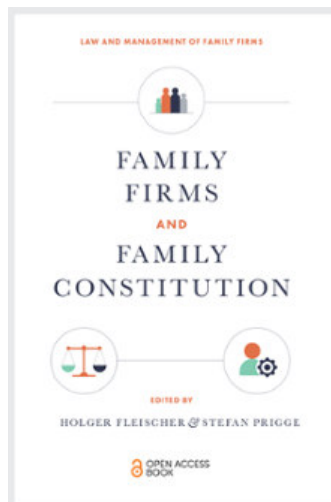
Self-Introduction

English Premiere League vs. Bundesliga

Vergleich der Corporate Governance

Dr. Alexander Juschus / Ralf Leister / Prof. Dr. Stefan Prigge /
Zach Dallas*

Zeitschrift für Corporate Governance, 2022



Family Firms and Family
Constitution. Eds. Holger
Fleischer and Stefan Prigge,
2024

Football stocks: a new asset class attractive to institutional investors? Empirical results and impulses for researching investor motivations beyond return

Football stocks

471

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Sport, Business and Management, 2020

PRIVATE EQUITY INVESTORS AND FAMILY FIRMS: THE ROLE OF EXIT INTENTIONS AND CONFLICTS

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Hamburg, Germany

Corporate Ownership and Control, 2019

Self-Introduction

Cofounder of Sports Governance e.V.

<https://sportsgovernance.net/de/>

Sports Governance Kodex

<http://www.sports-governance-kodex.org/>

Hamburger Abendblatt

HSV NEWS

Professor warnt: „Beim HSV gibt es Interessenskonflikte“



Aktualisiert: 06.09.2022, 05:55

| Lesedauer: 9 Minuten

Kai Schiller



Thomas Wüstefeld steht im Zentrum mutmaßlicher Interessenskonflikte.

Foto: IMAGO /
Philipp Szyza

Aufsichtsrat, Vorstand,
Anteilseigner – die Grenzen
im Volkspark
verschwimmen. Beispiel:
Jansen und Frömming sind
nicht transparent.

The World of Finance: Major/Minor

- Major Finance and Financial Markets
 - Minor Finance (2nd or 3rd year)
 - Module Grundlagen Finance
 - Module Finanzielles Risikomanagement
 - Module Mergers & Acquisitions (M&A)
 - Minor Financial Markets (3rd year)
 - Module Sustainable Finance
 - Module Data Science/ Künstliche Intelligenz/ Digitalisierung in Finance, FINTECHS und Start-Up-Financing
 - Module Finanzmärkte

The World of Finance: Sustainable Finance / HSBA Finance Club



[Link to study](#)

The World of Finance: Sustainable Finance / HSBA Finance Club

Abbildung 1: Sustainable Finance im engeren Sinne

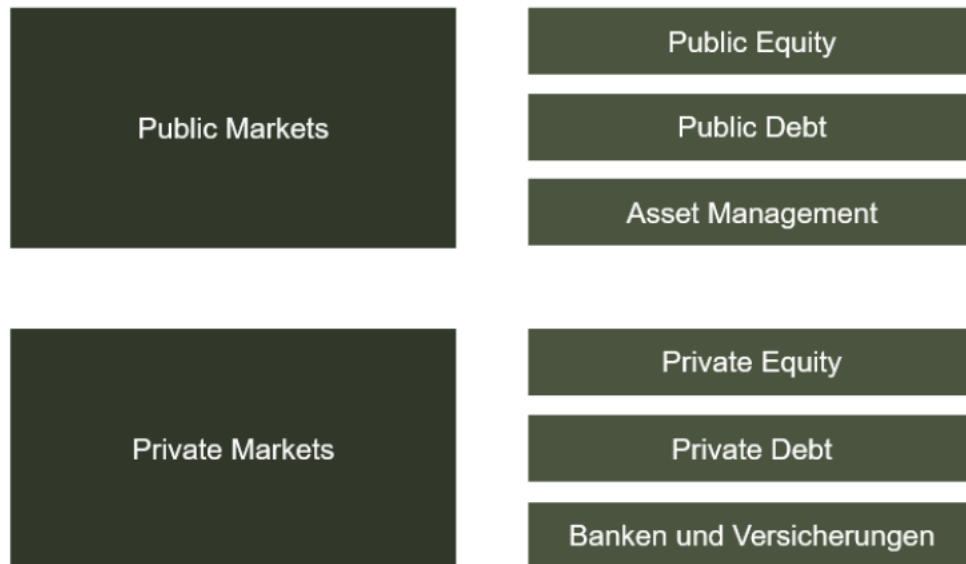


Abbildung 2: Das Sustainable-Finance-Ökosystem (Sustainable Finance im weiteren Sinne)



Seminare

- Vorträge von Experten und spannenden Persönlichkeiten aus der Finanzbranche
- Börsenplanspiel mit Musterdepots, Adventskalender mit vielen spannenden Wettbewerben (Aktien-Duelle, Börsen-Quiz), ...



HSBA
Finance
Club

Infos & Networking

- Komitee Meetings zum Austausch aktueller Themen
- Meinungsaustausch und Informationen zu Finanz-News über WhatsApp, Teams etc.
- Größere Networking Events und Wettbewerbe mit anderen Business Schools
- Bisherige Kooperationen u.a. mit EY, Airbus, M. M. Warburg & Co Bank, DealCircle, Aduva Capital, Carlsquare, PwC, Jungheinrich, ...

Social Media

Instagram: @hsbafinanceclub

LinkedIn: HSBA Finance Club

E-Mail: financeclub@myhsba.de



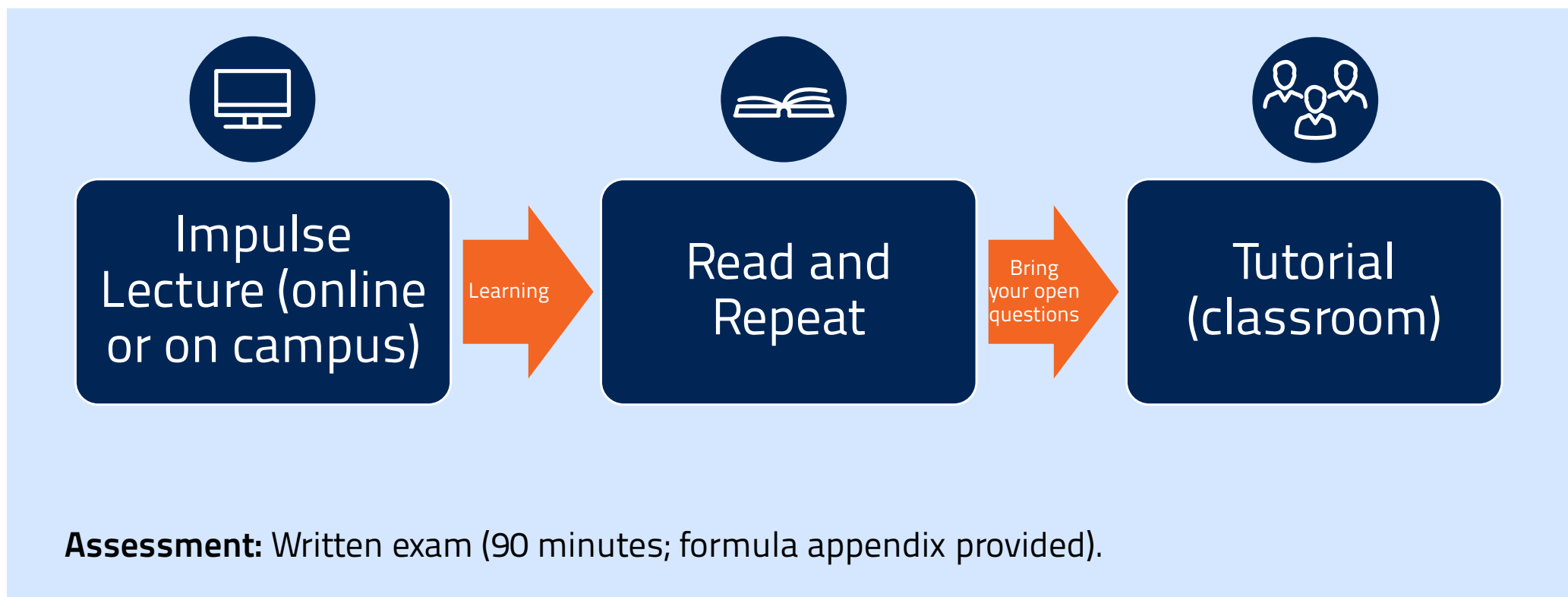
The World of Finance: Making the Link to Accounting

- Capital budgeting: In which real assets to invest?

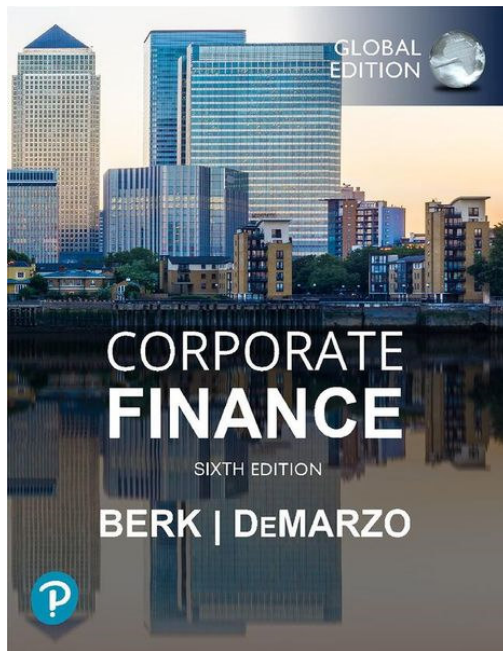
GLOBAL CONGLOMERATE CORPORATION			GLOBAL CONGLOMERATE CORPORATION		
Consolidated Balance Sheet Year Ended December 31 (in \$ million)			Consolidated Balance Sheet Year Ended December 31 (in \$ million)		
Assets	2018	2017	Liabilities and Stockholders' Equity	2018	2017
Current Assets			Current Liabilities		
Cash	21.2	19.5	Accounts payable	29.2	24.5
Accounts receivable	18.5	13.2	Notes payable/short-term debt	3.5	3.2
Inventories	15.3	14.3	Current maturities of long-term debt	13.3	12.3
Other current assets	2.0	1.0	Other current liabilities	2.0	4.0
Total current assets	57.0	48.0	Total current liabilities	48.0	44.0
Long-Term Assets			Long-Term Liabilities		
Land	22.2	20.7	Long-term debt	99.9	76.3
Buildings	36.5	30.5	Capital lease obligations	—	—
Equipment	39.7	33.2	Total debt	99.9	76.3
Less accumulated depreciation	(18.7)	(17.5)	Deferred taxes	7.6	7.4
Net property, plant, and equipment	79.7	66.9	Other long-term liabilities	—	—
Goodwill and intangible assets	20.0	20.0	Total long-term liabilities	107.5	83.7
Other long-term assets	21.0	14.0	Total Liabilities	155.5	127.7
Total long-term assets	120.7	100.9	Stockholders' Equity	22.2	21.2
Total Assets	177.7	148.9	Total Liabilities and Stockholders' Equity	177.7	148.9

- Market value of equity vs. book value of equity
- Cost of capital (debt, equity)
→ WAAC
(weighted average cost of capital)

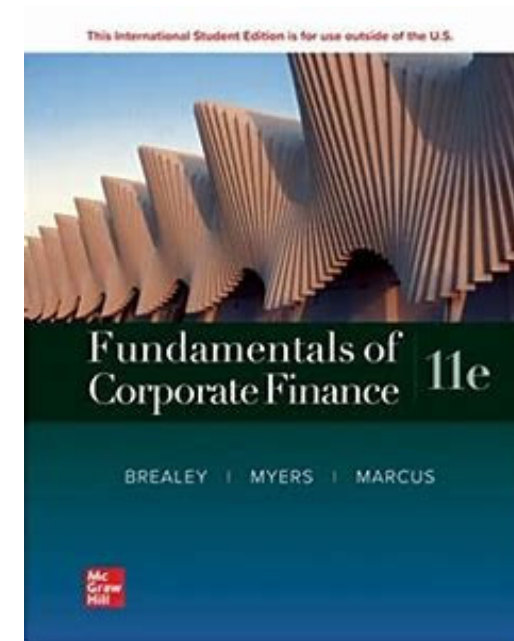
Principles of Finance – Teaching Concept



Basic Books

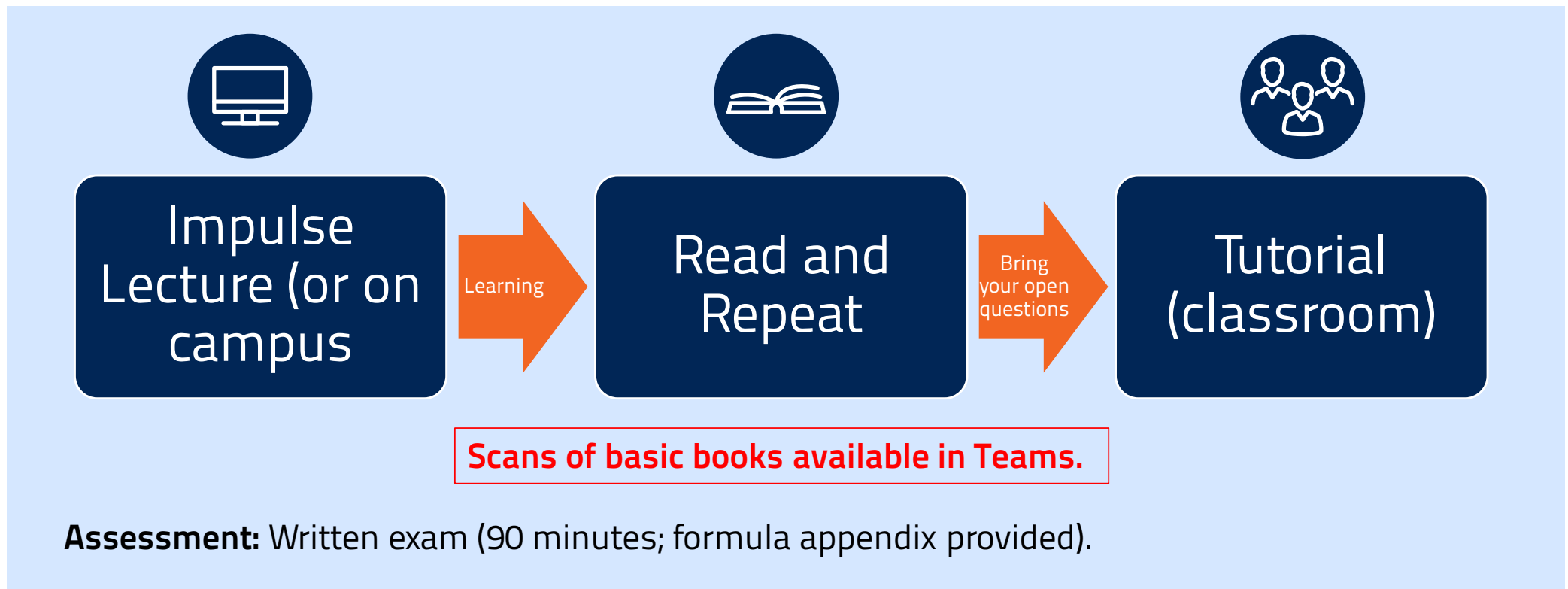


Berk/DeMarzo: Corporate Finance, 6th edition (global edition), Pearson, 2024.
BDM



Brealey/Myers/Marcus: Fundamentals of Corporate Finance, 11th edition (int. student edition), McGraw-Hill, 2023.
BMM

Principles of Finance – Teaching Concept



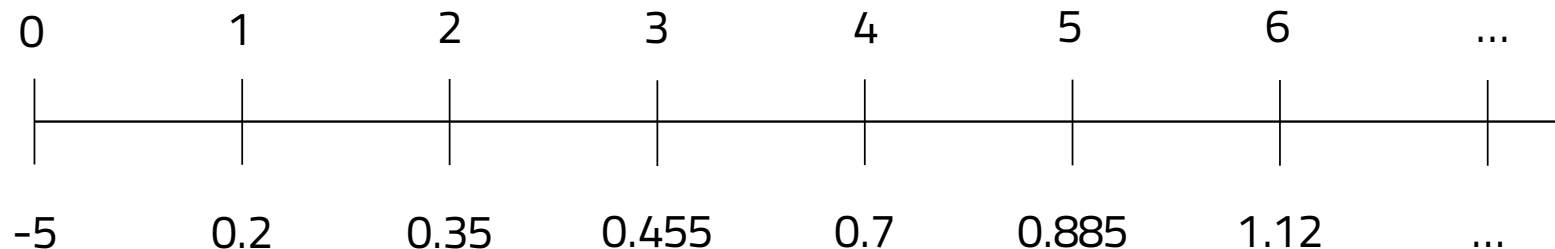
The Course Journey

VW considers to invest EUR5 billion in the development into a new line of e-cars.

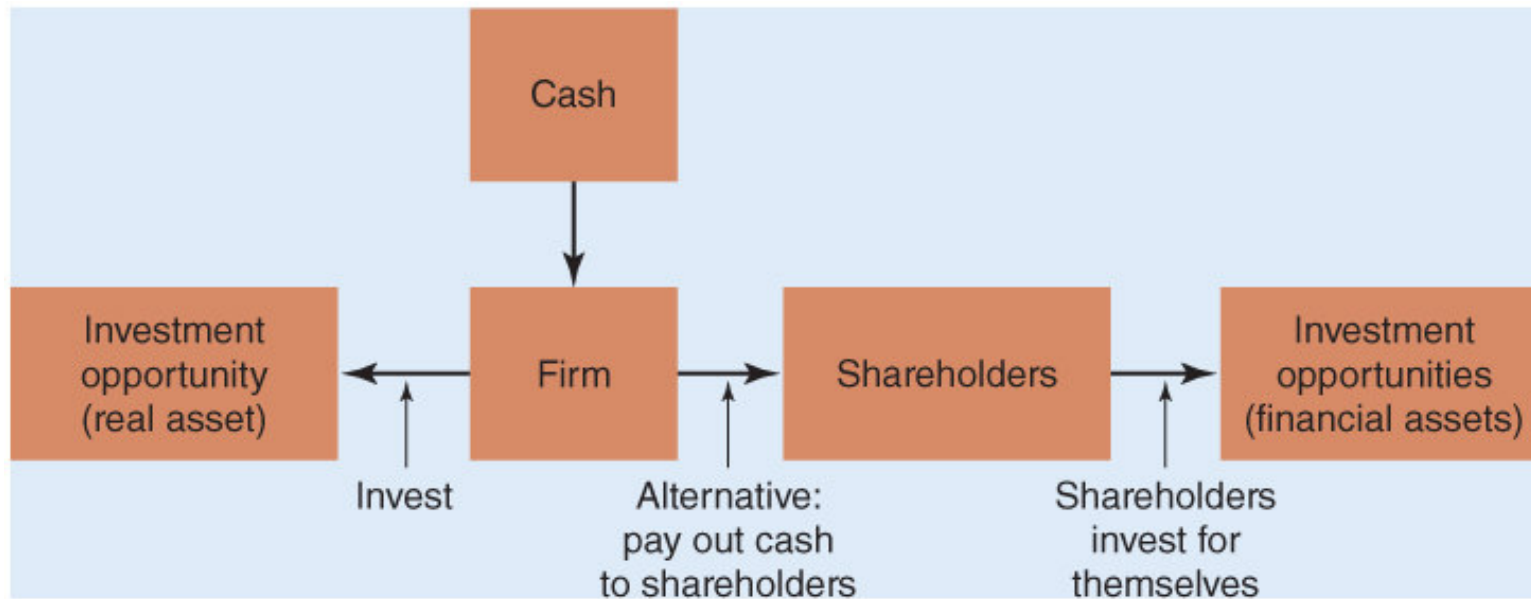
- Good idea?

Step 0 (not part of this course)

- Estimate cash flows caused by this project (done by, inter alia, management accounting and market research department). → **timeline**
- Note: All cash flows are **expected** values in EUR bn; they could be higher or lower.



The Course Journey



Taken from the slides accompanying Brealey/Myers/Marcus: Fundamentals of Corporate Finance, 11th edition.

new e-car line

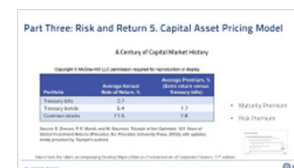
may include debt and equity

alternative use of money
→ opportunity cost of capital
→ benchmark

The Course Journey

Recap from Accounting: Main differences between debt and equity

- Shareholders are residual claimants, i.e., they are the last in line being served.
- Equity is the safety cushion protecting debt from losses.
 - In a given company, equity is riskier than debt from the capital providers' perspective.
 - Cost of equity > cost of debt
- Usually obligation to pay back debt, not so for equity.
- Often voting rights for equity.
- Addendum: Relevant forms of debt for this course are (bank) loans and bonds (=tradable debt).



The Course Journey

Main topic of the course: As a company, how to make a proper investment decision (=capital budgeting)?

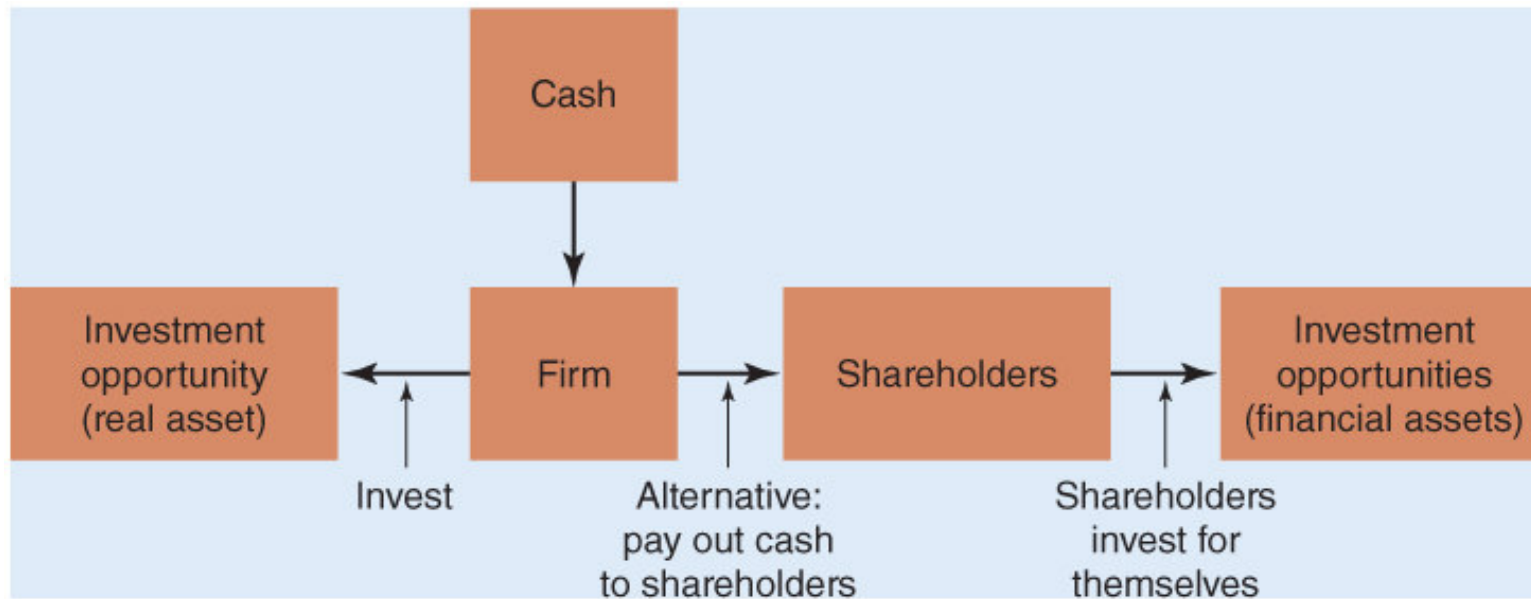
- Key task: determine the opportunity cost of capital from capital market data (=benchmark).
 - Make ourselves familiar with the bond market (debt) → **chapter 3**
 - Make ourselves familiar with the stock market (equity) → **chapter 4**
 - Make ourselves familiar with the link between risk and return → **chapters 5/6**
 - Calculate the cost of capital including tax advantage of debt = WACC (weighted average cost of capital) → **chapter 6**

The Course Journey

Main topic of the course: As a company, how to make a proper investment decision (=capital budgeting)?

- Before that
 - Projects, bonds, and shares → reasoning in streams of cash flows (timeline)
 - Recap of financial mathematics from 1st semester → **chapter 1**
 - Introduction to investment decision (capital budgeting) rules
 - Net Present Value (NPV), Internal Rate of Return (IRR), Payback Rule → **chapter 2**
 - After that refinement of benchmark in **chapters 3 to 6**

The Course Journey



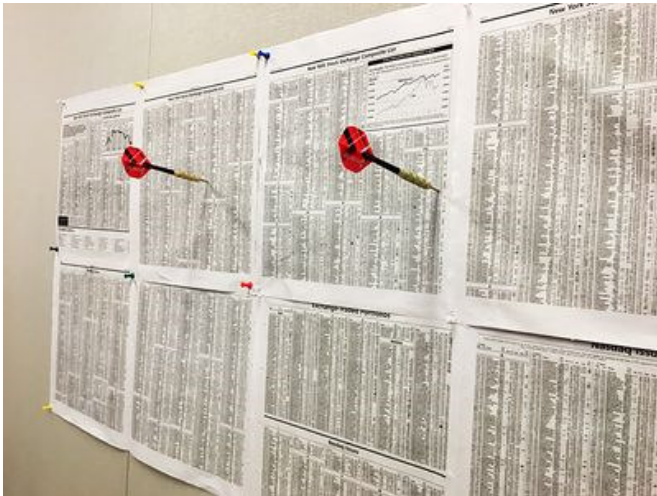
Taken from the slides accompanying Brealey/Myers/Marcus: Fundamentals of Corporate Finance, 11th edition.

capital
budgeting rules
(chapter 2)

- benchmark debt (ch. 3)
- benchmark equity (ch. 4)
- risk/return (ch. 5/6)
- WACC (ch. 6)

The Course Journey: Takeaways for Personal Finance

- How likely is it that you or a paid investment manager is able to achieve abnormal returns systematically? Spoiler: highly unlikely → **chapter 4**



The Wall Street Journal, Can Throwing Darts Beat Hedge-Fund Managers' Stock Picks?, 24.4.2018. <https://www.wsj.com/articles/sohn-stock-tips-may-not-be-worth-peanuts-1524592131>; last accessed 19.3.2024.

- There is a link between risk and return → **chapters 5/6**
 - Beware of high yield, "safe" investments!
 - Diversify!

Agenda

Part One: Time and Interest Rates

1. Time Value of Money 27.7.26

2. Investment Decision Rules 3./10.8.26

Part Two: Financial Instruments

3. Bonds 10./17.8.26

4. Stocks 17./24.8.26



Agenda

Part Three: Risk and Return

5. Capital Asset Pricing Model 31.8./7.9.26

6. Capital Market Based Cost of Capital 7./14.9.26



Agenda

Part One: Time and Interest Rates

1. Time Value of Money 13.4.26

2. Investment Decision Rules 20./27.4.26

Part Two: Financial Instruments

3. Bonds 27.4./4.5.26

4. Stocks 4./11.5.26



Part One: Time and Interest Rates 1. Time Value of Money: Literature

- BDM, chapter 4
- BMM, chapter 5

Part One: Time and Interest Rates 1. Time Value of Money

Terms and concepts you should be familiar with after chapter 1

- timeline
- three rules of time travel
- present value, net present value (NPV), future value, internal rate of return (IRR)
- perpetuity, annuity
- calculating NPV, future value, and IRR for a variety of different cash flow streams

Part One: Time and Interest Rates 1. Time Value of Money

Structure of Chapter 1

- Timeline
- Three rules of time travel
- Valuing a stream of cash flows
- Calculating the net present value (NPV)
- Perpetuities and annuities
- Internal rate of return (IRR)

Part One: Time and Interest Rates 1. Time Value of Money

Structure of Chapter 1

- **Timeline**
- Three rules of time travel
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Part One: Time and Interest Rates 1. Time Value of Money

Timeline

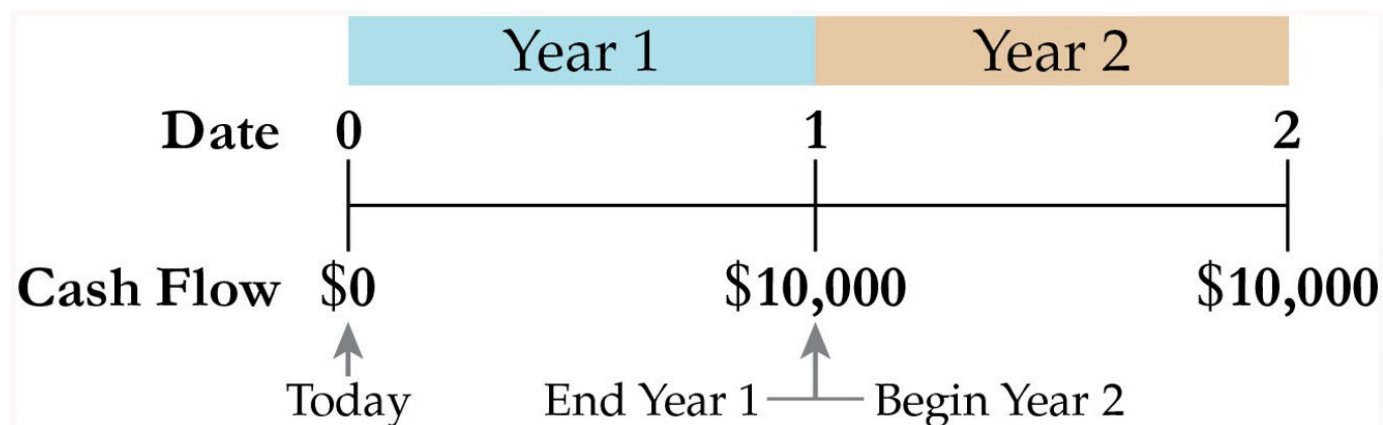
- A timeline is a linear representation of the timing of potential cash flows.
- Drawing a timeline of the cash flows will help you visualize the financial problem.

Taken from the slides accompanying Berk/DeMarzo: Corporate Finance, 6th edition.

Part One: Time and Interest Rates 1. Time Value of Money

Timeline

- Assume that you made a loan to a friend. You will be repaid in two payments, one at the end of each year over the next two years.



Taken from the slides accompanying Berk/DeMarzo: Corporate Finance, 6th edition.

Part One: Time and Interest Rates 1. Time Value of Money

Timeline

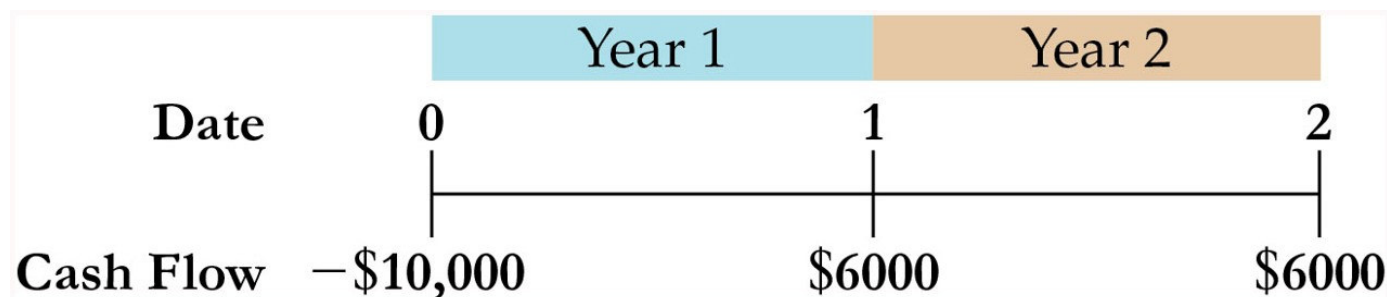
- Differentiate between two types of cash flows
 - **Inflows** are positive cash flows.
 - **Outflows** are negative cash flows, which are indicated with a – (minus) sign

Taken from the slides accompanying Berk/DeMarzo: Corporate Finance, 6th edition.

Part One: Time and Interest Rates 1. Time Value of Money

Timeline

- Assume that you are lending \$10,000 today and that the loan will be repaid in two annual \$6,000 payments.



- The first cash flow at date 0 (today) is represented as a negative sum because it is an outflow.
- Timelines can represent cash flows that take place at the end of any time period—a month, a week, a day, etc.

Taken from the slides accompanying Berk/DeMarzo: Corporate Finance, 6th edition.

Part One: Time and Interest Rates 1. Time Value of Money

Structure of Chapter 1

- Timeline
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Part One: Time and Interest Rates 1. Time Value of Money

The Three Rules of Time Travel

- Financial decisions often require combining cash flows or comparing values. Three rules govern these processes.

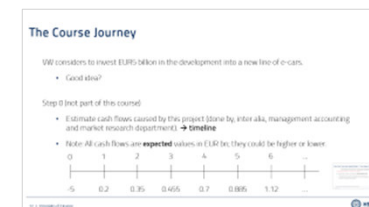
Rule 1	Only values at the same point in time can be compared or combined	
Rule 2	To move a cash flow forward in time, you must compound it.	Future value of a Cash flow $FV_n = C \times (1 + r)^n$
Rule 3	To move a cash flow backward in time, you must discount it.	Present value of a Cash flow $PV = C \div (1 + r)^n = \frac{C}{(1 + r)^n}$

Taken from the slides accompanying Berk/DeMarzo: Corporate Finance, 6th edition.

Part One: Time and Interest Rates 1. Time Value of Money

The Three Rules of Time Travel. Rule 1: Comparing and Combining Values

- A dollar today and a dollar in one year are not equivalent.
- It is only possible to compare or combine values **at the same point in time**.
 - Which would you prefer: A gift of \$1,000 today or \$1,210 at a later date?
 - To answer this, you will have to compare the alternatives to decide which is worth more. One factor to consider: How long is "later?"



Taken from the slides accompanying Berk/DeMarzo: Corporate Finance, 6th edition.

Part One: Time and Interest Rates 1. Time Value of Money

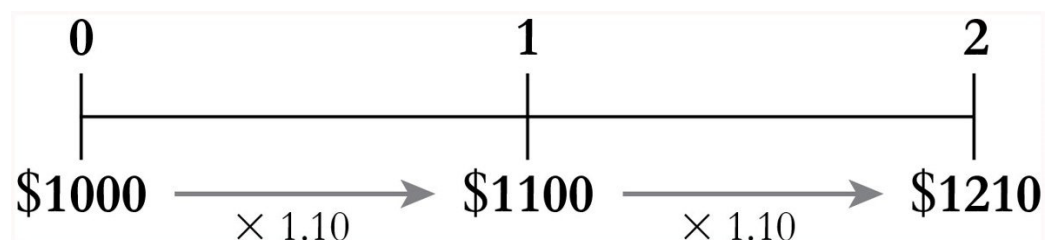
The Three Rules of Time Travel. Rule 2: Moving Cash Flows Forward in Time

- To move a cash flow forward in time, you must compound it.
 - Suppose you have a choice between receiving \$1,000 today or \$1,210 in two years. You believe you can earn 10% on the \$1,000 today but want to know what the \$1,000 will be worth in two years.

Taken from the slides accompanying Berk/DeMarzo: Corporate Finance, 6th edition.

Part One: Time and Interest Rates 1. Time Value of Money

The Three Rules of Time Travel. Rule 2: Moving Cash Flows Forward in Time



This is not a timeline
as introduced above!

- Future Value of a Cash Flow

$$FV_n = C \times (1 + r) \times (1 + r) \times \cdots \times (1 + r) = C \times (1 + r)^n$$

Taken from the slides accompanying Berk/DeMarzo: Corporate Finance, 6th edition.

Part One: Time and Interest Rates 1. Time Value of Money

The Three Rules of Time Travel. Rule 2: Moving Cash Flows Forward in Time

The Power of Compounding

- Suppose you invest \$1,000 in an account paying 10% interest per year. How much will you have in the account in seven years? In 20 years ? In 75 years?

$$7 \text{ years: } \$1000 \times (1.10)^7 = \$1,948.72$$

$$20 \text{ years: } \$1000 \times (1.10)^{20} = \$6,727.50$$

$$75 \text{ years: } \$1000 \times (1.10)^{75} = \$1,271,895.37$$

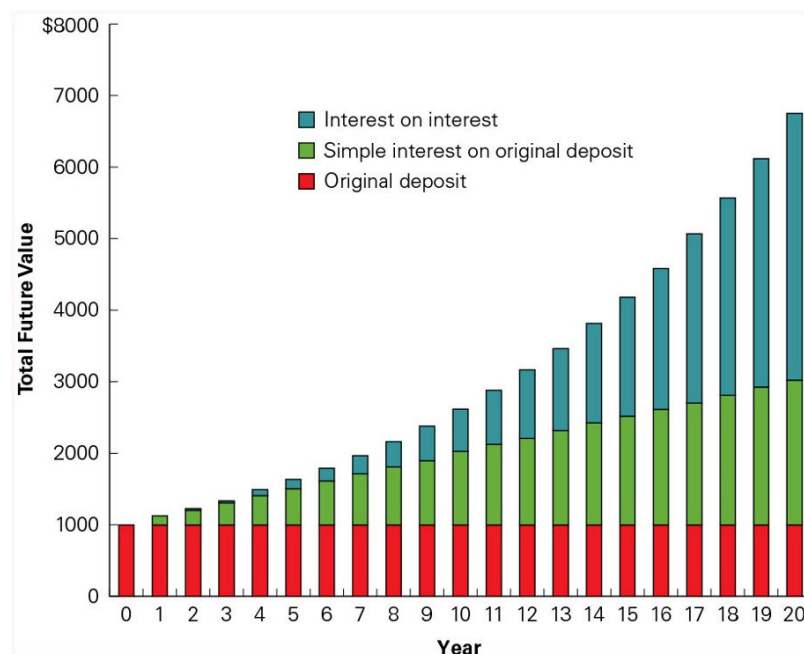
- Note that at 10% interest, your money will nearly double in 7 years. After 20 years, it will increase almost sevenfold. And if you invest for 75 years, you will be a millionaire !

Taken from the slides accompanying Berk/DeMarzo: Corporate Finance, 6th edition.

Part One: Time and Interest Rates 1. Time Value of Money

The Three Rules of Time Travel. Rule 2: Moving Cash Flows Forward in Time

The Power of Compounding



Interest rate: 10%

Taken from the slides accompanying Berk/DeMarzo: Corporate Finance, 6th edition.

Part One: Time and Interest Rates 1. Time Value of Money

The Three Rules of Time Travel. Rule 3: Moving Cash Flows Back in Time

- To move a cash flow backward in time, we must discount it.
- Present Value of a Cash Flow

$$PV = C \div (1 + r)^n = \frac{C}{(1 + r)^n}$$

Taken from the slides accompanying Berk/DeMarzo: Corporate Finance, 6th edition.

Part One: Time and Interest Rates 1. Time Value of Money

The Three Rules of Time Travel. Rule 3: Moving Cash Flows Back in Time

Present Value of a Single Future Cash Flow

- **Problem**
 - You are considering investing in a savings bond that will pay **\$15,000 in 10 years**.
 - If the competitive **market interest rate** is fixed at **6% per year**, what is the bond worth today?

Taken from the slides accompanying Berk/DeMarzo: Corporate Finance, 6th edition.

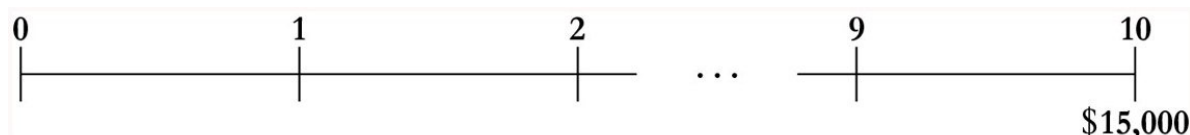
Part One: Time and Interest Rates 1. Time Value of Money

The Three Rules of Time Travel. Rule 3: Moving Cash Flows Back in Time

Present Value of a Single Future Cash Flow

- **Solution**

- The cash flows for this bond are represented by the following timeline:



- Thus, the bond is worth \$15,000 in 10 years. To determine the value today, we compute the present value:

$$PV = \frac{15,000}{1.06^{10}} = \$8,375.92 \text{ today}$$

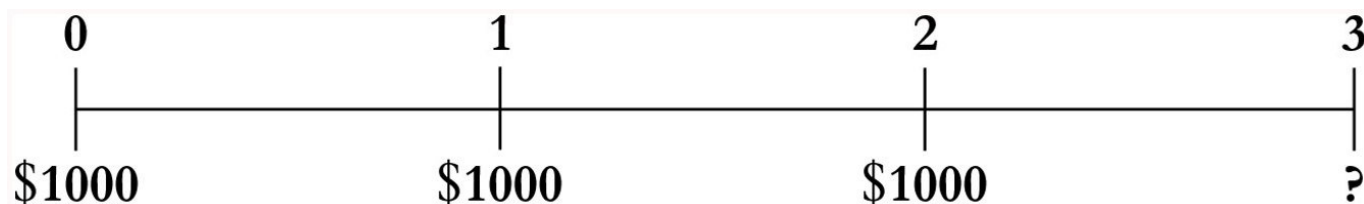
- The bond is worth much less today than its final payoff because of the **time value of money**.

Taken from the slides accompanying Berk/DeMarzo: Corporate Finance, 6th edition.

Part One: Time and Interest Rates 1. Time Value of Money

The Three Rules of Time Travel. Summarizing Example

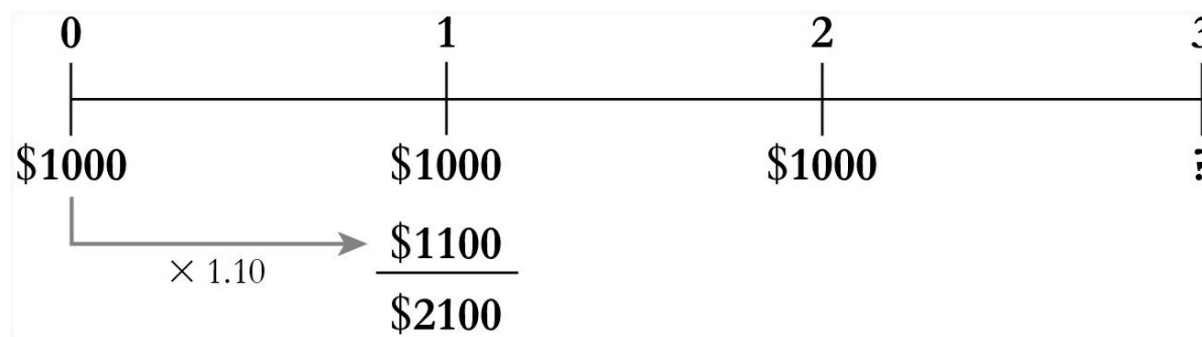
- Suppose we plan to save \$1,000 today, and \$1,000 at the end of each of the next two years. If we can earn a fixed 10% interest rate on our savings, how much will we have three years from today?



Taken from the slides accompanying Berk/DeMarzo: Corporate Finance, 5th edition.

Part One: Time and Interest Rates 1. Time Value of Money

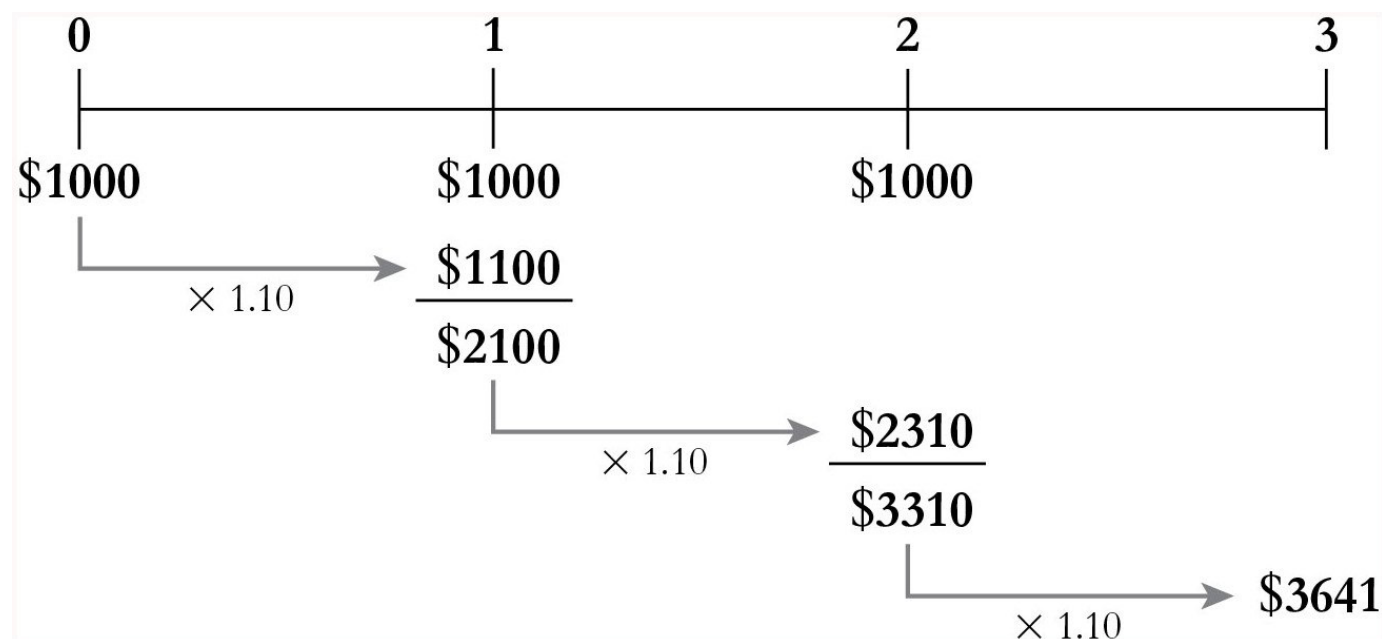
The Three Rules of Time Travel. Summarizing Example



Taken from the slides accompanying Berk/DeMarzo: Corporate Finance, 5th edition.

Part One: Time and Interest Rates 1. Time Value of Money

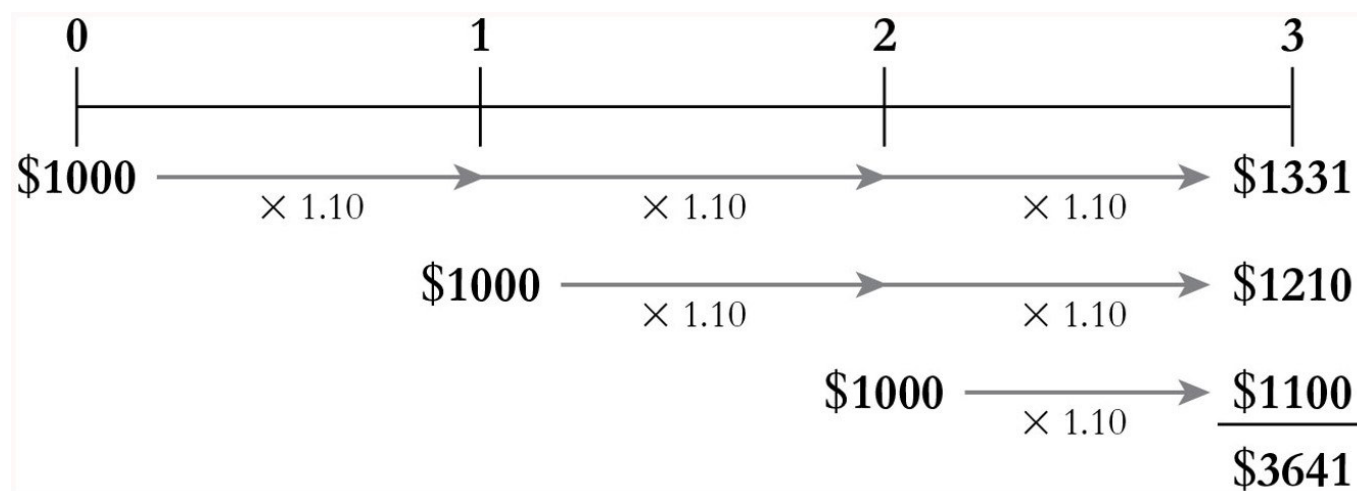
The Three Rules of Time Travel. Summarizing Example



Taken from the slides accompanying Berk/DeMarzo: Corporate Finance, 5th edition.

Part One: Time and Interest Rates 1. Time Value of Money

The Three Rules of Time Travel. Summarizing Example

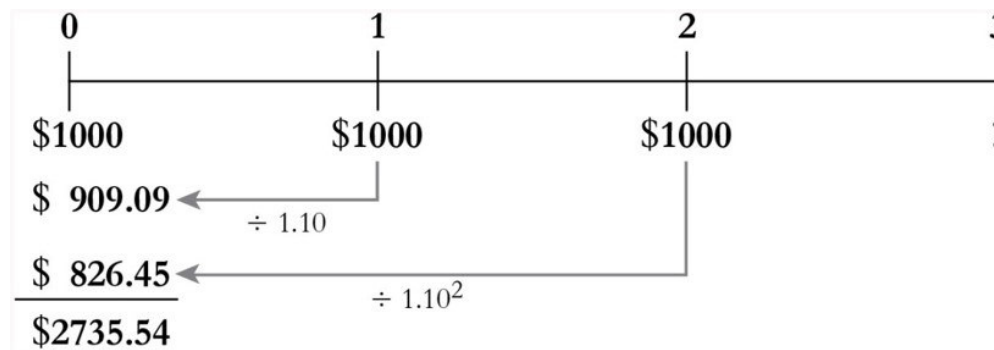


Taken from the slides accompanying Berk/DeMarzo: Corporate Finance, 5th edition.

Part One: Time and Interest Rates 1. Time Value of Money

The Three Rules of Time Travel. Summarizing Example

- Let's solve this problem in a different way than we did earlier. **First**, compute the **present value** of the cash flows. There are several ways to perform this calculation. Here we treat each cash flow separately and then combine the present values.

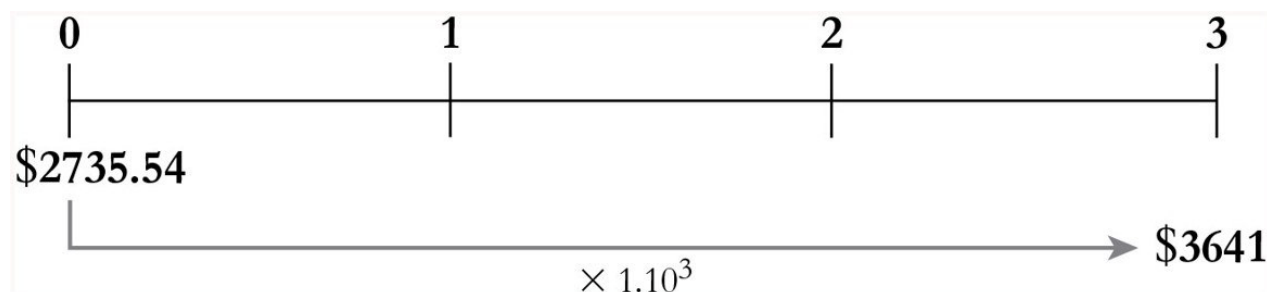


Taken from the slides accompanying Berk/DeMarzo: Corporate Finance, 5th edition.

Part One: Time and Interest Rates 1. Time Value of Money

The Three Rules of Time Travel. Summarizing Example

- Saving \$2,735.54 today is equivalent to saving \$1,000 per year for three years. Now let's compute its future value in year 3.
- This answer of \$3,641 is precisely the same result we found earlier. As long as we apply the three rules of time travel, we will always get the correct answer.



Taken from the slides accompanying Berk/DeMarzo: Corporate Finance, 5th edition.

Part One: Time and Interest Rates 1. Time Value of Money

Structure of Chapter 1

- Timeline
- Three rules of time travel
- Valuing a stream of cash flows
- Calculating the net present value (NPV)
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- Internal rate of return (IRR)

Part One: Time and Interest Rates 1. Time Value of Money

Calculating the Net Present Value

- Calculating the NPV of future cash flows allows us to evaluate an investment decision.
- Net Present Value compares the present value of cash inflows (benefits) to the present value of cash outflows (costs).

Taken from the slides accompanying Berk/DeMarzo: Corporate Finance, 6th edition.

Part One: Time and Interest Rates 1. Time Value of Money

Calculating the Net Present Value

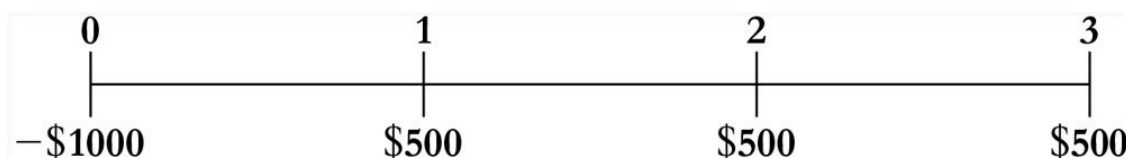
- **Problem: Net Present Value of an Investment Opportunity**
 - You have been offered the following investment opportunity: if you invest \$1,000 today, you will receive \$500 at the end of each of the next three years. If you could otherwise earn 10% per year on your money, should you undertake the investment opportunity?

Taken from the slides accompanying Berk/DeMarzo: Corporate Finance, 6th edition.

Part One: Time and Interest Rates 1. Time Value of Money

Calculating the Net Present Value

- **Solution: Net Present Value of an Investment Opportunity**
 - As always, we start with a timeline. We denote the upfront investment as a negative cash flow (because it is money we need to spend) and the money we receive as a positive cash flow.



- To decide whether we should accept this opportunity, we compute the NPV by computing the present value of the stream:

$$\text{NPV} = -1,000 + \frac{500}{1.10} + \frac{500}{1.10^2} + \frac{500}{1.10^3} = \$243.43$$

Taken from the slides accompanying Berk/DeMarzo: Corporate Finance, 6th edition.

Part One: Time and Interest Rates 1. Time Value of Money

Calculating the Net Present Value

- **Solution: Net Present Value of an Investment Opportunity**
 - Because the NPV is **positive**, the benefits exceed the costs and we should make the investment.
 - Indeed, the NPV tells us that taking this opportunity is like getting **an extra \$243.43 that you can spend today**. To illustrate, suppose you borrow \$1,000 to invest in the opportunity and an extra \$243.43 to spend today. How much would you owe on the **\$1,243.43 loan** in three years? At 10% interest, the amount you would owe would be

$$FV = (\$1,000 + \$243.43) \times (1.10)^3 = \$1,655 \text{ in three years}$$

Taken from the slides accompanying Berk/DeMarzo: Corporate Finance, 6th edition.

Part One: Time and Interest Rates 1. Time Value of Money

Calculating the Net Present Value

- **Solution: Net Present Value of an Investment Opportunity**
 - At the same time, the investment opportunity generates cash flows. If you put these cash flows into a bank account, how much will you have saved three years from now? The future value of the savings is

$$FV = (\$500 \times 1.10^2) + (\$500 \times 1.10) + \$500 = \$1,655 \text{ in three years}$$

- As you see, you can use your bank savings to repay the loan. Taking the opportunity therefore allows you to spend \$243.43 **today** at no extra cost.

Taken from the slides accompanying Berk/DeMarzo: Corporate Finance, 6th edition.

Part One: Time and Interest Rates 1. Time Value of Money

Structure of Chapter 1

- Timeline
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- Internal rate of return (IRR)

Part One: Time and Interest Rates 1. Time Value of Money

Perpetuities and Annuities: Perpetuities

- When a constant cash flow will occur at regular intervals forever it is called a perpetuity.



- The present value of a perpetuity is simply the cash flow divided by the interest rate.

$$PV(C \text{ in perpetuity}) = \frac{C}{r}$$

Taken from the slides accompanying Berk/DeMarzo: Corporate Finance, 6th edition.

Part One: Time and Interest Rates 1. Time Value of Money

Perpetuities and Annuities: Perpetuities

- **Problem**
 - You want to endow a chair for a female professor of finance at your alma mater. You'd like to attract a prestigious faculty member, so you'd like the endowment to add \$100,000 per year to the faculty member's resources (salary, travel, databases, etc.). If you expect to earn a rate of return of 4% annually on the endowment, how much will you need to donate to fund the chair?

Taken from the slides accompanying Berk/DeMarzo: Corporate Finance, 6th edition.

Part One: Time and Interest Rates 1. Time Value of Money

Perpetuities and Annuities: Perpetuities

- **Solution**

- The timeline of the cash flows looks like this:
- 

- This is a perpetuity of \$100,000 per year. The funding you would need to give is the present value of that perpetuity. From the formula:

$$PV = \frac{C}{r} = \frac{\$100,000}{.04} = \$2,500,000$$

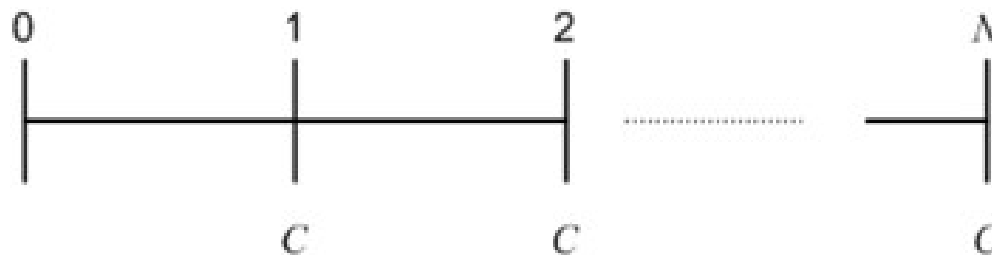
- You would need to donate \$2.5 million to endow the chair.

Taken from the slides accompanying Berk/DeMarzo: Corporate Finance, 6th edition.

Part One: Time and Interest Rates 1. Time Value of Money

Perpetuities and Annuities: Annuities

- When a constant cash flow will occur at regular intervals for a finite number of N periods, it is called an annuity.



- Present Value of an Annuity

$$PV = \frac{C}{(1+r)} + \frac{C}{(1+r)^2} + \frac{C}{(1+r)^3} + \cdots + \frac{C}{(1+r)^N} = \sum_{n=1}^N \frac{C}{(1+r)^n}$$

Taken from the slides accompanying Berk/DeMarzo: Corporate Finance, 6th edition.

Part One: Time and Interest Rates 1. Time Value of Money

Perpetuities and Annuities: Annuities

- Calculating the present value of an annuity this way can be rather burdensome.
- Fortunately, there is a **shortcut** (for a derivation see Berk/DeMarzo, p. 152-153):
- **PV of an annuity of constant payments C for N periods:**

$$\text{PV (annuity of C for N periods with interest rate } r) = \frac{C}{r} \left(1 - \frac{1}{(1 + r)^N} \right)$$

Taken from the slides accompanying Berk/DeMarzo: Corporate Finance, 6th edition.

Part One: Time and Interest Rates 1. Time Value of Money

Perpetuities and Annuities: Growing Annuity

- The present value of a growing annuity with the initial cash flow C, growth rate g, and interest rate r is defined as:
- Present Value of a Growing Annuity:

$$PV = C \times \frac{1}{(r - g)} \left(1 - \left(\frac{1 + g}{1 + r} \right)^N \right)$$

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Part One: Time and Interest Rates 1. Time Value of Money

Perpetuities and Annuities: Annuities

- **Problem: Present Value of a Lottery Prize Annuity**
 - You are the lucky winner of the \$30 million state lottery. You can take your prize money either as **(a)** 30 payments of \$1 million per year (**starting today**), or **(b)** \$15 million paid today. If the interest rate is 8%, which option should you take?

Taken from the slides accompanying Berk/DeMarzo: Corporate Finance, 6th edition.

Part One: Time and Interest Rates 1. Time Value of Money

Perpetuities and Annuities: Annuities

- **Solution: Present Value of a Lottery Prize Annuity**
 - Option (a) provides \$30 million of prize money but paid annually. In this case, the cash flows are an annuity in which the **first payment begins immediately**, sometimes called an **annuity due**.
 - Because the first payment is paid today, the last payment will occur in 29 years (for a total of 30 payments). We can compute the present value of the final 29 payments as a standard annuity of \$1 million per year using the annuity formula:

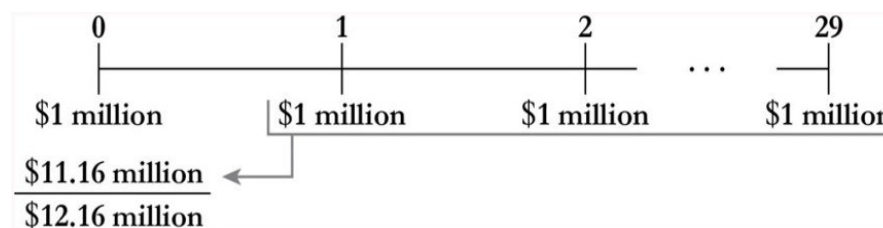
$$\begin{aligned} \text{PV} \left(29 \text{ yr annuity of } \$1 \frac{\text{million}}{\text{yr}} \right) &= \$1 \text{ million} \times \frac{1}{.08} \left(1 - \frac{1}{1.08^{29}} \right) \\ &= \$11.16 \text{ million today} \end{aligned}$$

Taken from the slides accompanying Berk/DeMarzo: Corporate Finance, 6th edition.

Part One: Time and Interest Rates 1. Time Value of Money

Perpetuities and Annuities: Annuities

- **Solution: Present Value of a Lottery Prize Annuity**
 - Adding the \$1 million we receive upfront, this option has a present value of \$12.16 million:



- Therefore, the present value of option (a) is only \$12.16 million, and so it is more valuable to take option (b) and receive \$15 million upfront—even though we receive only half the total cash amount. The difference, of course, is due to the time value of money.

Taken from the slides accompanying Berk/DeMarzo: Corporate Finance, 6th edition.

Part One: Time and Interest Rates 1. Time Value of Money

Perpetuities and Annuities: Annuities

- **Future Value of an Annuity**

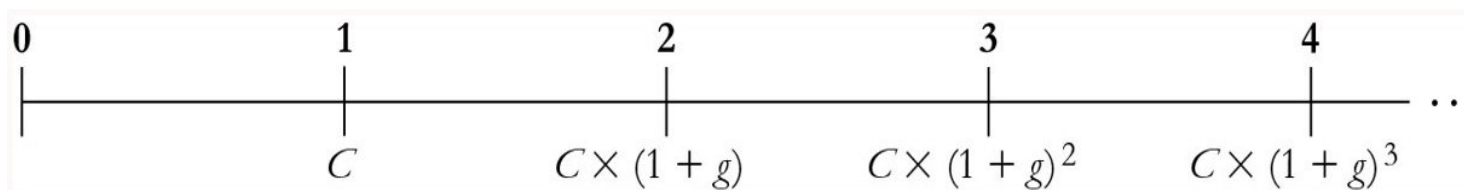
$$\begin{aligned}\text{FV (annuity)} &= \text{PV} \times (1 + r)^N \\ &= \frac{C}{r} \left(1 - \frac{1}{(1 + r)^N} \right) \times (1 + r)^N \\ &= C \times \frac{1}{r} \left((1 + r)^N - 1 \right)\end{aligned}$$

Taken from the slides accompanying Berk/DeMarzo: Corporate Finance, 6th edition.

Part One: Time and Interest Rates 1. Time Value of Money

Perpetuities and Annuities: Growing Perpetuity

- Assume you expect the amount of your perpetual payment to **increase at a constant rate, g** .



- Present Value of a Growing Perpetuity

$$\text{PV (growing perpetuity)} = \frac{C}{r - g}$$

Taken from the slides accompanying Berk/DeMarzo: Corporate Finance, 6th edition.

Part One: Time and Interest Rates 1. Time Value of Money

Perpetuities and Annuities: Growing Perpetuity

- **Problem: Endowing a Growing Perpetuity**

- Originally, you planned to donate money to your alma mater to fund an annual \$30,000 MBA graduation party. Given an interest rate of 8% per year, the required donation was the present value of

$$PV = \frac{\$30,000}{0.08} = \$375,000 \text{ today}$$

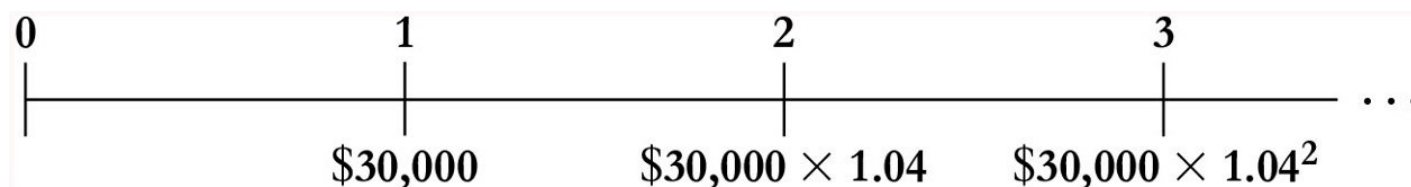
- Before accepting the money, however, the MBA student association has asked that you increase the donation to **account for the effect of inflation on the cost** of the party in future years. Although \$30,000 is adequate for next year's party, the students estimate that the party's cost will **rise by 4% per year thereafter**. To satisfy their request, how much do you need to donate now?

Taken from the slides accompanying Berk/DeMarzo: Corporate Finance, 6th edition.

Part One: Time and Interest Rates 1. Time Value of Money

Perpetuities and Annuities: Growing Perpetuity

- **Solution**



- The cost of the party next year is \$30,000, and the cost then increases 4% per year forever. From the timeline, we recognize the form of a growing perpetuity. To finance the growing cost, you need to provide the present value today of

$$PV = \frac{\$30,000}{0.08 - 0.04} = \$750,000 \text{ today}$$

- You need to double the size of your gift!

Taken from the slides accompanying Berk/DeMarzo: Corporate Finance, 6th edition.

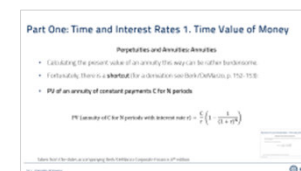
Part One: Time and Interest Rates 1. Time Value of Money

Perpetuities and Annuities: Growing Annuity

- The present value of a growing annuity with the initial cash flow C , growth rate g , and interest rate r is defined as:
- Present Value of a Growing Annuity

$$PV = C \times \frac{1}{(r - g)} \left(1 - \left(\frac{1 + g}{(1 + r)} \right)^N \right)$$

Taken from the slides accompanying Berk/DeMarzo: Corporate Finance, 6th edition.



Part One: Time and Interest Rates 1. Time Value of Money

Perpetuities and Annuities: Growing Annuity

- **Problem:**
 - You want to begin saving for your retirement. You plan to contribute \$12,000 to the account at the end of this year. **You anticipate you will be able to increase your annual contributions by 3% each year for the next 45 years.** If your expected annual return is 8%, how much do you expect to have in your retirement account when you retire in 45 years?

Taken from the slides accompanying Berk/DeMarzo: Corporate Finance, 6th edition.

Part One: Time and Interest Rates 1. Time Value of Money

Perpetuities and Annuities: Growing Annuity

- **Solution:**
 - The **present value** of the series of deposits is:

$$PV = \$12,000 \times \frac{1}{.08 - .03} \left[1 - \left[\frac{1 + .03}{1 + .08} \right]^{45} \right] = \$211,567$$

- The **future value** of the series of deposits is:

$$FV = \$211,567 \times (1.08)^{45} = \$6,753,314$$

Taken from the slides accompanying Berk/DeMarzo: Corporate Finance, 6th edition.

Part One: Time and Interest Rates 1. Time Value of Money

Structure of Chapter 1

- Timeline
- Three rules of time travel
- Valuing a stream of cash flows
- Calculating the net present value (NPV)
- Perpetuities and annuities
- Internal rate of return (IRR)

Part One: Time and Interest Rates 1. Time Value of Money

Structure of Chapter 1

- Timeline
- Three rules of time travel
- Valuing a stream of cash flows
- Calculating the net present value (NPV)
- Perpetuities and annuities
- Internal rate of return (IRR)

Part One: Time and Interest Rates 1. Time Value of Money

The Internal Rate of Return

- In some situations, you know the present value and cash flows of an investment opportunity, but you do not know the **internal rate of return (IRR)**, the interest rate that sets the net present value of the cash flows equal to zero.
- Calculation of the IRR is comparatively simple when there is only a **single future payment** or in cases of a **perpetuity**. Other cases require **approximation** of the IRR. We will apply **interpolation** in these cases.

Taken from the slides accompanying Berk/DeMarzo: Corporate Finance, 6th edition.

Part One: Time and Interest Rates 1. Time Value of Money

The Internal Rate of Return: Single Future Payment

- **Problem**
 - You can invest \$100,000 in an asset that promises you to pay back **\$150,000 after six years**. What is the IRR of this investment?
- **Solution**
 - Set up the NPV equation for this asset, set NPV=0 and solve for r:

$$\text{NPV} = -100,000 + \frac{150,000}{(1+r)^6} = 0$$

$$r = \sqrt[6]{\frac{150,000}{100,000}} - 1 = 0.0699 \approx 7\%$$

Part One: Time and Interest Rates 1. Time Value of Money

The Internal Rate of Return: Perpetuity

- **Problem**
 - Jessica has just graduated with her MBA . Rather than take the job she was offered at a prestigious investment bank—Baker, Bellingham, and Botts—she has decided to go into business for herself. She believes that her business will require an **initial investment of \$1 million. After that, it will generate a cash flow of \$100,000 at the end of one year, and this amount will grow by 4% per year thereafter.** What is the IRR of this investment opportunity?

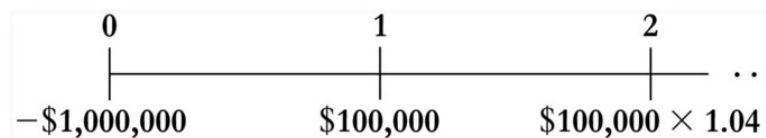
Taken from the slides accompanying Berk/DeMarzo: Corporate Finance, 6th edition.

Part One: Time and Interest Rates 1. Time Value of Money

The Internal Rate of Return: Perpetuity

- **Solution**

- The timeline is



- The timeline shows that the future cash flows are a growing perpetuity with a growth rate of 4%. Recall that the PV of a growing perpetuity is $\frac{C}{(r - g)}$.
- Thus, the NPV of this investment would equal zero if $1,000,000 = \frac{100,000}{r - 0.04}$
- We can solve this equation for r: $r = \frac{100,000}{1,000,000} + 0.04 = 0.14$
- So, the IRR on this investment is 14%.

Taken from the slides accompanying Berk/DeMarzo: Corporate Finance, 6th edition.

Part One: Time and Interest Rates 1. Time Value of Money

The Internal Rate of Return: Approximation via Interpolation

- **Problem**
 - Let us assume that you can invest \$10,000 into an asset promising cash flows of \$4,000, \$5,000, and \$3,000 at the end of year 1, year 2, and year 3, respectively.

Part One: Time and Interest Rates 1. Time Value of Money

The Internal Rate of Return: Approximation via Interpolation

- **Solution**

- Calculate an NPV_1 with an r_1 that leads to a positive NPV_1 .

$$NPV_1 (10\%) = -10,000 + \frac{4,000}{1.1} + \frac{5,000}{1.1^2} + \frac{3,000}{1.1^3} = 22.54$$

- Calculate an NPV_2 with an r_2 that leads to a negative NPV_2 .

$$NPV_2 (11\%) = -10,000 + \frac{4,000}{1.11} + \frac{5,000}{1.11^2} + \frac{3,000}{1.11^3} = -144.71$$

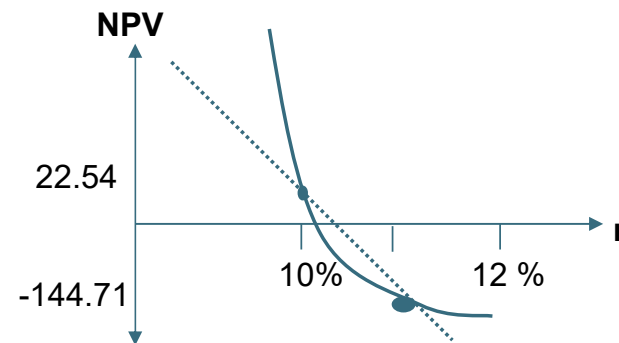
- Interpolate with the following formula
$$r \approx r_1 - \frac{NPV_1}{NPV_2 - NPV_1} \cdot (r_2 - r_1)$$

$$r \approx 0.1 - \frac{22.54}{-144.71 - 22.54} \cdot (0.11 - 0.1) = 0.1013 = 10.13\%$$

Part One: Time and Interest Rates 1. Time Value of Money

The Internal Rate of Return: Approximation via Interpolation

- Solution



Part One: Time and Interest Rates 1. Time Value of Money

Terms and concepts you should be familiar with after chapter 1

- timeline
- three rules of time travel
- present value, net present value (NPV), future value, internal rate of return (IRR)
- perpetuity, annuity
- calculating NPV, future value, and IRR for a variety of different cash flow streams

Part One: Time and Interest Rates 1. Time Value of Money

Formulas from chapter 1

- Future Value of a Cash Flow: $FV_n = C \times (1 + r) \times (1 + r) \times \dots \times (1 + r) = C \times (1 + r)^n$
- Present Value of a Cash Flow: $PV = C \div (1 + r)^n = \frac{C}{(1 + r)^n}$
- Net Present Value (NPV): compares the present value of cash inflows (benefits) to the present value of cash outflows (costs).
- Present Value of a Perpetuity (first payment at time 1): $PV(C \text{ in perpetuity}) = \frac{C}{r}$

Part One: Time and Interest Rates 1. Time Value of Money

Formulas from chapter 1

- Present Value of an Annuity: $PV (\text{annuity of } C \text{ for } N \text{ periods with interest rate } r) = \frac{C}{r} \left(1 - \frac{1}{(1 + r)^N} \right)$
- Future Value of an Annuity:
$$\begin{aligned} FV (\text{annuity}) &= PV \times (1 + r)^N \\ &= \frac{C}{r} \left(1 - \frac{1}{(1 + r)^N} \right) \times (1 + r)^N \\ &= C \times \frac{1}{r} ((1 + r)^N - 1) \end{aligned}$$

Part One: Time and Interest Rates 1. Time Value of Money

Formulas from chapter 1

- Present Value of a Growing Perpetuity: $PV (\text{growing perpetuity}) = \frac{C}{r - g}$
- Present Value of a Growing Annuity: $PV = C \times \frac{1}{(r - g)} \left(1 - \left(\frac{1 + g}{(1 + r)} \right)^N \right)$
- Approximate Internal Rate of Return (IRR) via Interpolation (with $NPV_1 > 0$ and $NPV_2 < 0$): $r \approx r_1 - \frac{NPV_1}{NPV_2 - NPV_1} \cdot (r_2 - r_1)$

Agenda

Part One: Time and Interest Rates

1. Time Value of Money 13.4.26

2. Investment Decision Rules 20./27.4.26

Part Two: Financial Instruments

3. Bonds 27.4./4.5.26

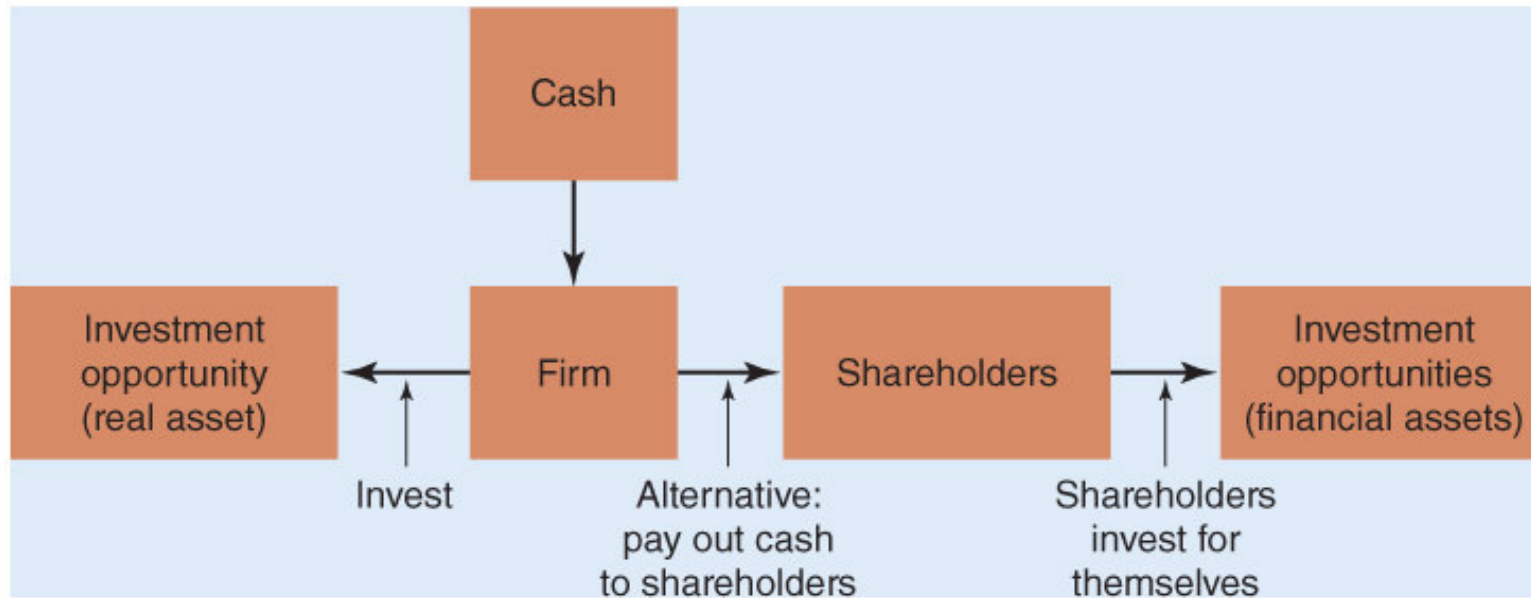
4. Stocks 4./11.5.26



Part One: Time and Interest Rates 2. Investment Decision Rules

- BDM, chapter 7
- BMM, chapter 8

The Course Journey



Taken from the slides accompanying Brealey/Myers/Marcus: Fundamentals of Corporate Finance, 11th edition.

capital
budgeting rules
(chapter 2)

- benchmark debt (ch. 3)
- benchmark equity (ch. 4)
- risk/return (ch. 5/6)
- WACC (ch. 6)

Part One: Time and Interest Rates 2. Investment Decision Rules

Terms and concepts you should be familiar with after chapter 2

- NPV and NPV rule
- IRR and pitfalls of IRR in capital budgeting
- Payback rule
- Profitability Index

Part One: Time and Interest Rates 2. Investment Decision Rules

Structure of Chapter 2

- Net Present Value (NPV) and Stand-Alone Projects
- The Internal Rate of Return (IRR)
- The Payback Rule
- Choosing between Projects
- Project Selection with Resource Constraints

Part One: Time and Interest Rates 2. Investment Decision Rules

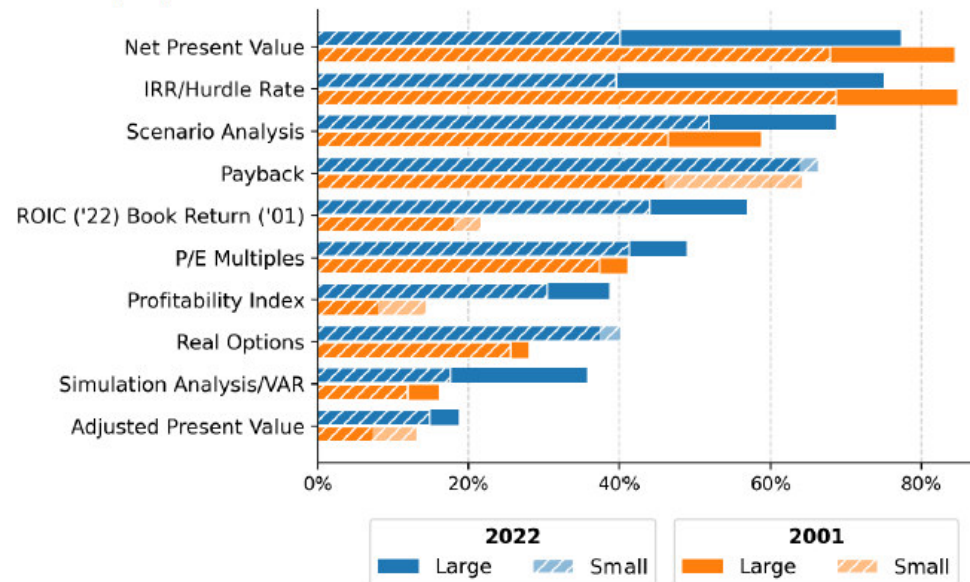
Structure of Chapter 2

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Part One: Time and Interest Rates 2. Investment Decision Rules

Figure 2. Capital Budgeting Techniques

This figure displays CFO responses to the following question about the capital budgeting methods their companies use to make investment decisions: *How frequently does your firm use the following techniques when deciding which projects or acquisitions to pursue?* {0 = Never, 1, 2, 3, 4 = Always}. The percentage of firms that answer 3 or 4 are shown. The blue bars display results for the 2022 Duke CFO survey (March 2020 wave); the orange bars display results from Graham and Harvey (2001). Within each blue and orange bar, the solid portion displays responses for large firms (revenue above \$1 billion), and the white, crosshatched portion displays responses for small firms (revenue below \$1 billion). For example, 77% of large firms (40% of small firms) say that Net Present Value was an important or very important decision rule in 2022.



Part One: Time and Interest Rates 2. Investment Decision Rules

The Payback Rule

- The payback period is the amount of time it takes to recover or pay back the initial investment.
- If the payback period is less than a pre-specified length of time, you accept the project.
- Otherwise, you reject the project.
- The payback rule is used by many companies because of its simplicity.

Learn from the data and apply the information to your own business.

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Graham, John R.: Presidential Address: Corporate Finance and Reality. In: Journal of Finance, vol. 77 (2022), p. 1975-2045, p. 1987.

Part One: Time and Interest Rates 2. Investment Decision Rules

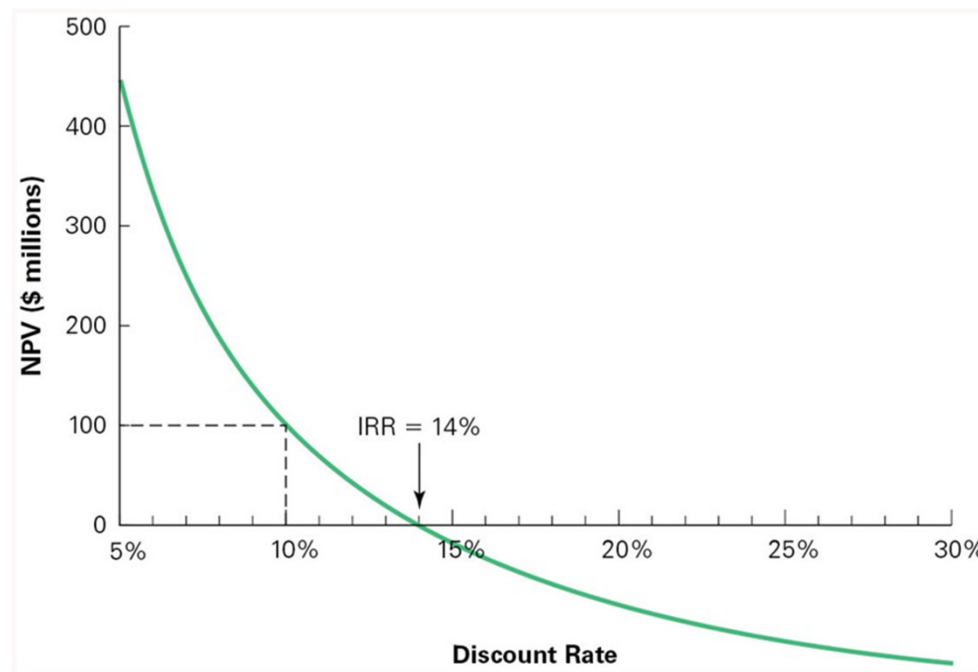
Net Present Value (NPV) and Stand-Alone Projects

- Consider a take-it-or-leave-it investment decision involving a single, stand-alone project for Fredrick's Feed and Farm (FFF).
 - The project costs \$250 million and is expected to generate cash flows of \$35 million per year, starting at the end of the first year and lasting forever.
- The NPV of the project is calculated as:
$$NPV = -250 + \frac{35}{r}$$
- The NPV is dependent on the discount rate.
- If FFF's cost of capital is 10%, the NPV is \$100 million, and they should undertake the investment.
- Generally: Take any investment where the NPV is positive.

Taken from the slides accompanying Berk/DeMarzo: Corporate Finance, 6th edition.

Part One: Time and Interest Rates 2. Investment Decision Rules

Net Present Value (NPV) and Stand-Alone Projects



Taken from the slides accompanying Berk/DeMarzo: Corporate Finance, 6th edition.

Part One: Time and Interest Rates 2. Investment Decision Rules

Net Present Value (NPV) and Stand-Alone Projects

- IRR measures the average return of the investment and the sensitivity of the NPV to any estimation error in the cost of capital.
- The difference between the cost of capital (=discount rate) and the IRR is the **maximum estimation error in the cost of capital** that can exist without altering the original decision.
 - In this example: discount rate 10%, IRR 14% → The NPV-based decision to accept this project is correct as long as our estimate of 10% is within 4 percentage points of the true cost of capital.

Taken from the slides accompanying Berk/DeMarzo: Corporate Finance, 6th edition.

Part One: Time and Interest Rates 2. Investment Decision Rules

Net Present Value (NPV) and Stand-Alone Projects

- Sometimes alternative investment rules may give the same answer as the NPV rule, but at other times they may disagree.
 - **When the rules conflict, the NPV decision rule should be followed.**

Taken from the slides accompanying Berk/DeMarzo: Corporate Finance, 6th edition.

Part One: Time and Interest Rates 2. Investment Decision Rules

Structure of Chapter 2

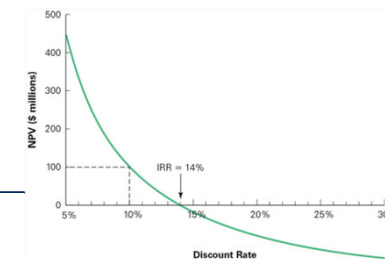
- Net Present Value (NPV) and Stand-Alone Projects
- The Internal Rate of Return (IRR)
- The Payback Rule
- Choosing between Projects
- Project Selection with Resource Constraints

Part One: Time and Interest Rates 2. Investment Decision Rules

Internal Rate of Return (IRR) Investment Rule

- Take any investment where the IRR exceeds the cost of capital.
- Turn down any investment whose IRR is less than the cost of capital.
- The IRR Investment Rule will give the same answer as the NPV rule **in many, but not all, situations.**
- In general, the IRR rule works for a stand-alone project **if all of the project's negative cash flows precede its positive cash flows.**
- In the preceding figure, whenever the cost of capital is below the IRR of 14%, the project has a positive NPV, and you should undertake the investment.

Taken from the slides accompanying Berk/DeMarzo: Corporate Finance, 6th edition.



Part One: Time and Interest Rates 2. Investment Decision Rules

Internal Rate of Return (IRR) Investment Rule: Applying the IRR Rule

- In other cases, the IRR rule may disagree with the NPV rule and thus be incorrect.
- **Situations where the IRR rule and NPV rule may be in conflict:**
 - Delayed Investments
 - Nonexistent IRR
 - Multiple IRRs

Taken from the slides accompanying Berk/DeMarzo: Corporate Finance, 6th edition.

Part One: Time and Interest Rates 2. Investment Decision Rules

Internal Rate of Return (IRR) Investment Rule: Pitfall 1: Delayed Investments

- Assume you have just retired as the CEO of a successful company. A major publisher has offered you a book deal. The publisher will pay you **\$1 million upfront** if you agree to write a book about your experiences. You estimate that it will take **three years** to write the book. The time you spend writing will cause you to give up speaking engagements amounting to **\$500,000 per year**. You estimate your **opportunity cost to be 10%**.

Taken from the slides accompanying Berk/DeMarzo: Corporate Finance, 6th edition.

Part One: Time and Interest Rates 2. Investment Decision Rules

Internal Rate of Return (IRR) Investment Rule: Pitfall 1: Delayed Investments

- Should you accept the deal? Calculate the IRR.
- IRR=23.38%
$$r \approx r_1 - \frac{NPV_1}{NPV_2 - NPV_1} \cdot (r_2 - r_1)$$
- The IRR is greater than the cost of capital. → Thus, the IRR rule indicates you should **accept** the deal.

Taken from the slides accompanying Berk/DeMarzo: Corporate Finance, 6th edition.

Part One: Time and Interest Rates 2. Investment Decision Rules

Internal Rate of Return (IRR) Investment Rule: Pitfall 1: Delayed Investments

- Should you accept the deal? Calculate the NPV.

$$\text{NPV} = 1,000,000 - \frac{500,000}{1.1} - \frac{500,000}{1.1^2} - \frac{500,000}{1.1^3} = -\$243,426$$

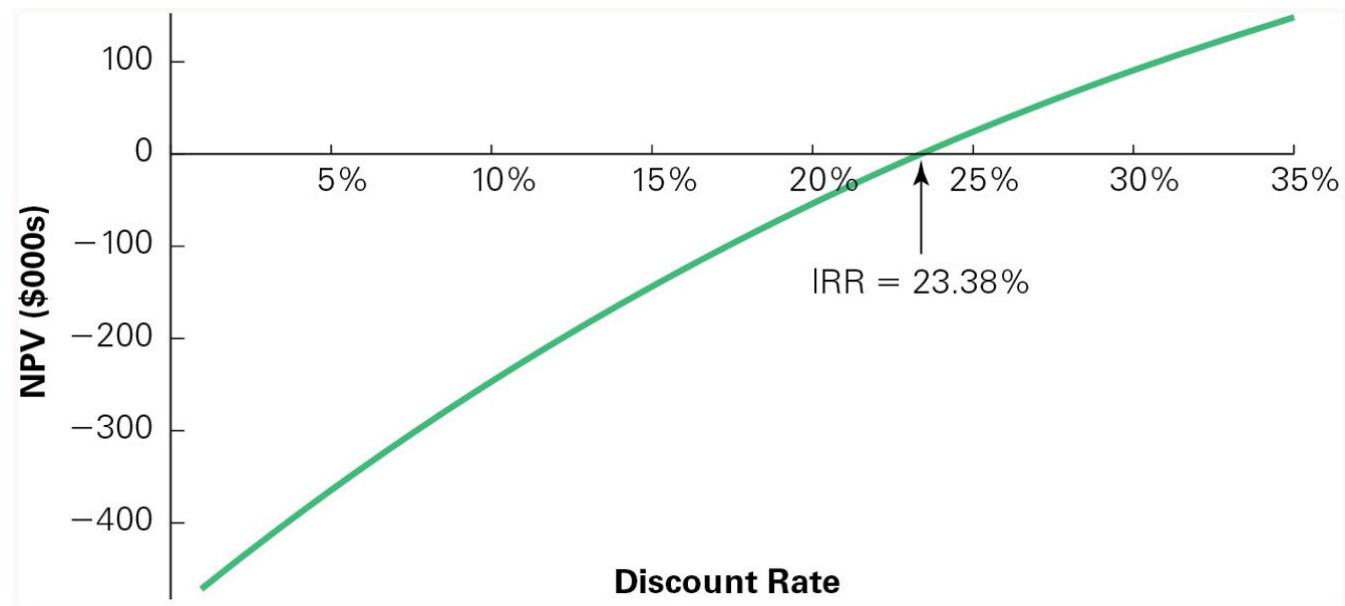
- Since the NPV is negative, the NPV rule indicates you should **reject** the deal.

Taken from the slides accompanying Berk/DeMarzo: Corporate Finance, 6th edition.

Part One: Time and Interest Rates 2. Investment Decision Rules

Internal Rate of Return (IRR) Investment Rule: Pitfall 1: Delayed Investments

- **When the benefits of an investment occur before the costs**, the NPV is an increasing function of the discount rate.

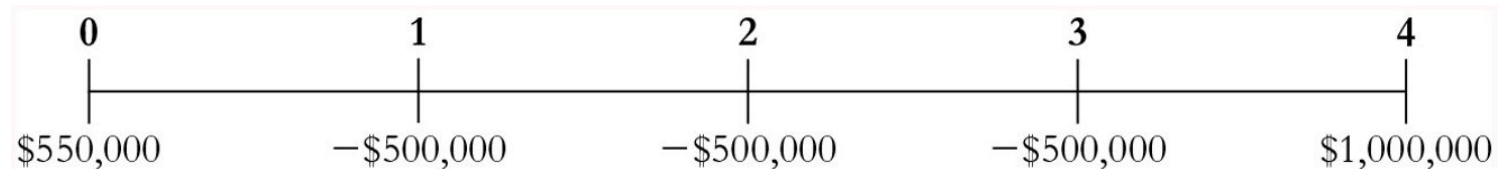


Taken from the slides accompanying Berk/DeMarzo: Corporate Finance, 6th edition.

Part One: Time and Interest Rates 2. Investment Decision Rules

Internal Rate of Return (IRR) Investment Rule: Pitfall 2: Multiple IRRs

- Suppose you inform the publisher that it needs to sweeten the deal before you will accept it. The publisher offers \$550,000 in advance and \$1,000,000 in four years when the book is published.
- Should you accept or reject the new offer?
- The cash flows would now look like

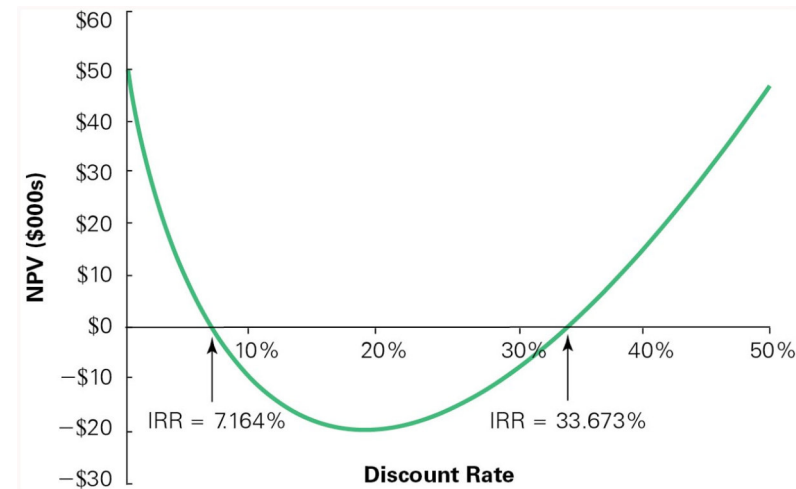


Taken from the slides accompanying Berk/DeMarzo: Corporate Finance, 6th edition.

Part One: Time and Interest Rates 2. Investment Decision Rules

Internal Rate of Return (IRR) Investment Rule: Pitfall 2: Multiple IRRs

- In this case, there are two IRRs : 7.164% and 33.673%.
- Because there is more than one IRR, the IRR rule **cannot** be applied.
- As seen in the figure, for your opportunity cost of capital of 10%, the deal has a **negative NPV** and, thus, you should reject the deal.

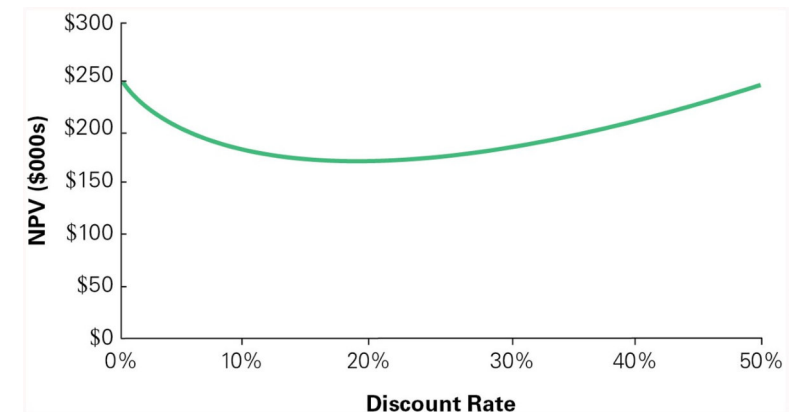


Taken from the slides accompanying Berk/DeMarzo: Corporate Finance, 6th edition.

Part One: Time and Interest Rates 2. Investment Decision Rules

Internal Rate of Return (IRR) Investment Rule: Pitfall 3: Nonexistent IRR

- Finally, you are able to get the publisher to increase his advance to \$750,000, in addition to the \$1 million when the book is published in four years.
- With these cash flows, **no IRR exists**; there is no discount rate that makes NPV equal to zero.
- No IRR exists because the NPV is positive for all values of the discount rate. Thus, the **IRR rule cannot be used**.



Taken from the slides accompanying Berk/DeMarzo: Corporate Finance, 6th edition.

Part One: Time and Interest Rates 2. Investment Decision Rules

Internal Rate of Return (IRR) Investment Rule: Problems with the IRR Rule

- **Problem**
 - Consider projects with the following cash flows:

Project	0	1	2
A	-375	-300	900
B	-22,222	50,000	-28,000
C	400	400	-1,056
D	-4,300	10,000	-6,000

- **For which of these projects is the IRR rule valid?** Which of these projects have an IRR close to 20%?

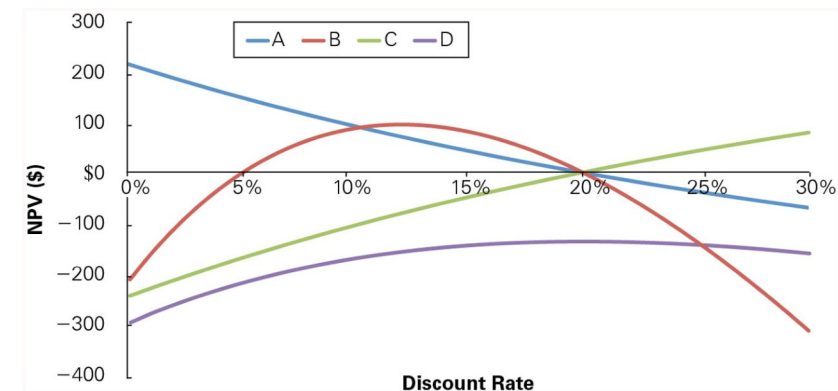
Taken from the slides accompanying Berk/DeMarzo: Corporate Finance, 6th edition.

Part One: Time and Interest Rates 2. Investment Decision Rules

Internal Rate of Return (IRR) Investment Rule: Problems with the IRR Rule

- **Solution**

- We plot the NPV profile for each project. From the NPV profiles, we can see that **projects A, B, and C** each have an IRR of approximately 20%, where **project D** has no IRR. Note also that **project B** has another IRR of 5%.
- The IRR rule is valid only if the project has a positive NPV for every discount rate below the IRR. **Thus, the IRR rule is only valid for project A.** this project is the only one for which all the negative cash flows precede the positive ones. **While the IRR Rule works for project A, it fails for each of the other projects.**



Project	0	1	2
A	-375	-300	900
B	-22,222	50,000	-28,000
C	400	400	-1,056
D	-4,300	10,000	-6,000

Taken from the slides accompanying Berk/DeMarzo: Corporate Finance, 6th edition.

Part One: Time and Interest Rates 2. Investment Decision Rules

Structure of Chapter 2

- Net Present Value (NPV) and Stand-Alone Projects
- The Internal Rate of Return (IRR)
- The Payback Rule
- Choosing between Projects
- Project Selection with Resource Constraints

Part One: Time and Interest Rates 2. Investment Decision Rules

The Payback Rule

- **The payback period is the amount of time it takes to recover or pay back the initial investment.**
- If the payback period is less than a pre-specified length of time, you accept the project.
- Otherwise, you reject the project.
- The payback rule is used by many companies because of its simplicity.



Taken from the slides accompanying Berk/DeMarzo: Corporate Finance, 6th edition.

Part One: Time and Interest Rates 2. Investment Decision Rules

The Payback Rule

- **Problem**
 - Assume Fredrick's requires all projects to have a payback period of five years or less. Would the firm undertake the fertilizer project under this rule?
 - The project costs \$250 million and is expected to generate cash flows of \$35 million per year, starting at the end of the first year and lasting forever.

Taken from the slides accompanying Berk/DeMarzo: Corporate Finance, 6th edition.

Part One: Time and Interest Rates 2. Investment Decision Rules

The Payback Rule

- **Solution**
 - Recall that the project requires an initial investment of \$250 million, and will generate \$35 million per year. The sum of the cash flows from year 1 to year 5 is $\$35 \times 5 = \175 million, which will not cover the initial investment of \$250 million. In fact, it will not be until year 8 that the initial investment will be paid back ($\$35 \times 8 = \280 million).
 - Because the payback period for this project exceeds five years, Fredrick's will reject the project

Taken from the slides accompanying Berk/DeMarzo: Corporate Finance, 6th edition.

Part One: Time and Interest Rates 2. Investment Decision Rules

The Payback Rule: Pitfalls

- Ignores the project's cost of capital and time value of money. (But this could be mitigated with the dynamic payback method that applies discounted cash flows.)
- Ignores cash flows after the payback period.
- Relies on an ad hoc decision criterion.

Taken from the slides accompanying Berk/DeMarzo: Corporate Finance, 6th edition.

Part One: Time and Interest Rates 2. Investment Decision Rules

Intermediate Exercise

You are considering opening a new plant. The plant will cost \$100 million upfront. After that, it is expected to produce profits of \$30 million at the end of every year. The cash flows are expected to last forever.

- Calculate the NPV of this investment opportunity if your cost of capital is 8%. Should you make the investment? (BDM 7.3)
- Does the IRR rule agree with the NPV rule? Calculate and explain. (BDM 7.11)

Part One: Time and Interest Rates 2. Investment Decision Rules

Structure of Chapter 2

- Net Present Value (NPV) and Stand-Alone Projects
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Part One: Time and Interest Rates 2. Investment Decision Rules

Choosing between Projects: Mutually Exclusive Projects

- When you must choose only one project among several possible projects, the choice is mutually exclusive.
- **NPV Rule**
 - Select the project with the highest positive NPV.
- **IRR Rule**
 - Selecting the project with the highest IRR may lead to mistakes.

Taken from the slides accompanying Berk/DeMarzo: Corporate Finance, 6th edition.

Part One: Time and Interest Rates 2. Investment Decision Rules

Choosing between Projects: NPV and Mutually Exclusive Projects

- **Problem**

- A small commercial property is for sale near your university. Given its location, you believe a student-oriented business would be very successful there. You have researched several possibilities and come up with the following cash flow estimates (including the cost of purchasing the property). Which investment should you choose?

Project	Initial Investment	First-Year Cash Flow	Growth Rate	Cost of Capital
Book store	\$300,000	\$63,000	3.0%	8%
Coffee shop	\$400,000	\$80,000	3.0%	8%
Music store	\$400,000	\$104,000	0.0%	8%
Electronic store	\$400,000	\$100,000	3.0%	11%

Taken from the slides accompanying Berk/DeMarzo: Corporate Finance, 6th edition.



Part One: Time and Interest Rates 2. Investment Decision Rules

Choosing between Projects: NPV and Mutually Exclusive Projects

- **Solution**

- Assuming each business lasts indefinitely, we can compute the present value of the cash flows from each as a constant growth perpetuity. The NPV of each project is

$$\text{NPV (Book Store)} = -300,000 + \frac{63,000}{8\% - 3\%} = \$960,000$$

$$\text{NPV (Coffe Shop)} = -400,000 + \frac{80,000}{8\% - 3\%} = \$1,200,000$$

$$\text{NPV (Music Store)} = -400,000 + \frac{104,000}{8\%} = \$900,000$$

$$\text{NPV (Electronic Store)} = -400,000 + \frac{100,000}{11\% - 3\%} = \$850,000$$

- Thus, all of the alternatives have a positive NPV. But, because we can only choose one, the coffee shop is the best alternative.

Taken from the slides accompanying Berk/DeMarzo: Corporate Finance, 6th edition.

Part One: Time and Interest Rates 2. Investment Decision Rules

Choosing between Projects: IRR Rule and Mutually Exclusive Investments

- **Differences in Scale**
 - If a project's size is doubled, its NPV will double.
 - This is not the case with IRR.
 - Thus, the IRR rule cannot be used to compare projects of different scales.

Taken from the slides accompanying Berk/DeMarzo: Corporate Finance, 6th edition.

Part One: Time and Interest Rates 2. Investment Decision Rules

Choosing between Projects: IRR Rule and Mutually Exclusive Investments

- **Differences in Scale**
 - Consider two of the projects from the previous example:

	Book store	Coffee shop
Initial Investment	\$300,000	\$400,000
Cash Flow _{year 1}	\$63,000	\$80,000
Annual Growth Rate	3%	3%
Cost of capital	8%	8%
IRR	24%	23%
NPV	\$960,000	\$1,200,000

Taken from the slides accompanying Berk/DeMarzo: Corporate Finance, 6th edition.

Part One: Time and Interest Rates 2. Investment Decision Rules

Choosing between Projects: IRR Rule and Mutually Exclusive Investments

- **Timing of Cash Flows**

- Another problem with the IRR is that it can be affected by changing the timing of the cash flows, even when the scale is the same.
- IRR is a return, but the dollar value of earning a given return depends on how long the return is earned.
- Consider again the **coffee shop** and the **music store** investment in the example above.
- Both have the same initial scale and the same horizon.
- The coffee shop has a lower IRR (23% vs. 26%), but a higher NPV because of its higher growth rate. The relatively delayed cash flows make the coffee shop effectively a longer-term investment.



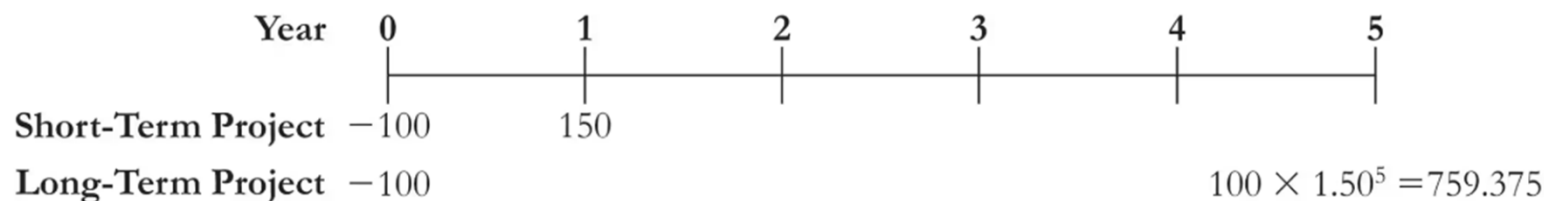
The thumbnail slide shows a table comparing two investment projects, likely the 'coffee shop' and 'music store' mentioned in the text. The table has columns for Year, Cash Flow, and NPV. The 'coffee shop' project has a lower IRR (23%) but a higher NPV compared to the 'music store' project (26% IRR).

Taken from the slides accompanying Berk/DeMarzo: Corporate Finance, 6th edition.

Part One: Time and Interest Rates 2. Investment Decision Rules

Choosing between Projects: IRR Rule and Mutually Exclusive Investments

- **Timing of Cash Flows: A Second Example**
 - Consider the following short-term and long-term projects:



- Both projects have an IRR of 50%, but one lasts for one year, while the other has a five-year horizon. If the cost of capital for both projects is 10%, the short-term project has an NPV of $-100 + 150/1.10 = \$36.36$, whereas the long-term project has an NPV of $-100 + 759.375/1.10^5 = \371.51 . Notice that despite having the same IRR, the long-term project is more than 10 times as valuable as the short-term project.

Taken from the slides accompanying Berk/DeMarzo: Corporate Finance, 6th edition.

Part One: Time and Interest Rates 2. Investment Decision Rules

Choosing between Projects: IRR Rule and Mutually Exclusive Investments

- **Differences in Risk**
 - An IRR that is attractive for a safe project need not be attractive for a riskier project.
 - Consider the investment in the **electronics store** from the example above.
 - The IRR is higher than those of the other investment opportunities (28%), yet the NPV is the lowest.
 - The higher cost of capital means a higher IRR is necessary to make the project attractive.



Part One: Time and Interest Rates 2. Investment Decision Rules

Choosing between Projects: IRR and Mutually Exclusive Investments

Project	Initial Investment	Year 1 Cash Flow	Year 2 Cash Flow	IRR	NPV (10%)
Safe Project	100	100	100	20%	10.00
Riskier Project	100	120	120	28%	5.00
Electronics Store	100	110	110	28%	2.00

Taken from the slides accompanying Berk/DeMarzo: Corporate Finance, 6th edition.

Part One: Time and Interest Rates 2. Investment Decision Rules

Structure of Chapter 2

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Part One: Time and Interest Rates 2. Investment Decision Rules

Project Selection with Resource Constraints

- Consider three possible projects with a \$100 million budget constraint:

Project	NPV(\$ millions)	Initial Investment (\$ millions)	Profitability Index NPV/Investment
I	110	100	1.1
II	70	50	1.4
III	60	50	1.2

- The **profitability index** can be used to identify the optimal combination of projects to undertake:

$$\text{Profitability Index} = \frac{\text{Value Created}}{\text{Resource Consumed}} = \frac{\text{NPV}}{\text{Resource Consumed}}$$

- It is better to take projects II and III together and forgo project I.

Taken from the slides accompanying Berk/DeMarzo: Corporate Finance, 6th edition.

Part One: Time and Interest Rates 2. Investment Decision Rules

Project Selection with Resource Constraints

- **Problem: Profitability Index with a Human Resource Constraint**
 - Your division at NetIt, a larger networking company, has put together a project proposal to develop a new home networking router. The expected NPV of the project is \$17.7 million, and the project will require 50 software engineers. NetIt has a total of 190 engineers available, and **the router project must compete with the following other projects for these engineers:**

Taken from the slides accompanying Berk/DeMarzo: Corporate Finance, 6th edition.

Part One: Time and Interest Rates 2. Investment Decision Rules

Project Selection with Resource Constraints

- **Problem: Profitability Index with a Human Resource Constraint**

Project	NPV (\$ millions)	Engineering Headcount
Router	17.7	50
Project A	22.7	47
Project B	8.1	44
Project C	14.0	40
Project D	11.5	61
Project E	20.6	58
Project F	12.9	32
Total	107.5	332

Taken from the slides accompanying Berk/DeMarzo: Corporate Finance, 6th edition.

Part One: Time and Interest Rates 2. Investment Decision Rules

Project Selection with Resource Constraints

- **Solution: Profitability Index with a Human Resource Constraint**
 - The goal is to maximize the total NPV we can create with 190 engineers (at most). We compute the profitability index for each project using Engineering Headcount in the denominator, and then sort projects based on the index:

Taken from the slides accompanying Berk/DeMarzo: Corporate Finance, 6th edition.

Part One: Time and Interest Rates 2. Investment Decision Rules

Project Selection with Resource Constraints

- **Solution: Profitability Index with a Human Resource Constraint**

Project	NPV (\$ millions)	Engineering Headcount (EHC)	Profitability Index (NPV per EHC)	Cumulative EHC Required
Project A	22.7	47	0.483	47
Project F	12.9	32	0.403	79
Project E	20.6	58	0.355	137
Router	17.7	50	0.354	187
Project C	14.0	40	0.350	
Project D	11.5	61	0.189	
Project B	8.1	44	0.184	

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Part One: Time and Interest Rates 2. Investment Decision Rules

Project Selection with Resource Constraints

- **Solution: Profitability Index with a Human Resource Constraint**
 - We now assign the resource to the projects in descending order according to the profitability index. The final column shows the cumulative use of the resource as each project is taken on until the resource is used up.
 - To **maximize NPV within the constraint of 190 engineers**, NetIt should choose the first four projects on the list. There is no other combination of projects that will create more value without using more engineers than we have.
 - Note, however, that the resource constraint **forces NetIt to forgo three otherwise valuable projects** (C, D, and B) with a total NPV of \$33.6 million.

Taken from the slides accompanying Berk/DeMarzo: Corporate Finance, 6th edition.

Part One: Time and Interest Rates 2. Investment Decision Rules

Project Selection with Resource Constraints

- **Shortcomings of the Profitability Index**
 - In some situations, the profitability Index does not give an accurate answer.
 - Suppose in the previous example that NetIt has an additional small project with an NPV of only \$120,000 that requires three engineers.
 - The profitability index in this case is $\frac{0.12}{3} = 0.04$, so this project would appear at the bottom of the ranking.
 - However, three of the 190 employees are not being used after the first four projects are selected. As a result, it would make sense to take on this project even though it would be ranked last.

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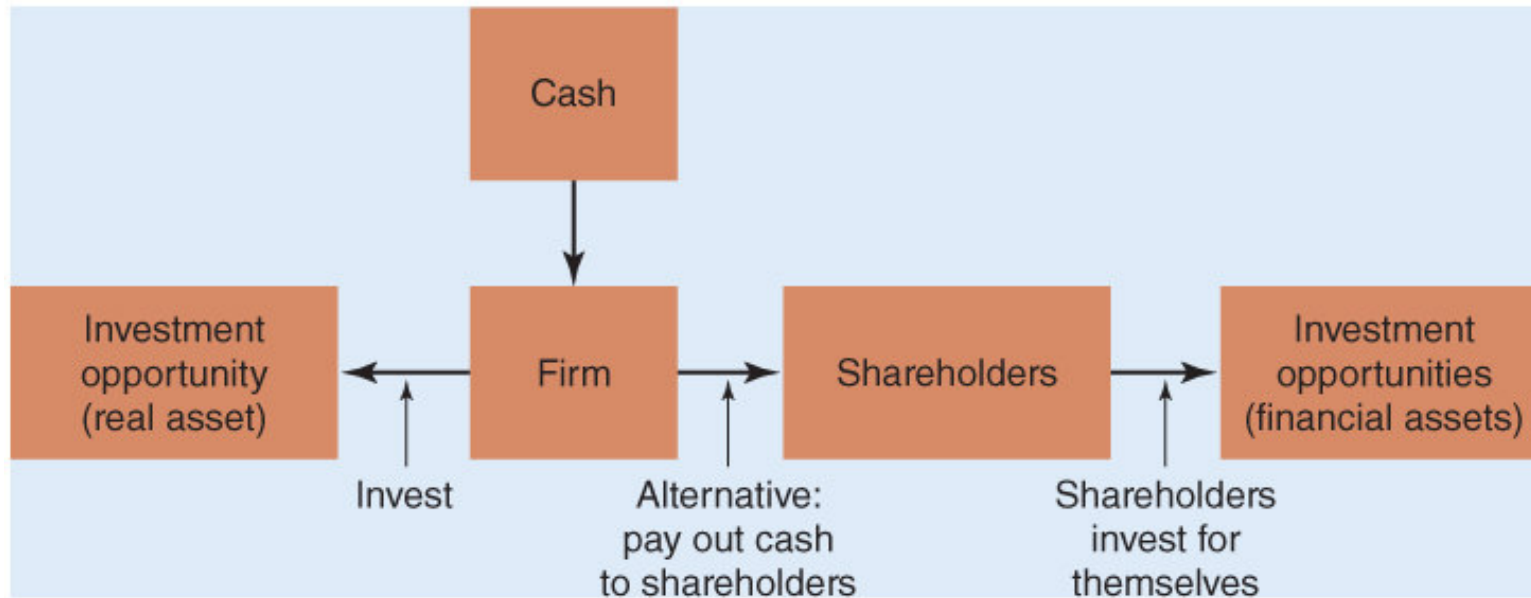
Part One: Time and Interest Rates 2. Investment Decision Rules

Project Selection with Resource Constraints

- **Shortcomings of the Profitability Index**
 - With multiple resource constraints, the profitability index can break down completely.

Taken from the slides accompanying Berk/DeMarzo: Corporate Finance, 6th edition.

The Course Journey



Taken from the slides accompanying Brealey/Myers/Marcus: Fundamentals of Corporate Finance, 11th edition.

capital
budgeting rules
(chapter 2)

- benchmark debt (ch. 3)
- benchmark equity (ch. 4)
- risk/return (ch. 5/6)
- WACC (ch. 6)

Part One: Time and Interest Rates 2. Investment Decision Rules

Terms and concepts you should be familiar with after chapter 2

- NPV and NPV rule
- IRR and pitfalls of IRR in capital budgeting
- Payback rule
- Profitability Index

Part One: Time and Interest Rates 2. Investment Decision Rules

Formulas from chapter 2

- Profitability Index = $\frac{\text{Value Created}}{\text{Resource Consumed}} = \frac{\text{NPV}}{\text{Resource Consumed}}$

Agenda

Part One: Time and Interest Rates

1. Time Value of Money 13.4.26

2. Investment Decision Rules 20./27.4.26

Part Two: Financial Instruments

3. Bonds 27.4./4.5.26

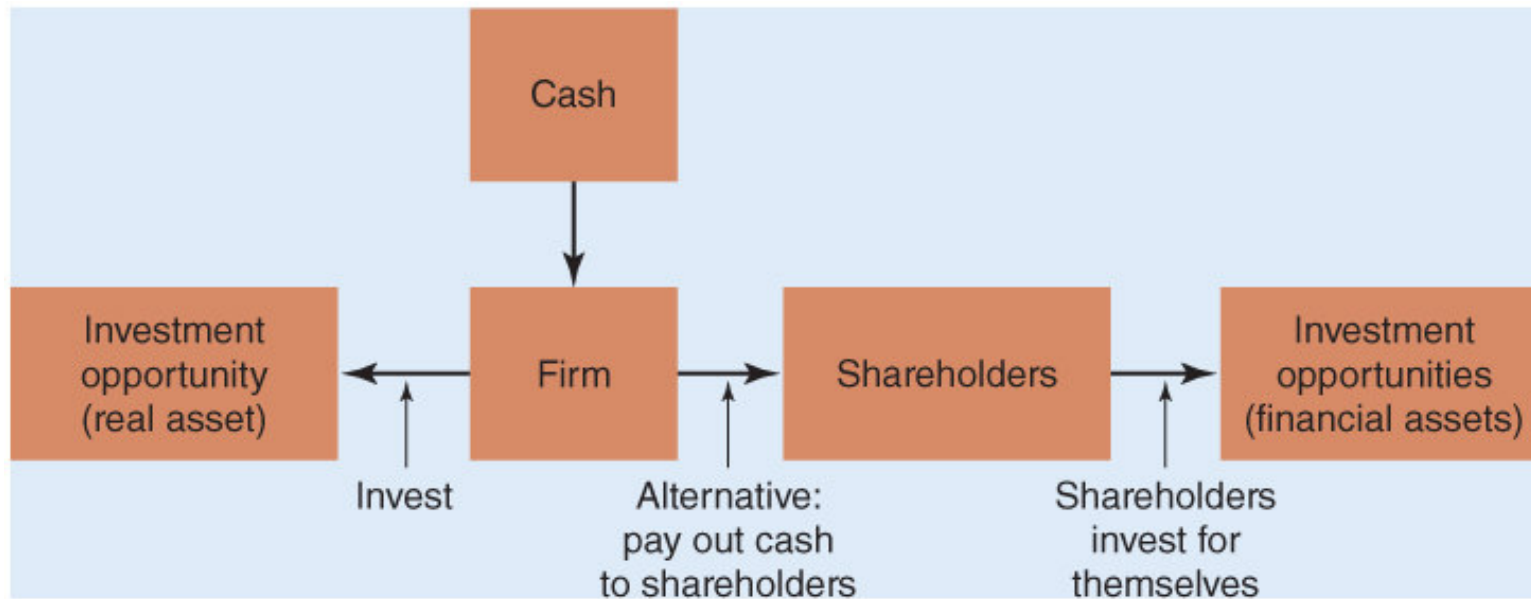
4. Stocks 4./11.5.26



Part Two: Financial Instruments 3. Bonds: Literature

- BDM, chapter 6
- BMM, chapter 6

The Course Journey



Taken from the slides accompanying Brealey/Myers/Marcus: Fundamentals of Corporate Finance, 11th edition.

capital
budgeting rules
(chapter 2)

- benchmark debt (ch. 3)
- benchmark equity (ch. 4)
- risk/return (ch. 5/6)
- WACC (ch. 6)

Part Two: Financial Instruments 3. Bonds

Terms and concepts you should be familiar with after chapter 3

- face value, coupon, coupon payment, coupon rate, coupon bond, zero-coupon bond
- bonds trading at a premium, at par, at a discount
- IRR of bonds
- reaction and sensitivity of bond prices to interest rate changes
- rating, default risk, default risk spread

Part Two: Financial Instruments 3. Bonds

Structure of Chapter 3

- Terminology
- Zero-Coupon Bonds
- Coupon Bonds
- Dynamic Behavior of Bond Prices
 - Discounts and Premiums
 - The Effect of Time on Bond Prices
 - Interest Rate Changes and Bond Prices
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Part Two: Financial Instruments 3. Bonds

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Part Two: Financial Instruments 3. Bonds

Terminology

- **Bond Certificate:** States the terms of the bond
- **Maturity Date:** Final repayment date
- **Term (Time to Maturity):** The time remaining until the repayment date
- **Coupon:** Promised interest payments
- **Face Value:** Notional amount used to compute the interest payments
- **Coupon Rate:** Determines the amount of each coupon payment, expressed as an APR (annual percentage rate)
- **Coupon Payment:**
$$CPN = \frac{\text{Coupon Rate} \times \text{Face Value}}{\text{Number of Coupon Payments per Year}}$$

Taken from the slides accompanying Berk/DeMarzo: Corporate Finance, 6th edition.



Part Two: Financial Instruments 3. Bonds

Terminology

- **Coupon Bonds**
 - Pay face value at maturity
 - Pay regular coupon interest payments
- **Zero-Coupon Bonds**
 - Do not make coupon payments
 - Always sell at a discount (a price lower than face value), so they are also called pure discount bonds
 - Treasury Bills are U.S. government zero-coupon bonds with a maturity of up to one year.
Germany: Bubills = unverzinsliche Schatzanweisungen

Taken from the slides accompanying Berk/DeMarzo: Corporate Finance, 6th edition.

Part Two: Financial Instruments 3. Bonds

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Part Two: Financial Instruments 3. Bonds

Zero-Coupon Bonds

- Suppose that a one-year, risk-free, zero-coupon bond with a \$100,000 face value has an initial price of \$96,618.36. The cash flows would be



- Although the bond pays no “interest,” your **compensation** is the difference between the initial price and the face value.

Taken from the slides accompanying Berk/DeMarzo: Corporate Finance, 6th edition.

Part Two: Financial Instruments 3. Bonds

Zero-Coupon Bonds

- **Yield to Maturity**

- The discount rate that sets the present value of the promised bond payments equal to the current market price of the bond (=IRR when buying the bond at price P and holding it until maturity, receiving the face value).

- Price of a Zero-Coupon Bond (FV=face value) $P = \frac{FV}{(1 + YTM)^n}$

- For the one-year zero coupon bond: $96,618.36 = \frac{100,000}{(1 + YTM)}$

$$1 + YTM = \frac{100,000}{96,618.36} = 1.035$$

- Thus, the YTM is 3.5%.

Taken from the slides accompanying Berk/DeMarzo: Corporate Finance, 6th edition.

Part Two: Financial Instruments 3. Bonds

Zero-Coupon Bonds

- Yield to Maturity of an n-Year Zero-Coupon Bond

$$\text{YTM} = \left(\frac{\text{FV}}{\text{P}} \right)^{\frac{1}{n}} - 1$$

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Part Two: Financial Instruments 3. Bonds

Structure of Chapter 3

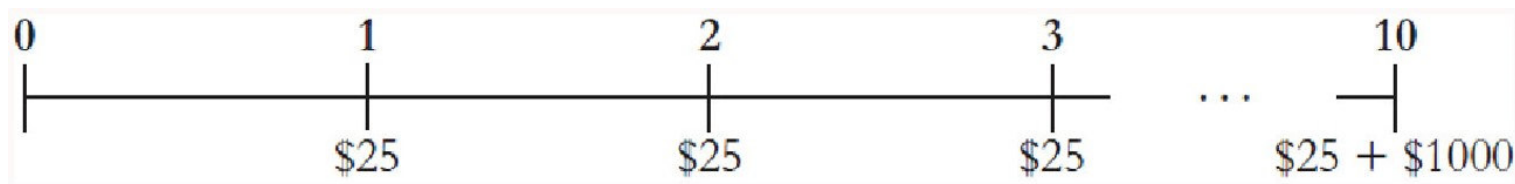
- Terminology
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- **Coupon Bonds**
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Part Two: Financial Instruments 3. Bonds

Coupon Bonds: Cash Flows

- **Problem**
 - The U.S. Treasury has just issued a five-year, \$1,000 bond with a 5% coupon rate and semiannual coupons. What cash flows will you receive if you hold this bond until maturity?

$$\text{CPN} = \$1,000 \times \frac{5\%}{2} = \$25.$$

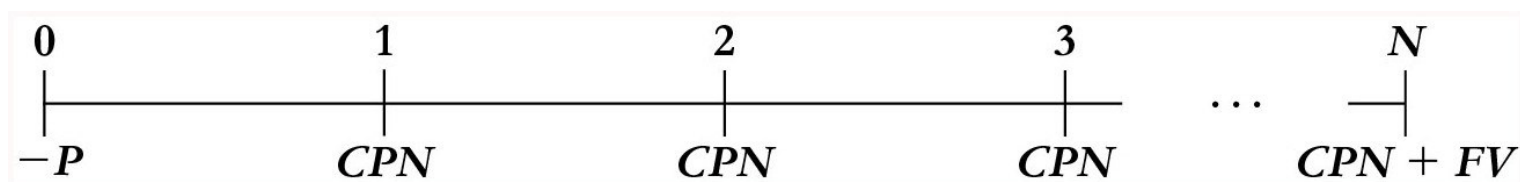


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Part Two: Financial Instruments 3. Bonds

Coupon Bonds: Yield to Maturity

- The YTM is the single discount rate that equates the present value of the bond's remaining cash flows to its current price.



- Yield to Maturity of a Coupon Bond:

$$P = CPN \times \frac{1}{YTM} \left(1 - \frac{1}{(1 + YTM)^N} \right) + \frac{FV}{(1 + YTM)^N}$$

Taken from the slides accompanying Berk/DeMarzo: Corporate Finance, 6th edition.



Part Two: Financial Instruments 3. Bonds

Coupon Bonds: Yield to Maturity

- **Problem**
 - Consider the five-year, \$1,000 bond with a 5% coupon rate and semiannual coupons described in the example above. If this bond is currently trading for a price of \$957.35, what is the bond's yield to maturity?

Taken from the slides accompanying Berk/DeMarzo: Corporate Finance, 6th edition.

Part Two: Financial Instruments 3. Bonds

Coupon Bonds: Yield to Maturity

- **Solution**

- Because the bond has 10 remaining coupon payments, we compute its yield y by solving:

$$957.35 = 25 \times \frac{1}{\text{YTM}} \left(1 - \frac{1}{(1 + \text{YTM})^{10}} \right) + \frac{1,000}{(1 + \text{YTM})^{10}}$$

- Approximation via interpolation requires calculating two NPVs, one positive and one negative:

$$\text{NPV} = -957.35 + 25 \times \frac{1}{\text{YTM}} \left(1 - \frac{1}{(1 + \text{YTM})^{10}} \right) + \frac{1,000}{(1 + \text{YTM})^{10}}$$

- $\text{NPV}_1(2\%) = 87.56$; $\text{NPV}_2(4\%) = -79.01$

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Part Two: Financial Instruments 3. Bonds

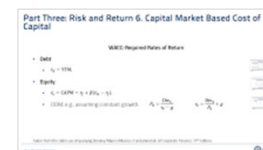
Coupon Bonds: Yield to Maturity

- **Solution**

- Interpolation formula:
$$r \approx r_1 - \frac{NPV_1}{NPV_2 - NPV_1} \cdot (r_2 - r_1)$$

- $r = y \approx 3.05\%$

- Because the bond pays coupons semiannually, this yield is for a six-month period. We convert it to an APR (annual percentage rate) by multiplying by the number of coupon payments per year. Thus, the bond has a yield to maturity equal to a **6.1% APR** with semiannual compounding.



Taken from the slides accompanying Berk/DeMarzo: Corporate Finance, 6th edition.

Part Two: Financial Instruments 3. Bonds

Intermediate Exercise

You buy the fixed coupon bond issued by Superdebt AG at a market price of EUR102. The bond has the following terms:

- Face value: EUR100
- Coupon rate: 6%
- Remaining time to maturity: 4 years

Draw the timeline of your investment assuming you hold the bond until it matures in four years.

What is the yield to maturity (IRR) that you will have earned after the four years? Write down the basic equation ($NPV=0$). If there is time, calculate the IRR.

Part Two: Financial Instruments 3. Bonds

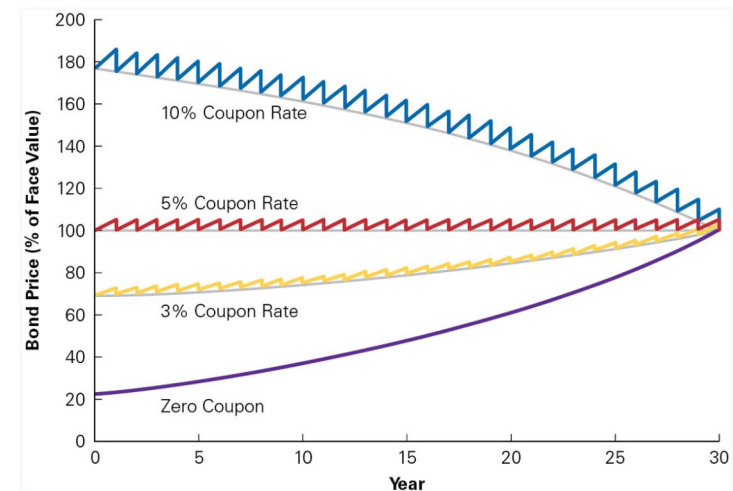
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Part Two: Financial Instruments 3. Bonds

Dynamic Behavior of Bond Prices

- **Discount:** A bond is selling at a **discount** if the price is less than the face value.
- **Par:** A bond is selling at **par** if the price is equal to the face value.
- **Premium:** A bond is selling at a **premium** if the price is greater than the face value.

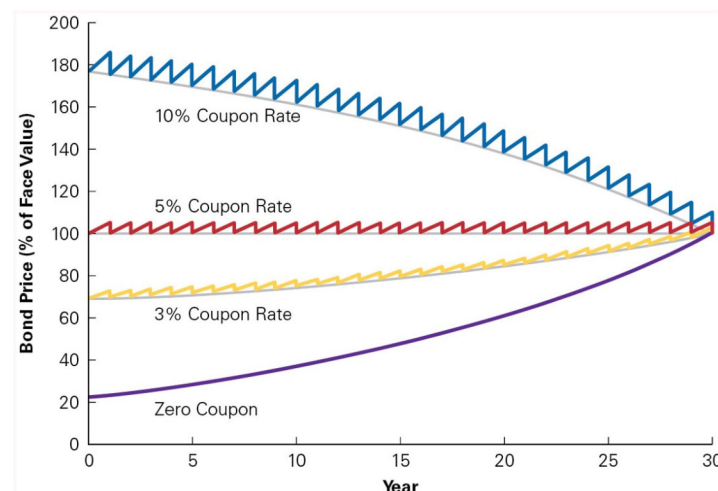


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Part Two: Financial Instruments 3. Bonds

Dynamic Behavior of Bond Prices: Discounts and Premiums

- If a coupon bond trades at a **discount**, an investor will earn a return both from receiving the coupons and from receiving a face value that exceeds the price paid for the bond.
- If a bond trades at a discount, its yield to maturity will exceed its coupon rate.

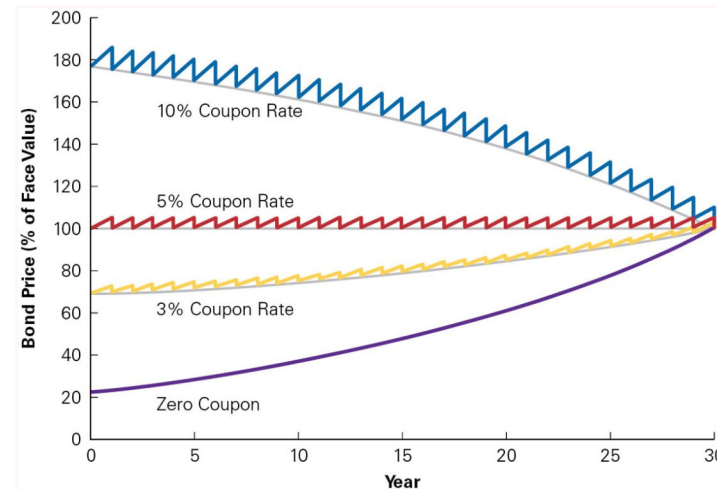


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Part Two: Financial Instruments 3. Bonds

Dynamic Behavior of Bond Prices: Discounts and Premiums

- If a coupon bond trades at a **premium**, it will earn a return from receiving the coupons, but this return will be diminished by receiving a face value less than the price paid for the bond.
- Most coupon bonds have a coupon rate so that the bonds **will initially trade at, or very close to, par.**

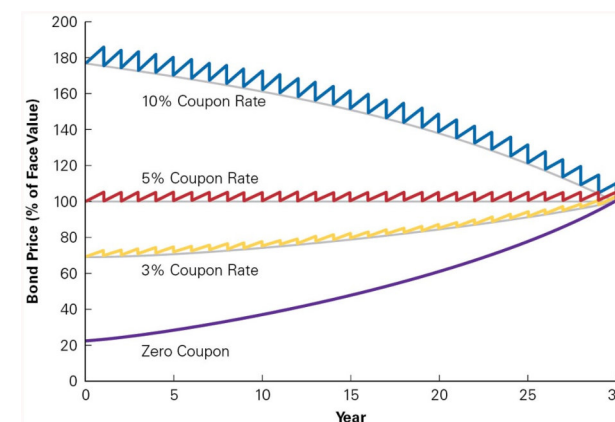


Taken from the slides accompanying Berk/DeMarzo: Corporate Finance, 6th edition.

Part Two: Financial Instruments 3. Bonds

Dynamic Behavior of Bond Prices: Discounts and Premiums

When the bond price is	We say the bond trades	This occurs when
greater than the face value	"above par" or "at a premium"	Coupon Rate > Yield to Maturity
equal to the face value	"at par"	Coupon Rate = Yield to Maturity
less than the face value	"below par" or "at a discount"	Coupon Rate < Yield to Maturity



Taken from the slides accompanying Berk/DeMarzo: Corporate Finance, 6th edition.

Part Two: Financial Instruments 3. Bonds

Dynamic Behavior of Bond Prices: Discounts and Premiums

- **Problem: Determining the Discount or Premium of a Coupon Bond**
 - Consider three 30-year bonds with annual coupon payments. One bond has a **10% coupon rate**, one has a **5% coupon rate**, and one has a **3% coupon rate**. If the **yield to maturity of each bond is 5%**, what is the price of each bond per \$100 face value? Which bond trades at a premium, which trades at a discount, and which trades at par?

Taken from the slides accompanying Berk/DeMarzo: Corporate Finance, 6th edition.

Part Two: Financial Instruments 3. Bonds

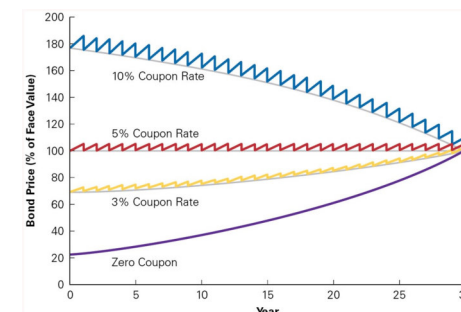
Dynamic Behavior of Bond Prices: Discounts and Premiums

- **Problem: Determining the Discount or Premium of a Coupon Bond**
 - We can compute the price of each bond:

$$P(10\% \text{ coupon}) = 10 \times \frac{1}{0.05} \left(1 - \frac{1}{1.05^{30}} \right) + \frac{100}{1.05^{30}} = \$176.86 \quad (\text{trades at a premium})$$

$$P(5\% \text{ coupon}) = 5 \times \frac{1}{0.05} \left(1 - \frac{1}{1.05^{30}} \right) + \frac{100}{1.05^{30}} = \$100.00 \quad (\text{trades at par})$$

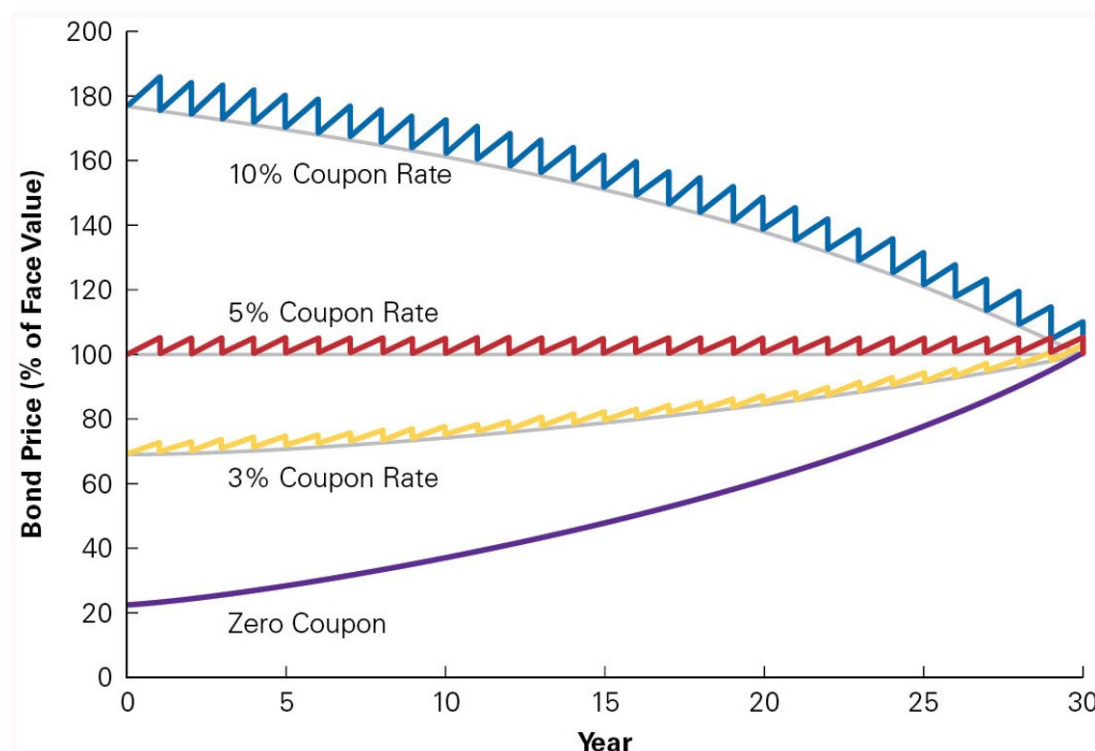
$$P(3\% \text{ coupon}) = 3 \times \frac{1}{0.05} \left(1 - \frac{1}{1.05^{30}} \right) + \frac{100}{1.05^{30}} = \$69.26 \quad (\text{trades at a discount})$$



Taken from the slides accompanying Berk/DeMarzo: Corporate Finance, 6th edition.

Part Two: Financial Instruments 3. Bonds

Dynamic Behavior of Bond Prices: The Effect of Time on Bond Prices



Taken from the slides accompanying Berk/DeMarzo: Corporate Finance, 6th edition.

Part Two: Financial Instruments 3. Bonds

Dynamic Behavior of Bond Prices: Interest Rate Changes and Bond Prices

- There is an **inverse relationship between interest rates and bond prices**.
 - As interest rates and bond yields rise, bond prices fall.
 - As interest rates and bond yields fall, bond prices rise.

Taken from the slides accompanying Berk/DeMarzo: Corporate Finance, 6th edition.

Part Two: Financial Instruments 3. Bonds

Dynamic Behavior of Bond Prices: Interest Rate Changes and Bond Prices

- The sensitivity of a bond's price to changes in interest rates depends on the bond's coupon rate and time to maturity.
 - Bonds with **lower coupon rates** are, all other things being equal, **more sensitive** to interest rate changes.
 - Bonds with **longer times to maturity** are, all other things being equal, **more sensitive** to interest rate changes.

Taken from the slides accompanying Berk/DeMarzo: Corporate Finance, 6th edition.

Part Two: Financial Instruments 3. Bonds

Dynamic Behavior of Bond Prices: Interest Rate Changes and Bond Prices

- **Problem**
 - Consider a **15-year zero-coupon bond** and a **30-year coupon bond with 10% annual coupons**. By what percentage will the price of each bond change if its yield to maturity **increases from 5% to 6%**?

Taken from the slides accompanying Berk/DeMarzo: Corporate Finance, 6th edition.

Part Two: Financial Instruments 3. Bonds

Dynamic Behavior of Bond Prices: Interest Rate Changes and Bond Prices

- **Solution**
 - First, we compute the price of each bond for each yield to maturity:

Yield to Maturity	15-Year, Zero-Coupon Bond	30-Year, 10% Annual Coupon Bond
5%	$\frac{100}{1.05^{15}} = \48.10	$10 \times \frac{1}{0.05} \left(1 - \frac{1}{1.05^{30}} \right) + \frac{100}{1.05^{30}} = \176.86
6%	$\frac{100}{1.06^{15}} = \41.73	$10 \times \frac{1}{0.06} \left(1 - \frac{1}{1.06^{30}} \right) + \frac{100}{1.06^{30}} = \155.06

Taken from the slides accompanying Berk/DeMarzo: Corporate Finance, 6th edition.

Part Two: Financial Instruments 3. Bonds

Dynamic Behavior of Bond Prices: Interest Rate Changes and Bond Prices

- **Solution**

- The price of the 15-year zero-coupon bond changes by $\frac{(41.73 - 48.10)}{48.10} = -13.2\%$
- For the 30-year bond with 10% annual coupons, the price change is $\frac{(155.06 - 176.86)}{176.86} = -12.3\%$.
- Even though the 30-year bond has a **longer maturity**, because of its **high coupon rate**, its sensitivity to a change in yield is actually less than that of the 15-year zero coupon bond.

Taken from the slides accompanying Berk/DeMarzo: Corporate Finance, 6th edition.

Part Two: Financial Instruments 3. Bonds

Intermediate Exercise

You buy the fixed coupon bond issued by Superdebt AG at a market price of EUR102. The bond has the following terms:

- Face value: EUR100
- Coupon rate: 6%
- Remaining time to maturity: 4 years

Why will your YTM after four years only be at 5.44% although the coupon rate of your bond is at 6%? How is this kind of bond called? What do you know about the general market interest rate level on your purchasing day?

Part Two: Financial Instruments 3. Bonds

Structure of Chapter 3

- Terminology
- Zero-Coupon Bonds
- Coupon Bonds
- Dynamic Behavior of Bond Prices
 - Discounts and Premiums
 - The Effect of Time on Bond Prices
 - Interest Rate Changes and Bond Prices
- **Default Risk**

Part Two: Financial Instruments 3. Bonds

Bond Issuers

- **Corporate Bonds**
 - Issued by corporations
- **Sovereign Bonds**
 - U.S. Treasury securities are generally considered to be (almost) default free. Same applies to German government bonds.

Taken from the slides accompanying Berk/DeMarzo: Corporate Finance, 6th edition.

Part Two: Financial Instruments 3. Bonds

Default Risk

- Neither sovereign nor corporate bonds are default-free,
 - e.g., Greece defaulted on its outstanding debt in 2012.
- Investors pay less for bonds with credit risk than they would for an otherwise identical default-free bond. (**price**)
- The **yield** of bonds with credit risk will be higher than that of otherwise identical default-free bonds.
- Default risk is measured in the form of **ratings**. Global market leaders: **Standard & Poor's** and **Moody's**.

Taken from the slides accompanying Berk/DeMarzo: Corporate Finance, 6th edition.

Part Two: Financial Instruments 3. Bonds

Default Risk: Rating

- **Investment Grade Bonds**
- **Speculative Bonds**, also known as **Junk Bonds** or **High-Yield Bonds**

Taken from the slides accompanying Berk/DeMarzo: Corporate Finance, 6th edition.

Part Two: Financial Instruments 3. Bonds

Default Risk: Rating

Rating*	Description (Moody's)
Investment Grade Debt	
Aaa/AAA	Judged to be of the best quality. They carry the smallest degree of investment risk and are generally referred to as “gilt edged.” Interest payments are protected by a large or an exceptionally stable margin and principal is secure. While the various protective elements are likely to change, such changes as can be visualized are most unlikely to impair the fundamentally strong position of such issues.
Aa/AA	Judged to be of high quality by all standards. Together with the Aaa group, they constitute what are generally known as high-grade bonds. They are rated lower than the best bonds because margins of protection may not be as large as in Aaa securities or fluctuation of protective elements may be of greater amplitude or there may be other elements present that make the long-term risk appear somewhat larger than the Aaa securities.
A/A	Possess many favorable investment attributes and are considered as upper-medium-grade obligations. Factors giving security to principal and interest are considered adequate, but elements may be present that suggest a susceptibility to impairment some time in the future.
Baa/BBB	Are considered as medium-grade obligations (i.e., they are neither highly protected nor poorly secured). Interest payments and principal security appear adequate for the present but certain protective elements may be lacking or may be characteristically unreliable over any great length of time. Such bonds lack outstanding investment characteristics and, in fact, have speculative characteristics as well.

Taken from the slides accompanying Berk/DeMarzo: Corporate Finance, 6th edition.

Part Two: Financial Instruments 3. Bonds

Default Risk: Rating

Speculative Bonds

Ba/BB	Judged to have speculative elements; their future cannot be considered as well assured. Often the protection of interest and principal payments may be very moderate, and thereby not well safeguarded during both good and bad times over the future. Uncertainty of position characterizes bonds in this class.
B/B	Generally lack characteristics of the desirable investment. Assurance of interest and principal payments of maintenance of other terms of the contract over any long period of time may be small.
Caa/CCC	Are of poor standing. Such issues may be in default or there may be present elements of danger with respect to principal or interest.
Ca/CC	Are speculative in a high degree. Such issues are often in default or have other marked shortcomings.
C/C, D	Lowest-rated class of bonds, and issues so rated can be regarded as having extremely poor prospects of ever attaining any real investment standing.

- Ratings: Moody's/Standard & Poor's

Taken from the slides accompanying Berk/DeMarzo: Corporate Finance, 6th edition.

Part Two: Financial Instruments 3. Bonds

Default Risk: Default Spread

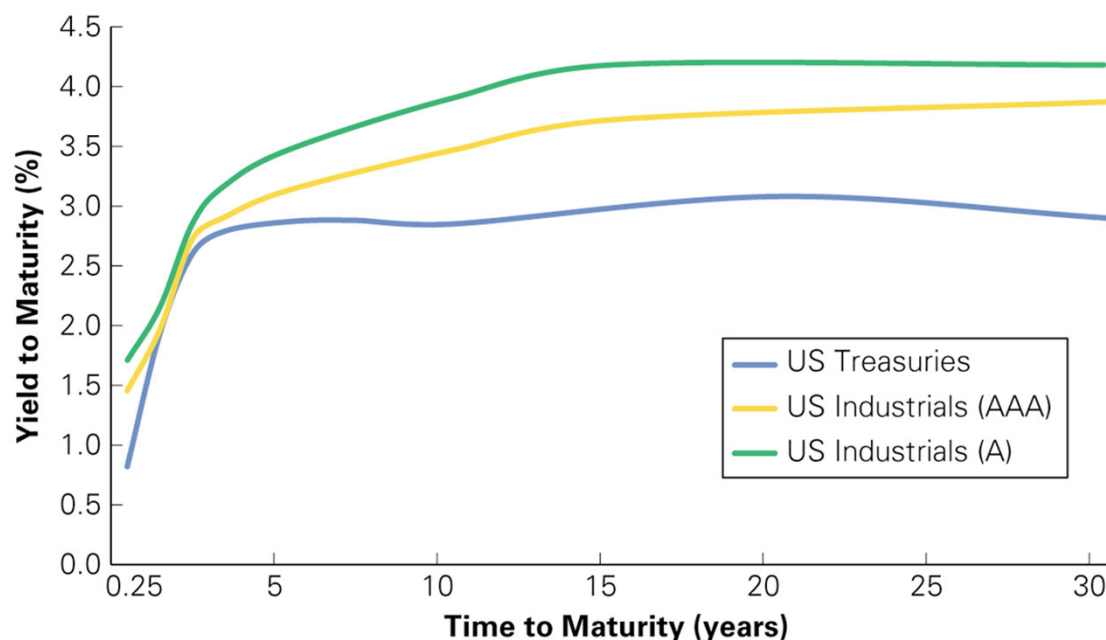
- **Default Spread (Credit Spread)**
 - The difference between the yield on corporate bonds and Treasury yields (or another benchmark considered to be (almost) risk-free)

Taken from the slides accompanying Berk/DeMarzo: Corporate Finance, 6th edition.

Part Two: Financial Instruments 3. Bonds

Default Risk: Default Spread

- Corporate Yield Curves for Various Ratings, April 2022



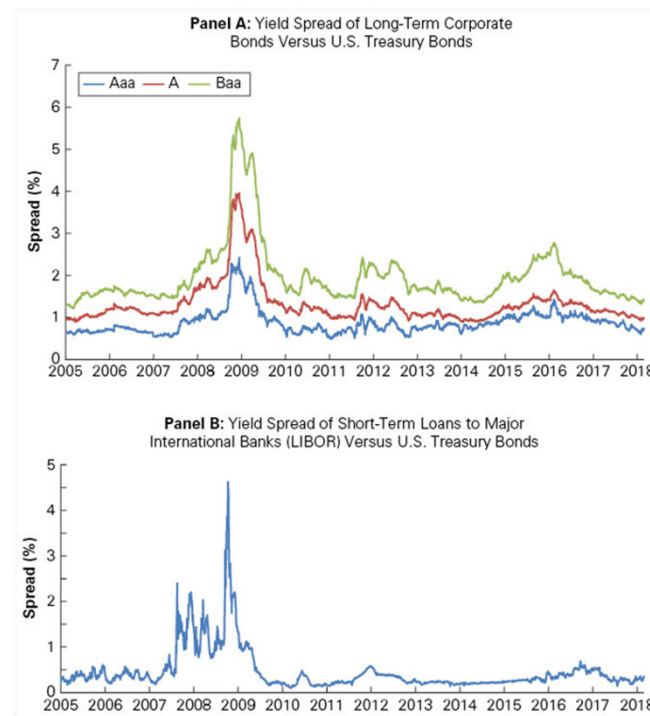
Source: Bloomberg.

Taken from the slides accompanying Berk/DeMarzo: Corporate Finance, 6th edition.

Part Two: Financial Instruments 3. Bonds

Default Risk: Default Spread

- Yield Spreads and the Financial Crisis



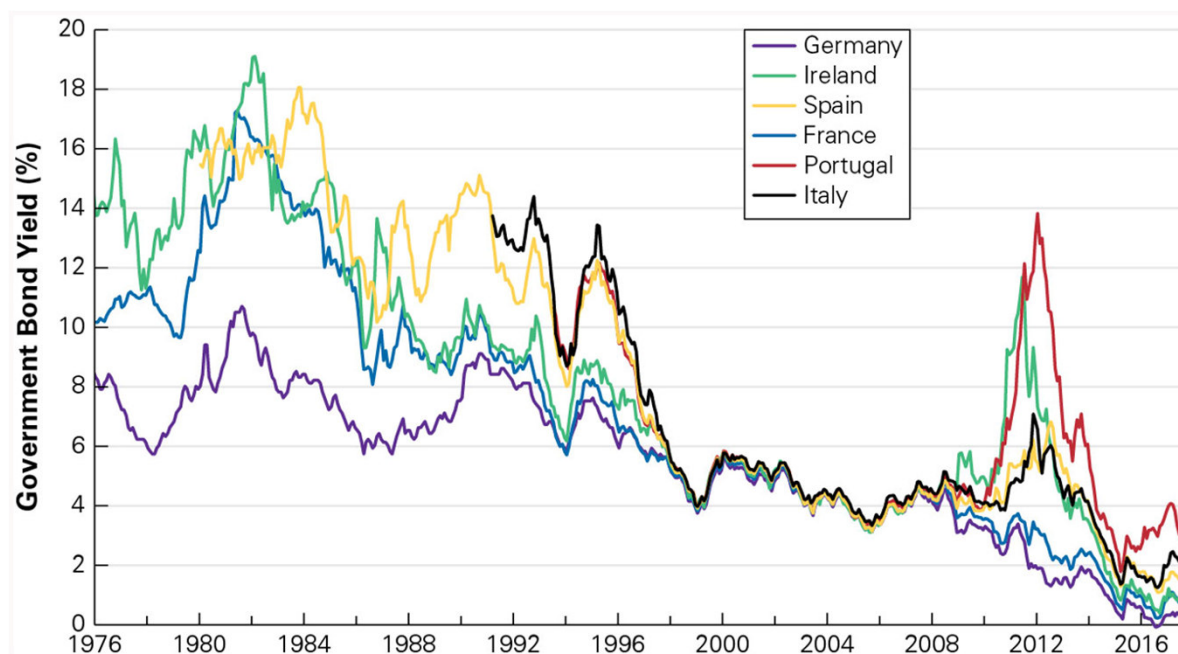
Source: Bloomberg.

Taken from the slides accompanying Berk/DeMarzo: Corporate Finance, 5th edition.

Part Two: Financial Instruments 3. Bonds

Default Risk: Default Spread

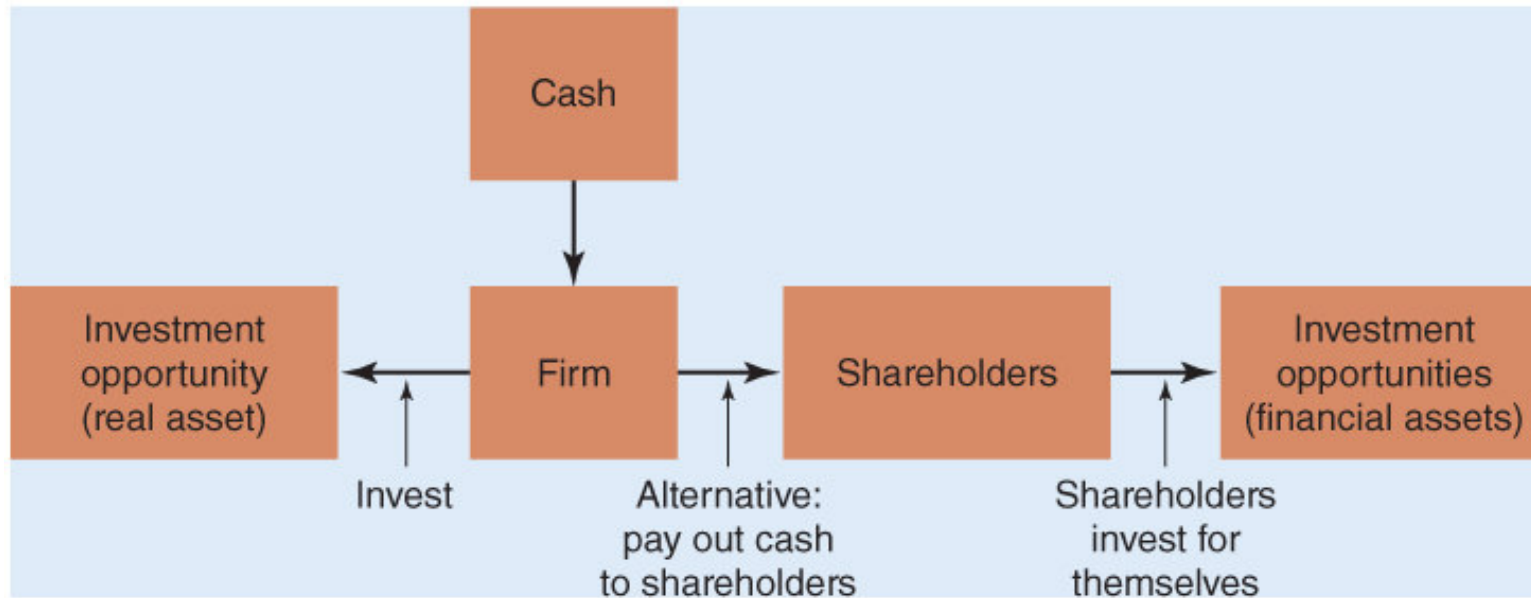
- European Government Bond Yields, 1976–2018



Source: Federal Reserve Economic Data.

Taken from the slides accompanying Berk/DeMarzo: Corporate Finance, 5th edition.

The Course Journey



Taken from the slides accompanying Brealey/Myers/Marcus: Fundamentals of Corporate Finance, 11th edition.

capital
budgeting rules
(chapter 2)

- benchmark debt (ch. 3)
- benchmark equity (ch. 4)
- risk/return (ch. 5/6)
- WACC (ch. 6)

Part Two: Financial Instruments 3. Bonds

Terms and concepts you should be familiar with after chapter 3

- face value, coupon, coupon payment, coupon rate, coupon bond, zero-coupon bond
- bonds trading at a premium, at par, at a discount
- IRR of bonds
- reaction and sensitivity of bond prices to interest rate changes
- rating, default risk, default risk spread

Part Two: Financial Instruments 3. Bonds

Formulas from chapter 3

- Coupon Payment: $CPN = \frac{\text{Coupon Rate} \times \text{Face Value}}{\text{Number of Coupon Payments per Year}}$
- Price of a Zero-Coupon Bond (FV=face value): $P = \frac{FV}{(1 + YTM)^n}$
- Yield to Maturity of an n-Year Zero-Coupon Bond: $YTM = \left(\frac{FV}{P}\right)^{\frac{1}{n}} - 1$
- Yield to Maturity of a Coupon Bond: discount rate y , that equates the PV of the bond's future cash flows with its price (=IRR):
$$P = CPN \times \frac{1}{YTM} \left(1 - \frac{1}{(1 + YTM)^N}\right) + \frac{FV}{(1 + YTM)^N}$$

Agenda

Part One: Time and Interest Rates

1. Time Value of Money 13.4.26

2. Investment Decision Rules 20./27.4.26

Part Two: Financial Instruments

3. Bonds 27.4./4.5.26

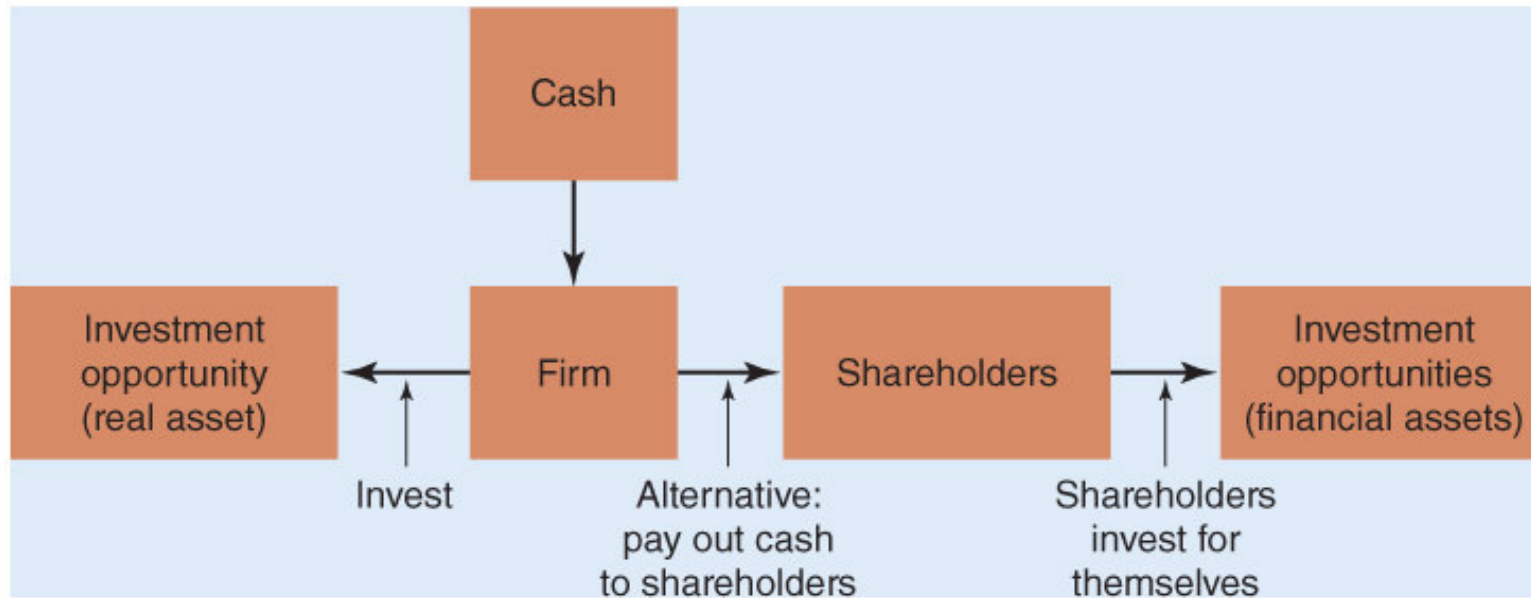
4. Stocks 4./11.5.26



Part Two: Financial Instruments 4. Stocks: Literature

- BDM, chapter 9.1-9.2, 9.5
- BMM, chapter 7.6-7.7

The Course Journey



Taken from the slides accompanying Brealey/Myers/Marcus: Fundamentals of Corporate Finance, 11th edition.

capital
budgeting rules
(chapter 2)

- benchmark debt (ch. 3)
- benchmark equity (ch. 4)
- risk/return (ch. 5/6)
- WACC (ch. 6)

Part Two: Financial Instruments 4. Stocks

Terms and concepts you should be familiar with after chapter 4

- dividend-discount model (DDM)
- dividend yield, capital gain, total return
- effect of profitable growth on market value
- payout rate, retention rate, earnings per share (EPS)
- market efficiency: weak form, semi-strong form, strong form
- random walk
- technical analysis, fundamental analysis

Part Two: Financial Instruments 4. Stocks

Structure of Chapter 4

- Dividend-Discount Model (DDM)
- Applying the DDM
- Market Efficiency

Part Two: Financial Instruments 4. Stocks

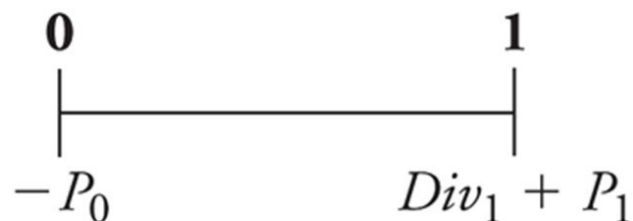
Structure of Chapter 4

- Dividend-Discount Model (DDM)
- Applying the DDM
- Market Efficiency

Part Two: Financial Instruments 4. Stocks

Dividend-Discout Model (DDM): A One-Year Investor

- Potential Cash Flows
 - Dividend
 - Sale of Stock
- Timeline for One-Year Investor



- Since the cash flows are risky, we must discount them at the equity cost of capital.

Taken from the slides accompanying Berk/DeMarzo: Corporate Finance, 6th edition.

Part Two: Financial Instruments 4. Stocks

Dividend-Discount Model (DDM): A One-Year Investor

$$P_0 = \left(\frac{Div_1 + P_1}{1 + r_E} \right)$$

- If the current stock price were less than this amount, expect investors to rush in and buy it, driving up the stock's price.
- If the stock price exceeded this amount, selling it would cause the stock price to quickly fall.

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Part Two: Financial Instruments 4. Stocks

Dividend-Discount Model (DDM): A One-Year Investor

- **Dividend Yield**
 - **Capital Gain**
 - Capital Gain Rate
 - **Total Return**
 - Dividend Yield + Capital Gain Rate
 - The expected total return of the stock should equal the expected return of other investments available in the market with equivalent risk.
- $$r_E = \frac{Div_1 + P_1}{P_0} - 1 = \underbrace{\frac{Div_1}{P_0}}_{\text{Dividend Yield}} + \underbrace{\frac{P_1 - P_0}{P_0}}_{\text{Capital Gain Rate}}$$

Taken from the slides accompanying Berk/DeMarzo: Corporate Finance, 6th edition.

Part Two: Financial Instruments 4. Stocks

Dividend-Discount Model (DDM): A One-Year Investor_Example

- **Problem**
 - Suppose you expect Walgreens Boots Alliance (a drugstore chain) to pay dividends of \$1.60 per share and trade for \$70 per share at the end of the year. If investments with equivalent risk to Walgreen's stock have an expected return of 8.5%, what is the most you would pay today for Walgreen's stock? What dividend yield and capital gain rate would you expect at this price?

Taken from the slides accompanying Berk/DeMarzo: Corporate Finance, 6th edition.

Part Two: Financial Instruments 4. Stocks

Dividend-Discount Model (DDM): A One-Year Investor_Example

- **Solution**

- Using the preceding equation, we have $P_0 = \frac{Div_1 + p_1}{1 + r_E} = \frac{1.60 + 70.00}{1.085} = \65.99

- At this price, Walgreen's **dividend yield** is $\frac{Div_1}{P_0} = \frac{1.60}{65.99} = 2.42\%$.

per share, for a **capital gain rate** of $\frac{4.01}{65.99} = 6.08\%$.

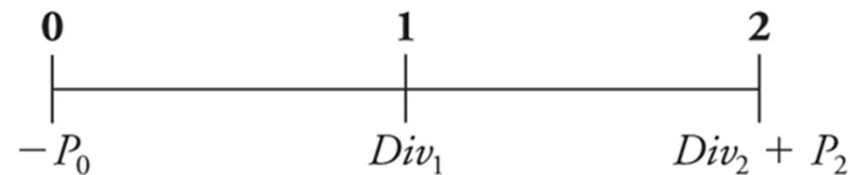
- Therefore, at this price, Walgreen's **expected total return** is $2.42\% + 6.08\% = 8.5\%$, which is equal to its **equity cost of capital**.

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Part Two: Financial Instruments 4. Stocks

Dividend-Discount Model (DDM): A Multi-Year Investor

- What is the price if we plan on holding the stock for two years?



$$P_0 = \frac{Div_1}{1 + r_E} + \frac{Div_2 + P_2}{(1 + r_E)^2}$$

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Part Two: Financial Instruments 4. Stocks

Dividend-Discount Model (DDM): A Multi-Year Investor

- The formula for the two-year investor differs from that of the one-year investor. The formula for the two-year investor includes dividend and price in year 2, the formula for the one-year investor does not. **Does this imply, that the two investors will value the stock differently?**

$$P_0 = \left(\frac{Div_1 + P_1}{1 + r_E} \right)$$

$$P_0 = \frac{Div_1}{1 + r_E} + \frac{Div_2 + P_2}{(1 + r_E)^2}$$

Taken from the slides accompanying Berk/DeMarzo: Corporate Finance, 6th edition.

Part Two: Financial Instruments 4. Stocks

Dividend-Discount Model (DDM): A Multi-Year Investor

- **The answer is no.** While a one-year investor does not care about the dividend and stock price in year 2 **directly**, she will care about them **indirectly** because they will affect the price for which she can sell the stock at the end of year 1.

$$P_1 = \left(\frac{Div_2 + P_2}{1 + r_E} \right)$$

- Substituting this formula in $P_0 = \left(\frac{Div_1 + P_1}{1 + r_E} \right)$

$$\text{yields } P_0 = \frac{Div_1}{1 + r_E} + \frac{Div_2 + P_2}{(1 + r_E)^2}$$

Taken from the slides accompanying Berk/DeMarzo: Corporate Finance, 6th edition.

Part Two: Financial Instruments 4. Stocks

Dividend-Discount Model (DDM): A Multi-Year Investor

- **Generalizing:** The selling price for each stockholder depends on the dividends expected by **all following stockholders**.
 - Also for the current stockholder, who does not plan to hold the stock for long, the dividends to be paid in, let's say, 10, 20 or 30 years matter.
 - Valuation does not depend on the intended holding period.

Taken from the slides accompanying Berk/DeMarzo: Corporate Finance, 6th edition.

Part Two: Financial Instruments 4. Stocks

Dividend-Discount Model (DDM): The DDM Equation

- What is the price if we plan on holding the stock for N years?

$$P_0 = \frac{Div_1}{1 + r_E} + \frac{Div_2}{(1 + r_E)^2} + \dots + \frac{Div_N}{(1 + r_E)^N} + \frac{P_N}{(1 + r_E)^N}$$

- This is known as the **Dividend-Discount Model**.
- Note that the above equation holds for any horizon N.
- Thus, all investors (with the same beliefs) will attach the same value to the stock, independent of their investment horizons.

Taken from the slides accompanying Berk/DeMarzo: Corporate Finance, 6th edition.

Part Two: Financial Instruments 4. Stocks

Dividend-Discount Model (DDM): The DDM Equation

$$P_0 = \frac{Div_1}{1 + r_E} + \frac{Div_2}{(1 + r_E)^2} + \frac{Div_3}{(1 + r_E)^3} + \dots = \sum_{n=1}^{\infty} \frac{Div_n}{(1 + r_E)^n}$$

- If we let go N to infinity, the price of any stock is equal to the present value of the expected future dividends it will pay.

Taken from the slides accompanying Berk/DeMarzo: Corporate Finance, 6th edition.

Part Two: Financial Instruments 4. Stocks

Intermediate Exercise

Krell Industries has a share price of \$21.44 today. If Krell is expected to pay a dividend of \$0.78 this year, and its stock price is expected to grow to \$24.63 at the end of the year, what is Krell's dividend yield and equity cost of capital?

Part Two: Financial Instruments 4. Stocks

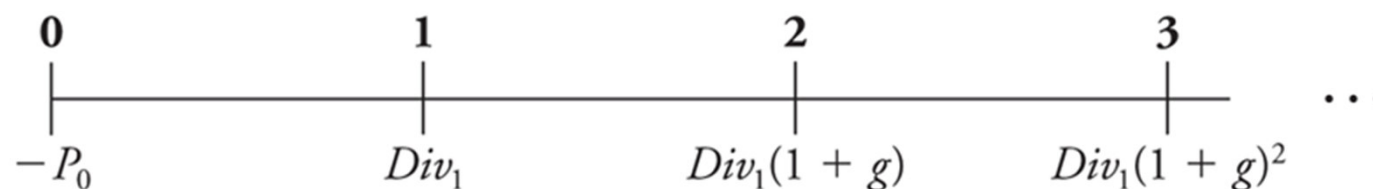
Structure of Chapter 4

- Dividend-Discount Model (DDM)
- Applying the DDM
- Market Efficiency

Part Two: Financial Instruments 4. Stocks

Applying the DDM: Constant Dividend Growth

- The simplest forecast for the firm's future dividends states that they will grow at a constant rate, g , forever.



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Part Two: Financial Instruments 4. Stocks

Applying the DDM: Constant Dividend Growth

- The value of the firm depends on the current dividend level, the cost of equity, and the growth rate.

$$P_0 = \frac{Div_1}{r_E - g}$$

$$r_E = \frac{Div_1}{P_0} + g$$

Taken from the slides accompanying Berk/DeMarzo: Corporate Finance, 6th edition.

Part Two: Financial Instruments 4. Stocks

Applying the DDM: Constant Dividend Growth

- **Problem: Valuing a Firm with Constant Dividend Growth**
 - Consolidated Edison, Inc. (Con Edison), is a regulated utility company that services the New York City area. Suppose Con Edison plans to pay \$3.00 per share in dividends in the coming year. If its equity cost of capital is 6% and dividends are expected to grow by 2% per year in the future, estimate the value of Con Edison's stock.

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Part Two: Financial Instruments 4. Stocks

Applying the DDM: Constant Dividend Growth

- **Solution: Valuing a Firm with Constant Dividend Growth**
 - If dividends are expected to grow perpetually at a rate of 2% per year, we can use the preceding equation to calculate the price of a share of Con Edison stock:

$$P_0 = \frac{Div_1}{r_E - g} = \frac{\$3.00}{0.06 - 0.02} = \$75$$

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Part Two: Financial Instruments 4. Stocks

Applying the DDM: Constant Dividend Growth

- The **effect of growth** demonstrated by assuming the **no-growth case**.

$$P_0 = \frac{Div1}{r_E - g} = \frac{\$3.00}{0.06 - 0.00} = \$50$$



Taken from the slides accompanying Berk/DeMarzo: Corporate Finance, 6th edition.

Part Two: Financial Instruments 4. Stocks

Applying the DDM: Dividends Versus Investment and Growth_A Simple Model of Growth

- **Dividend Payout Ratio:** The fraction of earnings paid as dividends each year.

$$Div_t = \underbrace{\frac{\text{Earnings}_t}{\text{Shares Outstanding}_t}}_{EPS_t} \times \text{Dividend Payout Rate}_t$$

- Assuming the number of shares outstanding is constant, the firm can do two things to increase its dividend:
 - Increase its earnings (net income).
 - Increase its dividend payout rate.

Taken from the slides accompanying Berk/DeMarzo: Corporate Finance, 6th edition.

Part Two: Financial Instruments 4. Stocks

Applying the DDM: Dividends Versus Investment and Growth_A Simple Model of Growth

- A firm can do one of two things with its earnings:
 - It can pay them out to investors.
 - It can retain and reinvest them.
 - **Retention Rate:** Fraction of current earnings that the firm retains.

Taken from the slides accompanying Berk/DeMarzo: Corporate Finance, 6th edition.

Part Two: Financial Instruments 4. Stocks

Applying the DDM: Dividends Versus Investment and Growth_A Simple Model of Growth

Change in Earnings = New Investment $\times$ Return on New Investment

New Investment = Earnings $\times$ Retention Rate

$$\rightarrow \text{Earnings} = \frac{\text{New Investment}}{\text{Retention Rate}}$$

$$\begin{aligned}\text{Earnings Growth Rate} &= \frac{\text{Change in Earnings}}{\text{Earnings}} \\ &= \text{Retention Rate} \times \text{Return on New Investment}\end{aligned}$$

$$g = \text{Retention Rate} \times \text{Return on New Investment}$$

- If the firm keeps its retention rate constant, then the growth rate in dividends will equal the growth rate of earnings.

Taken from the slides accompanying Berk/DeMarzo: Corporate Finance, 6th edition.

Part Two: Financial Instruments 4. Stocks

Applying the DDM: Dividends Versus Investment and Growth_A Simple Model of Growth

- **Profitable Growth**
 - If a firm wants to increase its share price, should it cut its dividend and invest more, or should it cut investment and increase its dividend?
 - The answer will depend on the profitability of the firm's investments.
 - **Cutting the firm's dividend to increase investment will raise the stock price if, and only if, the new investments have a positive NPV.**

Taken from the slides accompanying Berk/DeMarzo: Corporate Finance, 6th edition.

Part Two: Financial Instruments 4. Stocks

Applying the DDM: Dividends Versus Investment and Growth_A Simple Model of Growth

- **Problem: Cutting Dividends for Profitable Growth**
 - Crane Sporting Goods expect to have **earnings per share of \$6 in the coming year**. Rather than reinvest these earnings and grow, the firm plans to **pay out all of its earnings** as a dividend. With these **expectations of no growth**, Crane's **current share price is \$60**.
 - Suppose Crane could **cut its dividend payout rate to 75%** for the foreseeable future and use the retained earnings to open new stores. The **return on its investment in these stores is expected to be 12%**.
 - Assuming its equity cost of capital is unchanged, what effect would this new policy have on Crane's stock price?

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Part Two: Financial Instruments 4. Stocks

Applying the DDM: Dividends Versus Investment and Growth_A Simple Model of Growth

- **Solution: Cutting Dividends for Profitable Growth**
 - First, let's estimate Crane's **equity cost of capital**. Currently, Crane plans to pay a **dividend** equal to its earnings of \$6 per share. Given a share price of \$60, Crane's **dividend yield** is $\frac{\$6}{\$60} = 10\%$. With **no expected growth ($g = 0$)**, we can use the preceding equation to estimate r_E :
$$r_E = \frac{Div_1}{P_0} + g = 10\% + 0\% = 10\%$$
 - In other words, to justify Crane's stock price under its current policy, the **expected return of other stocks in the market with equivalent risk must be 10%**.

Taken from the slides accompanying Berk/DeMarzo: Corporate Finance, 6th edition.

Part Two: Financial Instruments 4. Stocks

Applying the DDM: Dividends Versus Investment and Growth_A Simple Model of Growth

- **Solution: Cutting Dividends for Profitable Growth**
 - Next, we consider the consequences of the new policy. If Crane reduces its dividend payout rate to 75%, then from the preceding equation

$$Div_t = \underbrace{\frac{\text{Earnings}_t}{\text{Shares Outstanding}_t}}_{\text{EPS}_t} \times \text{Dividend Payout Rate}_t$$

its **dividend** this coming year will fall to $Div_1 = EPS_1 \times 75\% = \$6 \times 75\% = \$4.50$.

Taken from the slides accompanying Berk/DeMarzo: Corporate Finance, 6th edition.

Part Two: Financial Instruments 4. Stocks

Applying the DDM: Dividends Versus Investment and Growth_A Simple Model of Growth

- **Solution: Cutting Dividends for Profitable Growth**

- At the same time, because the firm will now retain 25% of its earnings to invest in new stores, from the preceding equation $g = \text{Retention Rate} \times \text{Return on New Investment}$

its growth rate will increase to $g = 25\% \times 12\% = 3\%$.

- Assuming Crane can continue to grow at this rate, we can compute its **share price** under the new policy using the **constant dividend growth model**:

$$P_0 = \frac{Div_1}{r_E - g} = \frac{\$4.50}{0.10 - 0.03} = \$64.29$$

Taken from the slides accompanying Berk/DeMarzo: Corporate Finance, 6th edition.

Part Two: Financial Instruments 4. Stocks

Applying the DDM: Dividends Versus Investment and Growth_A Simple Model of Growth

- **Solution: Cutting Dividends for Profitable Growth**
 - Thus, Crane's share price should rise from \$60 to \$64.29 if it cuts its dividend to **invest in projects that offer a return (12%) greater than their cost of capital** (which we assume remains **10%**). These **projects are positive NPV**, and so by taking them Crane has created value for its shareholders.

Taken from the slides accompanying Berk/DeMarzo: Corporate Finance, 6th edition.

Part Two: Financial Instruments 4. Stocks

Applying the DDM: Dividends Versus Investment and Growth_A Simple Model of Growth

- **Solution: Cutting Dividends for Unprofitable Growth**
 - Suppose Crane Sporting Goods decides to cut its dividend payout rate to 75% to invest in new stores, as in the previous example but now suppose that the **return on these new investments is 8%, rather than 12%**.
 - Given its **expected earnings per share this year of \$6** and its **equity cost of capital of 10%**, what will happen to Crane's current share price in this case?

Taken from the slides accompanying Berk/DeMarzo: Corporate Finance, 6th edition.

Part Two: Financial Instruments 4. Stocks

Applying the DDM: Dividends Versus Investment and Growth_A Simple Model of Growth

- **Solution: Cutting Dividends for Unprofitable Growth**
 - Just as in the previous example, Crane's **dividend** will fall to $\$6 \times 75\% = \4.50 .
 - Its **growth rate** under the new policy, given the lower return on new investment, will now be $g = 25\% \times 8\% = 2\%$.
 - The **new share price** is therefore $P_0 = \frac{Div_1}{r_E - g} = \frac{\$4.50}{0.10 - 0.02} = \$56.25$
 - Thus, even though Crane will grow under the new policy, the new investments have negative NPV. Crane's share price will fall if it cuts its dividend to make new investments with a return of only 8% when its investors can earn 10% on other investments with comparable risk.

Taken from the slides accompanying Berk/DeMarzo: Corporate Finance, 6th edition.

Part Two: Financial Instruments 4. Stocks

Applying the DDM: Changing Growth Rates

- We cannot use the constant dividend growth model to value a stock if the growth rate is not constant.
- For example, young firms often have very high initial earnings growth rates.
- During this period of high growth, these firms often retain 100% of their earnings to exploit profitable investment opportunities.
- As they mature, their growth slows.
- At some point, their earnings exceed their investment needs, and they begin to pay dividends.

Taken from the slides accompanying Berk/DeMarzo: Corporate Finance, 6th edition.

Part Two: Financial Instruments 4. Stocks

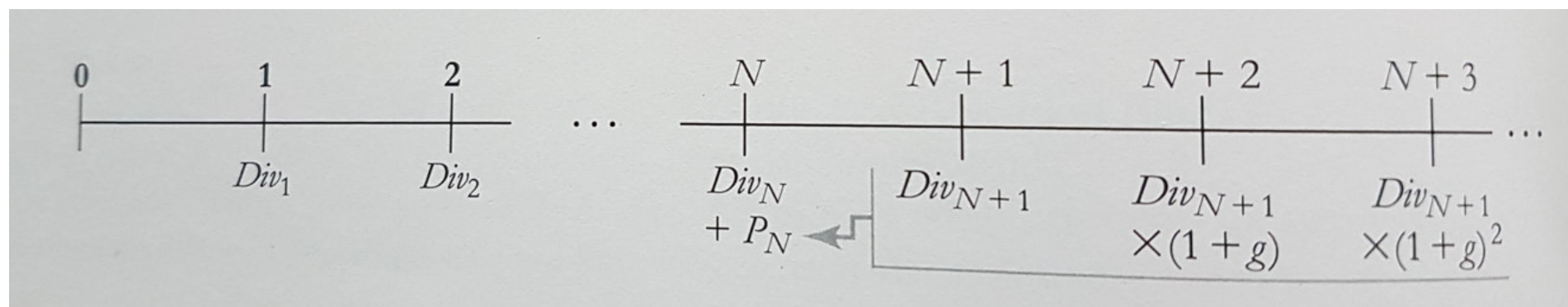
Applying the DDM: Changing Growth Rates

- Although we cannot use the constant dividend growth model directly when growth is not constant, we can use the general form of the model to value a firm **by applying the constant growth model to calculate the future share price of the stock once the expected growth rate stabilizes.**

Taken from the slides accompanying Berk/DeMarzo: Corporate Finance, 6th edition.

Part Two: Financial Instruments 4. Stocks

Applying the DDM: Changing Growth Rates



$$P_N = \frac{Div_{N+1}}{r_E - g}$$

$$P_0 = \frac{Div_1}{1 + r_E} + \frac{Div_2}{(1 + r_E)^2} + \dots + \frac{Div_N}{(1 + r_E)^N} + \frac{1}{(1 + r_E)^N} \left(\frac{Div_{N+1}}{r_E - g} \right)$$

Taken from the slides accompanying Berk/DeMarzo: Corporate Finance, 6th edition.

Part Two: Financial Instruments 4. Stocks

Applying the DDM: Changing Growth Rates

- **Problem: Valuing a Firm with Two Different Growth Rates**
 - Small Fry, Inc., has just invented a potato chip that looks and tastes like a French fry. Given the phenomenal market response to this product, Small Fry is reinvesting all of its earnings to expand its operations. Earnings were \$2 per share this past year and are expected **to grow at a rate of 20% per year until the end of year 4**. At that point, other companies are likely to bring out competing products. Analysts project that at the end of year 4, Small Fry will cut investment and begin paying 60% of its earnings as dividends and its **growth will slow to a long-run rate of 4%**. If Small Fry's equity cost of capital is 8%, what is the value of a share today?

Taken from the slides accompanying Berk/DeMarzo: Corporate Finance, 6th edition.

Part Two: Financial Instruments 4. Stocks

Applying the DDM: Changing Growth Rates

- **Solution: Valuing a Firm with Two Different Growth Rates**
 - We can use Small Fry's projected earnings growth rate and payout rate to forecast its future earnings and dividends as shown in the following spreadsheet:

	Year	0	1	2	3	4	5	6
Earnings								
1	EPS Growth Rate (versus prior year)		20%	20%	20%	20%	4%	4%
2	EPS	\$2.00	\$2.40	\$2.88	\$3.46	\$4.15	\$4.31	\$4.49
Dividends								
3	Dividend Payout Rate		0%	0%	0%	60%	60%	60%
4	Dividend		\$ —	\$ —	\$ —	\$2.49	\$2.59	\$2.69

- Starting from \$2.00 in year 0, EPS grows by 20% per year until year 4, after which growth slows to 4%. Small Fry's dividend payout rate is zero until year 4, when competition reduces its investment opportunities, and its payout rate rises to 60%. Multiplying EPS by the dividend payout ratio, we project Small Fry's future dividends in line 4.

Taken from the slides accompanying Berk/DeMarzo: Corporate Finance, 6th edition.

Part Two: Financial Instruments 4. Stocks

Applying the DDM: Changing Growth Rates

- **Problem: Valuing a Firm with Two Different Growth Rates**
 - From year 4 onward, Small Fry's dividends will grow at the expected long-run rate of 4% per year. Thus, we can use the constant dividend growth model to project Small Fry's share price at the end of year 3. Given its equity cost of capital of 8%,

$$P_3 = \frac{Div_4}{r_E - g} = \frac{\$2.49}{0.08 - 0.04} = \$62.25$$

- We then apply the dividend-discount model with this terminal value:

$$P_0 = \frac{Div_1}{1 + r_E} + \frac{Div_2}{(1 + r_E)^2} + \frac{Div_3}{(1 + r_E)^3} + \frac{P_3}{(1 + r_E)^3} = \frac{\$62.25}{(1.08)^3} = \$49.42$$

- As this example illustrates, the dividend-discount model is flexible enough to handle any forecasted pattern of dividends.

Taken from the slides accompanying Berk/DeMarzo: Corporate Finance, 6th edition.

Part Two: Financial Instruments 4. Stocks

Intermediate Exercise

What should the price of a share of stock be if dividends are projected at \$5/share, the discount rate is 10%, and growth is 0%?

What should the price of a share of stock be if the firm will pay a \$5 dividend in 1 year that is expected to grow at a constant rate of 5%? Assume a discount rate of 10%.

Part Two: Financial Instruments 4. Stocks

Structure of Chapter 4

- Dividend-Discount Model (DDM)
- Applying the DDM
- Market Efficiency

Part Two: Financial Instruments 4. Stocks

Market Efficiency

- Information efficiency actually is a broad concept, applicable to all kinds of markets. It is typically discussed on the example of the stock market.
- Markets are information aggregators, prices and rates are the resulting signals for resource allocation. "Wisdom of the crowd."
- An efficient market is a market in which prices reflect all relevant information.

Taken from the slides accompanying Berk/DeMarzo: Corporate Finance, 6th edition.

Part Two: Financial Instruments 4. Stocks

Market Efficiency: Three Forms

- **Weak Form Efficiency:** Market prices reflect all historical information (past price data).
- **Semi-Strong Form Efficiency:** Market prices reflect all publicly available information.
- **Strong Form Efficiency:** Market prices reflect all information, both public and private.
- Implication: Trading based on the analysis of the respective information set would not lead to a superior investment result ("beating the market").

Taken from the slides accompanying Brealey/Myers/Marcus: Fundamentals of Corporate Finance, 11th edition.

Part Two: Financial Instruments 4. Stocks

Market Efficiency: Weak Form

- In a weak-form efficient market, trading based on the analysis of past prices (technical analysis) would not lead to a superior investment result ("beating the market").
- **Technical Analysts**
 - Investors who attempt to identify undervalued stocks by searching for patterns in past stock prices.
 - Forecast stock prices based on watching the fluctuations in historical prices.
- Strategy would not be successful if prices follow a **random walk**.

Taken from the slides accompanying Brealey/Myers/Marcus: Fundamentals of Corporate Finance, 11th edition.

Part Two: Financial Instruments 4. Stocks

Market Efficiency: Weak Form

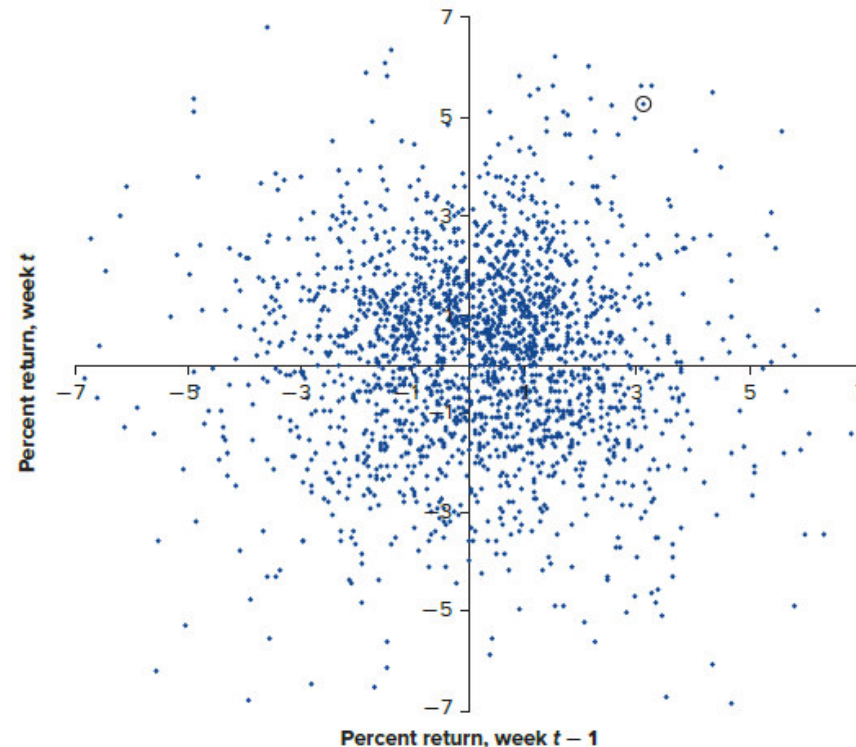
- **Random Walk Theory**
 - Security prices change randomly, with no predictable trends or patterns.
 - Statistically speaking, the movement of stock prices is random.
 - Skewed positive over the long term.

Taken from the slides accompanying Brealey/Myers/Marcus: Fundamentals of Corporate Finance, 11th edition.

Part Two: Financial Instruments 4. Stocks

Market Efficiency: Weak Form

- **Testing the weak form:**
Successive Weekly Returns of
New York Composite Index over a
40-Year Period (Scatter Plots):
Where's the pattern?



Taken from the slides accompanying Brealey/Myers/Marcus: Fundamentals of Corporate Finance, 11th edition.

Part Two: Financial Instruments 4. Stocks

Market Efficiency: Semi-Strong Form

- In a semi-strong form efficient market, trading based on the analysis of publicly available information (fundamental analysis) would not lead to a superior investment result ("beating the market").
- **Fundamental Analysts**
 - Investors who attempt to find mispriced securities by analyzing fundamental information, such as accounting data and business prospects.
 - Research the value of stocks using NPV and other measurements of cash flow.

Taken from the slides accompanying Brealey/Myers/Marcus: Fundamentals of Corporate Finance, 11th edition.

Part Two: Financial Instruments 4. Stocks

Market Efficiency: Semi-Strong Form

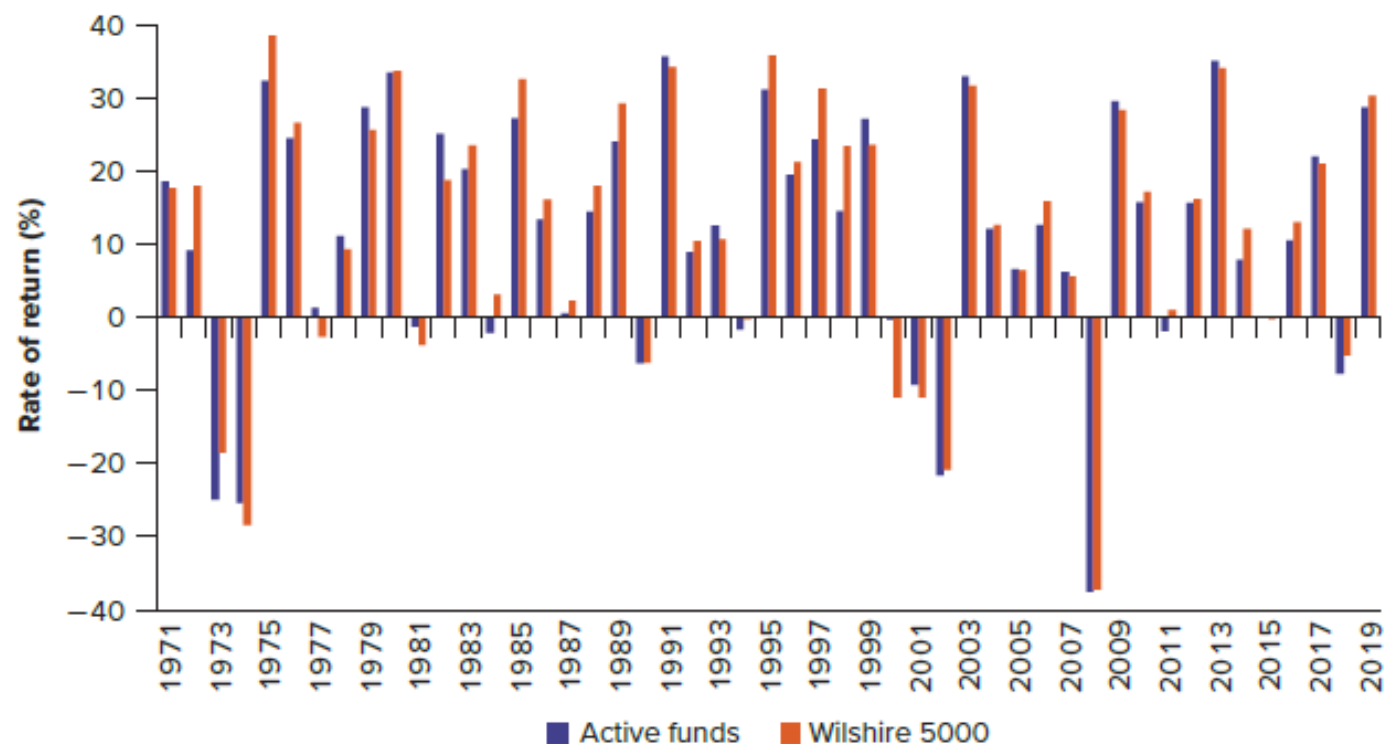
- **Testing the semi-strong form:** Do active investors, i.e., those that analyze publicly available information and select stocks based on their analysis, have a superior investment result in comparison with passive investors, i.e., investors that do not pick stocks but hold the complete market?

Taken from the slides accompanying Brealey/Myers/Marcus: Fundamentals of Corporate Finance, 11th edition.

Part Two: Financial Instruments 4. Stocks

- Performance of Fund Managers

Market Efficiency: Semi-Strong Form



Taken from the slides accompanying Brealey/Myers/Marcus: Fundamentals of Corporate Finance, 11th edition.

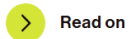
Part Two: Financial Instruments 4. Stocks

Market Efficiency: Semi-Strong Form

- Testing the semi-strong form: Do the prices react immediately when new information becomes known?
- **Biontech 10.3.26**

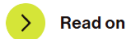
10 March 2026

BioNTech Announces Fourth Quarter and Full Year 2025 Financial Results and Corporate Update



10 March 2026

BioNTech and Co-Founders Announce Plan to Pursue Next-Generation mRNA Innovations in Co-Founders-Led New Company as BioNTech Advances Toward Becoming a Multi-Product Company by 2030



<https://investors.biontech.de/news/press-releases>; last access 7.4.2026.

Part Two: Financial Instruments 4. Stocks

Market Efficiency: Semi-Strong Form



Biontech Corporate News 10.3.26

Taken from Comdirect.

Part Three: Risk and Return 5. Capital Asset Pricing Model

Beta: Event Study of Bayer's Announcement to Acquire Monsanto on 19 May 2016

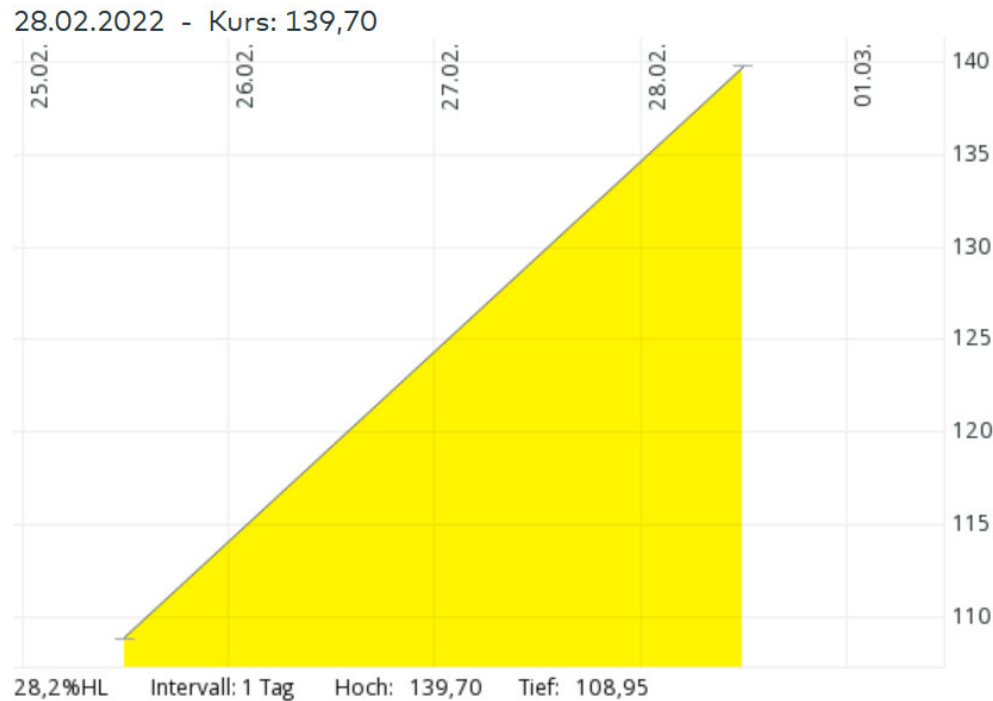
Beta (60 days)	0.8118009
Return Bayer 19.5.16	-0.0780015
Return DAX 19.5.16	-0.0148181
Normal Return Bayer	-0.012059
Abnormal Return Bayer	-0.0659425

Part Two: Financial Instruments 4. Stocks

Rheinmetall Aktie

Market Efficiency: Semi-Strong Form

WKN: 703000 ISIN: DE0007030009



**Rheinmetall Share 25.2.
(Friday) to 28.2.22 (Monday)**

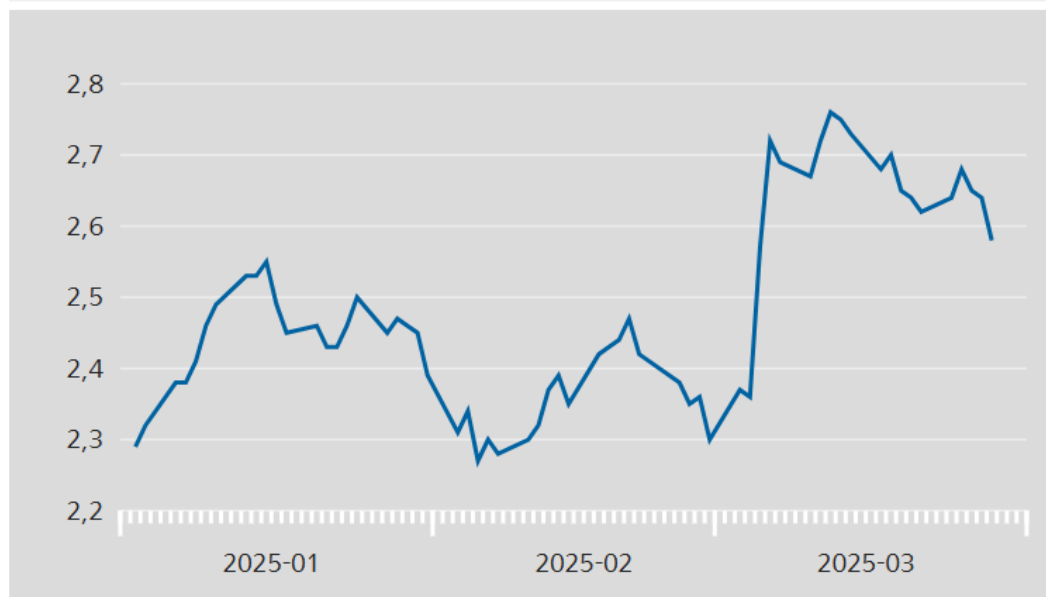
Taken from Comdirect.

Part Two: Financial Instruments 4. Stocks

Market Efficiency: Semi-Strong Form, the Case of Bonds

Umlaufrenditen inländischer Inhaberschuldverschreibungen /
Börsennotierte Bundeswertpapiere / Tageswerte

Prozent



Taken from
<https://www.bundesbank.de/de/statistiken/geld-und-kapitalmaerkte/zinssaetze-und-renditen/umlaufrenditen/umlaufrenditen-772416^>;
last access 28.3.25.

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Stand vom: 28.03.2025

Part Two: Financial Instruments 4. Stocks

Market Efficiency: Strong Form

- In a strong-form efficient market, trading based on private information, e.g., information of management board members not disclosed to the public, would not lead to superior investment returns.

Taken from the slides accompanying Berk/DeMarzo: Corporate Finance, 6th edition.

Part Two: Financial Instruments 4. Stocks

Market Efficiency

- Major markets with a good level of activity and liquidity are usually semi-strong form efficient. Strong-form efficiency is more a theoretical ideal.
- Some features of financial markets make it more likely that financial markets have a good degree of efficiency in comparison with other markets for goods and services:
 - high transparency standards
 - strong competition between market participants
 - rather global markets
 - rather homogeneous “goods” traded
 - low transaction costs

Part Two: Financial Instruments 4. Stocks

Market Efficiency

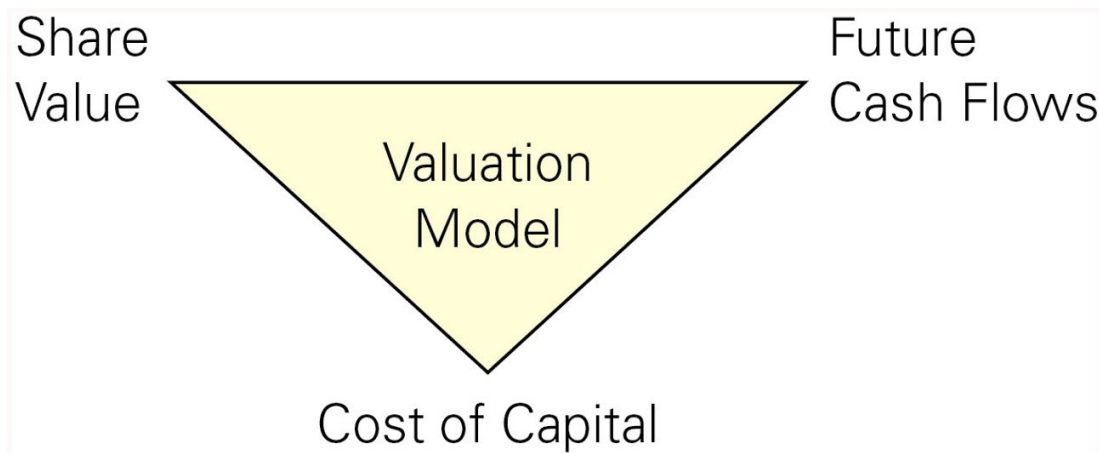
Abnormal returns are achievable from exploiting information sets not fully reflected in the market prices.

Information reflected in market prices	Degree of market efficiency		
	weak	semi-strong	strong
past price data	fully	fully	fully
all publicly available info	not fully	fully	fully
private info	not fully	not fully	fully

Part Two: Financial Instruments 4. Stocks

Market Efficiency: Using Information from Efficient Markets

- Our valuation model links the firm's future cash flows, its cost of capital, and its share price.
- Given accurate information about any two of these variables, a valuation model allows us to make inferences about the third variable.



The Valuation Triad

Taken from the slides accompanying Berk/DeMarzo: Corporate Finance, 6th edition.

Part Two: Financial Instruments 4. Stocks

Market Efficiency: Using Information from Efficient Markets

- For a publicly traded firm, its current stock price should already provide very accurate information, aggregated from a multitude of investors, regarding the true value of its shares.
- Based on its current stock price, a valuation model will tell us something about the firm's future cash flows or cost of capital.

Taken from the slides accompanying Berk/DeMarzo: Corporate Finance, 6th edition.

Part Two: Financial Instruments 4. Stocks

Market Efficiency: Using Information from Efficient Markets

- **Problem**
 - Suppose Tecnor Industries will pay a dividend this year of \$5 per share. Its equity cost of capital is 10%, and you expect its dividends to grow at a rate of about 4% per year, though you are somewhat unsure of the precise growth rate. If Tecnor's stock is currently trading for \$76.92 per share, how would you update your beliefs about its dividend growth rate?

Taken from the slides accompanying Berk/DeMarzo: Corporate Finance, 6th edition.

Part Two: Financial Instruments 4. Stocks

Market Efficiency: Using Information from Efficient Markets

- **Solution**

- If we apply the constant dividend growth model based on a 4% growth rate, we would estimate a stock price of $P_0 = \frac{5}{(0.10 - 0.04)} = \83.33 per share.
- The market price of \$76.92, however, implies that most investors expect dividends to grow at a somewhat slower rate. If we continue to assume a constant growth rate, we can solve for the growth rate consistent with the current market price:

$$r_E = \frac{Div_1}{P_0} + g \qquad g = r_E - \frac{Div_1}{P_0} = 10\% - \frac{5}{76.92} = 3.5\%$$

- Thus, given this market price for the stock, we should lower our expectations for the dividend growth rate unless we have very strong reasons to trust our own estimate.

Taken from the slides accompanying Berk/DeMarzo: Corporate Finance, 6th edition.

Part Two: Financial Instruments 4. Stocks

Market Efficiency: Conclusions for Companies and Investors

- If stocks are fairly priced, then **investors** who buy stocks can expect to receive future cash flows that fairly compensate them for the risk of their investment.
- In such cases, the average **investor** can invest with confidence, even if they are not fully informed. However, they should not expect to outperform the market systematically unless they dispose of private information.
- Implications for **corporate managers**: Focus on generating NPV with profitable projects on the assets' side of the balance sheet (capital budgeting). Financing can be expected to be a zero-NPV transactions as participants on efficient markets would not be willing to buy overvalued securities.

Taken from the slides accompanying Berk/DeMarzo: Corporate Finance, 6th edition.

Part Two: Financial Instruments 4. Stocks

Market Efficiency

- **A final reconsideration:**
 - In an efficient market, security prices change randomly, with no predictable trends or patterns. **Statistically speaking**, the movement of stock prices is random.
 - This randomness is of a completely different nature than that in the case of, e.g., roulette or throwing dices.
 - In an efficient market, security prices change for a reason, i.e., because of new relevant information. Information is only new when it was not expected/forecasted. Thus, security price changes cannot be predicted. Prices change randomly because new information is random.

Taken from the slides accompanying Brealey/Myers/Marcus: Fundamentals of Corporate Finance, 11th edition.

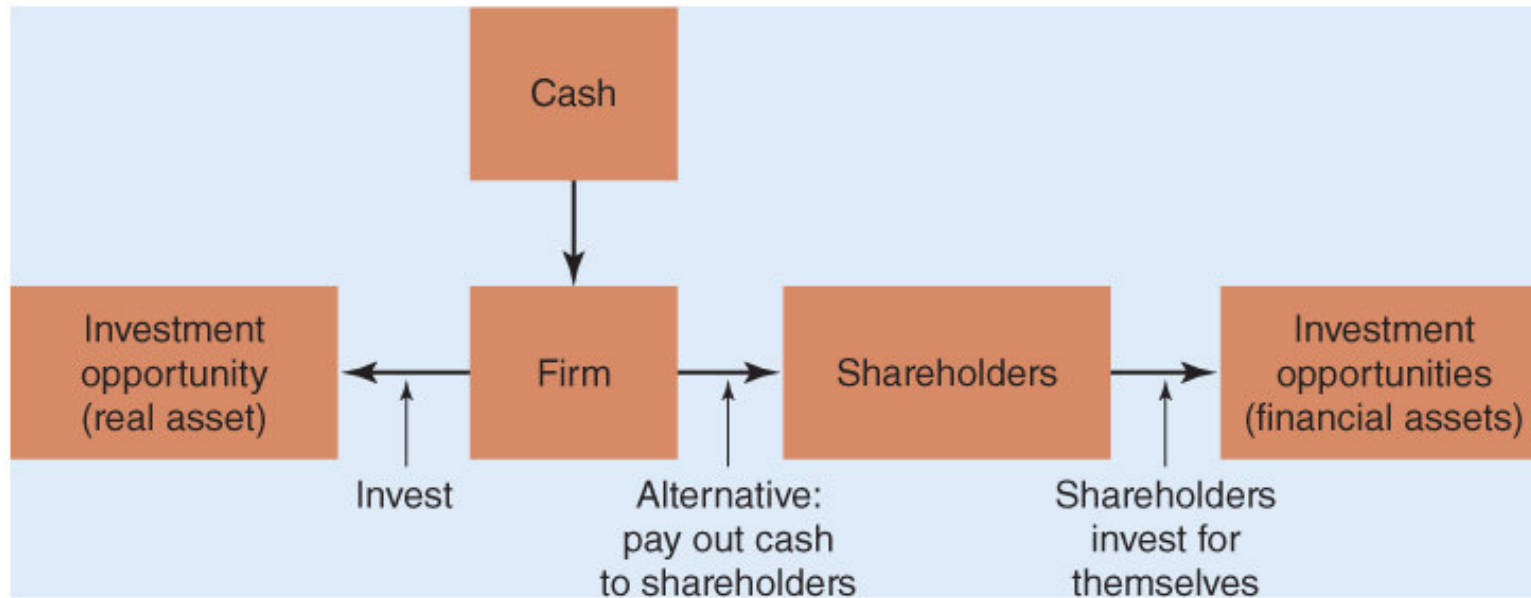
Part Two: Financial Instruments 4. Stocks

Intermediate Exercise

As a hobby, you analyze companies and do some trading in their shares. Most recently, you started to follow Diamond Inc. and read their reports and also other reports about this company. How likely is it, that you can earn abnormal returns with your trading activities in the Diamond share?

Would your answer change for the aunt of a good friend of yours? She is a member of the top management team of Diamond Inc.

The Course Journey



Taken from the slides accompanying Brealey/Myers/Marcus: Fundamentals of Corporate Finance, 11th edition.

capital
budgeting rules
(chapter 2)

- benchmark debt (ch. 3)
- benchmark equity (ch. 4)
- risk/return (ch. 5/6)
- WACC (ch. 6)

Part Two: Financial Instruments 4. Stocks

Terms and concepts you should be familiar with after chapter 4

- dividend-discount model (DDM)
- dividend yield, capital gain, total return
- effect of profitable growth on market value
- payout rate, retention rate, earnings per share (EPS)
- market efficiency: weak form, semi-strong form, strong form
- random walk
- technical analysis, fundamental analysis

Part Two: Financial Instruments 4. Stocks

Formulas from chapter 4

- Dividend-discount model (1 year): $P_0 = \left(\frac{Div_1 + P_1}{1 + r_E} \right)$
- Total return, capital gain, dividend yield: $r_E = \frac{Div_1 + P_1}{P_0} - 1 = \underbrace{\frac{Div_1}{P_0}}_{\text{Dividend Yield}} + \underbrace{\frac{P_1 - P_0}{P_0}}_{\text{Capital Gain Rate}}$
- Dividend-discount model (2 years): $P_0 = \frac{Div_1}{1 + r_E} + \frac{Div_2 + P_2}{(1 + r_E)^2}$

Part Two: Financial Instruments 4. Stocks

Formulas from chapter 4

- Dividend-discount model (n years):
$$P_0 = \frac{Div_1}{1 + r_E} + \frac{Div_2}{(1 + r_E)^2} + \dots + \frac{Div_N}{(1 + r_E)^N} + \frac{P_N}{(1 + r_E)^N}$$
$$P_0 = \frac{Div_1}{1 + r_E} + \frac{Div_2}{(1 + r_E)^2} + \frac{Div_3}{(1 + r_E)^3} + \dots = \sum_{n=1}^{\infty} \frac{Div_n}{(1 + r_E)^n}$$
- DDM, constant growth:
$$P_0 = \frac{Div_1}{r_E - g} \quad r_E = \frac{Div_1}{P_0} + g$$
- DDM, changing growth rates:
$$P_0 = \frac{Div_1}{1 + r_E} + \frac{Div_2}{(1 + r_E)^2} + \dots + \frac{Div_N}{(1 + r_E)^N} + \frac{1}{(1 + r_E)^N} \left(\frac{Div_{N+1}}{r_E - g} \right)$$

Part Two: Financial Instruments 4. Stocks

Formulas from chapter 4

- Change in Earnings = New Investment $\times$ Return on New Investment
- New Investment = Earnings $\times$ Retention Rate
- Earnings Growth Rate = $\frac{\text{Change in Earnings}}{\text{Earnings}}$
= Retention Rate $\times$ Return on New Investment
- $g = \text{Retention Rate} \times \text{Return on New Investment}$
- Dividend: $Div_t = \underbrace{\frac{\text{Earnings}_t}{\text{Shares Outstanding}_t}}_{EPS_t} \times \text{Dividend Payout Rate}_t$

Agenda

Part Three: Risk and Return

5. Capital Asset Pricing Model 18./26.5.26

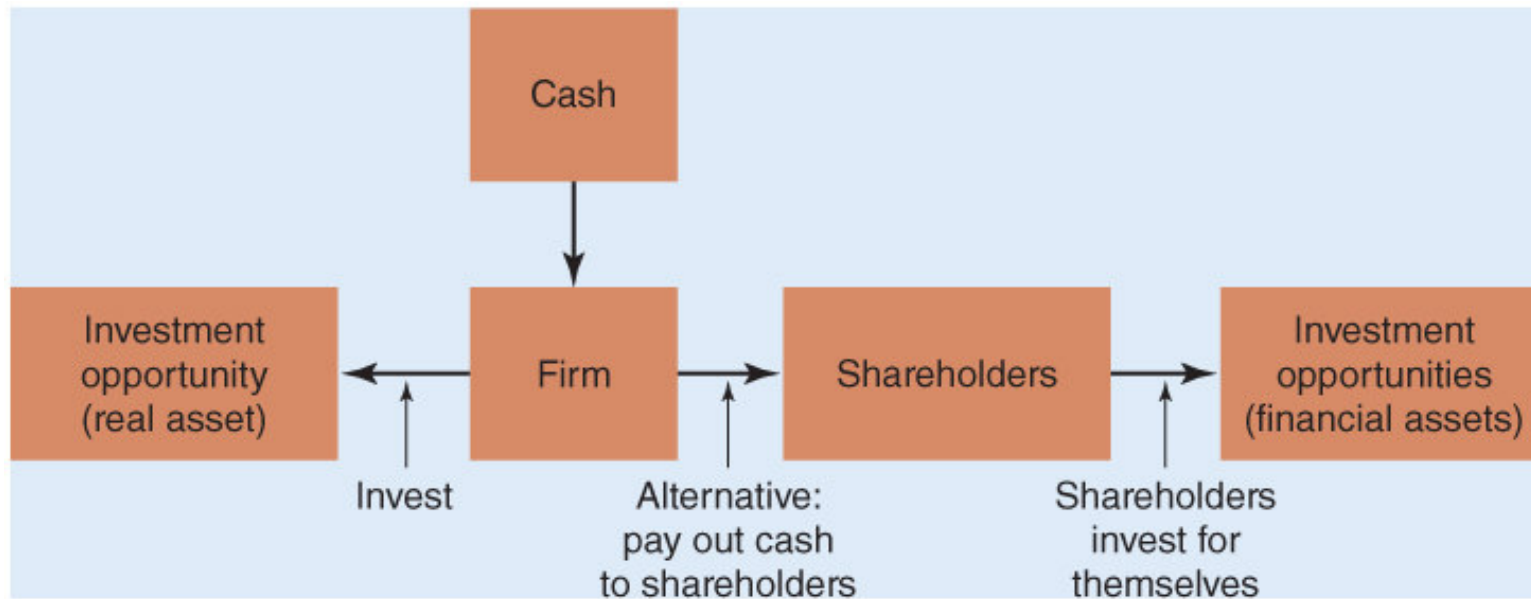
6. Capital Market Based Cost of Capital 26.5./1.6.26



Part Three: Risk and Return 5. Capital Asset Pricing Model: Literature

- BMM, chapter 11-12.3
- BDM, chapter 10

The Course Journey



Taken from the slides accompanying Brealey/Myers/Marcus: Fundamentals of Corporate Finance, 11th edition.

capital
budgeting rules
(chapter 2)

- benchmark debt (ch. 3)
- benchmark equity (ch. 4)
- risk/return (ch. 5/6)
- WACC (ch. 6)

Part Three: Risk and Return 5. Capital Asset Pricing Model

Terms and concepts you should be familiar with after chapter 5

- Maturity premium, risk premium
- Variance, standard deviation, correlation
- Total risk, systematic risk, specific risk, diversification
- Beta
- Capital Asset Pricing Model (CAPM), Security Market Line (SML)
- Market risk premium

Part Three: Risk and Return 5. Capital Asset Pricing Model

Structure of Chapter 5

- Introductory Remarks
- Recap Returns
- A Century of Capital Market History
- Recap (from Statistics): Measuring Risk
- Risk and Diversification
- Interim Summary
- Beta
- Capital Asset Pricing Model (CAPM)

Part Three: Risk and Return 5. Capital Asset Pricing Model

Structure of Chapter 5

- **Introductory Remarks**
- Recap Returns
- A Century of Capital Market History
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- Capital Asset Pricing Model (CAPM)

Part Three: Risk and Return 5. Capital Asset Pricing Model

Introductory Remarks: The Goal of Part Three

- Quotes from Chapter 4:
 - “The **expected total return of the stock** should equal the **expected return of other investments available in the market with equivalent risk.**”
 - “Thus, **Crane’s share price should rise** from \$60 to \$64.29 if it cuts its dividend to **invest in projects that offer a return (12%) greater than their cost of capital** (which we assume remains **10%**). These **projects are positive NPV**, and so by taking them Crane has created value for its shareholders.”
 - “Thus, even though Crane will grow under the new policy, the new investments have **negative NPV**. **Crane’s share price will fall** if it cuts its dividend to make **new investments with a return of only 8% when its investors can earn 10% on other investments with comparable risk.**”

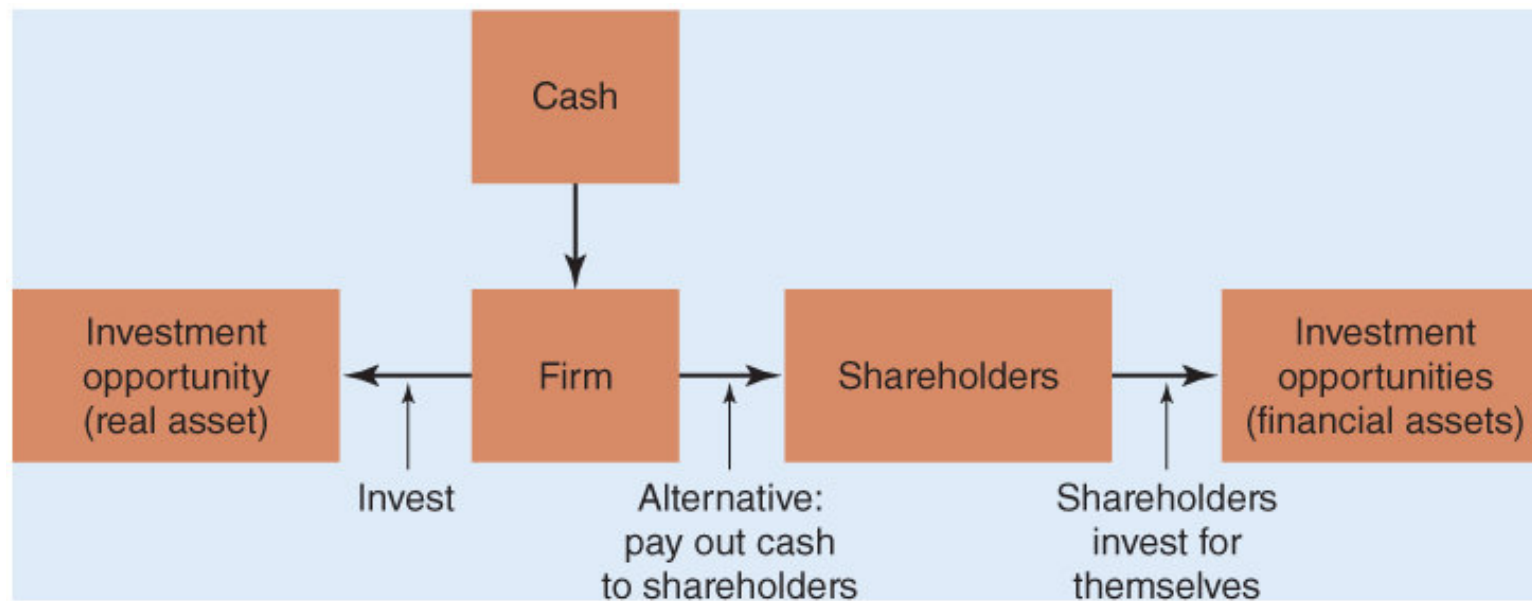
Part Three: Risk and Return 5. Capital Asset Pricing Model

Introductory Remarks: The Goal of Part Three

- → What does equivalent risk mean? How could we measure risk?
- → How do we know the expected return of financial instruments, assuming that we know their risk?
- **The goal of part three: deriving a market-based r .**

Part Three: Risk and Return 5. Capital Asset Pricing Model

Introductory Remarks: The Goal of Part Three



Part Three: Risk and Return 6. Capital Market Based Cost of Capital

The CAPM and the Opportunity Cost of Capital

- **Company cost of capital**
 - Opportunity cost of capital for investment in the entire firm.
 - The company cost of capital is the appropriate discount rate for an average risk investment project undertaken by the firm.
- **The project cost of capital** depends on the use of the capital.
 - It depends on the risk of the project and not the risk of the company.

Adapted from the slides accompanying Brealey/Myers/Marcus: Fundamentals of Corporate Finance, 11th edition.

Taken from the slides accompanying Brealey/Myers/Marcus: Fundamentals of Corporate Finance, 11th edition.

Part Three: Risk and Return 5. Capital Asset Pricing Model

Introductory Remarks: The Storyline

- A link between **expected** returns and the risk born by investors is intuitive and plausible (forward-looking perspective).
- Can we observe such a link for **realized** returns (backward-looking perspective)?
- For which part of total risk (**standard deviation**) can we expect a return (**systematic/market/macro risk**), and how can we measure this part of the risk (**beta**)? Is there a theoretical foundation (**portfolio theory, CAPM**)?

Part Three: Risk and Return 5. Capital Asset Pricing Model

Structure of Chapter 5

- Introductory Remarks
- **Recap Returns**
- A Century of Capital Market History
- Recap (from Statistics): Measuring Risk
- Risk and Diversification
- Interim Summary
- Beta
- Capital Asset Pricing Model (CAPM)

Part Three: Risk and Return 5. Capital Asset Pricing Model

Recap Returns

- The return from holding a security might consist of two components:
 - **payments** received while holding the security, e.g., coupon or dividend payments
 - the **difference between selling and buying price**
- To get a percentage return, payments and capital gain or loss are **related to the buying price**.
- In the simple **example of holding a stock for one year**:

$$\text{Percentage return} = \frac{\text{capital gain} + \text{dividend}}{\text{initial share price}}$$

Taken from the slides accompanying Brealey/Myers/Marcus: Fundamentals of Corporate Finance, 11th edition.

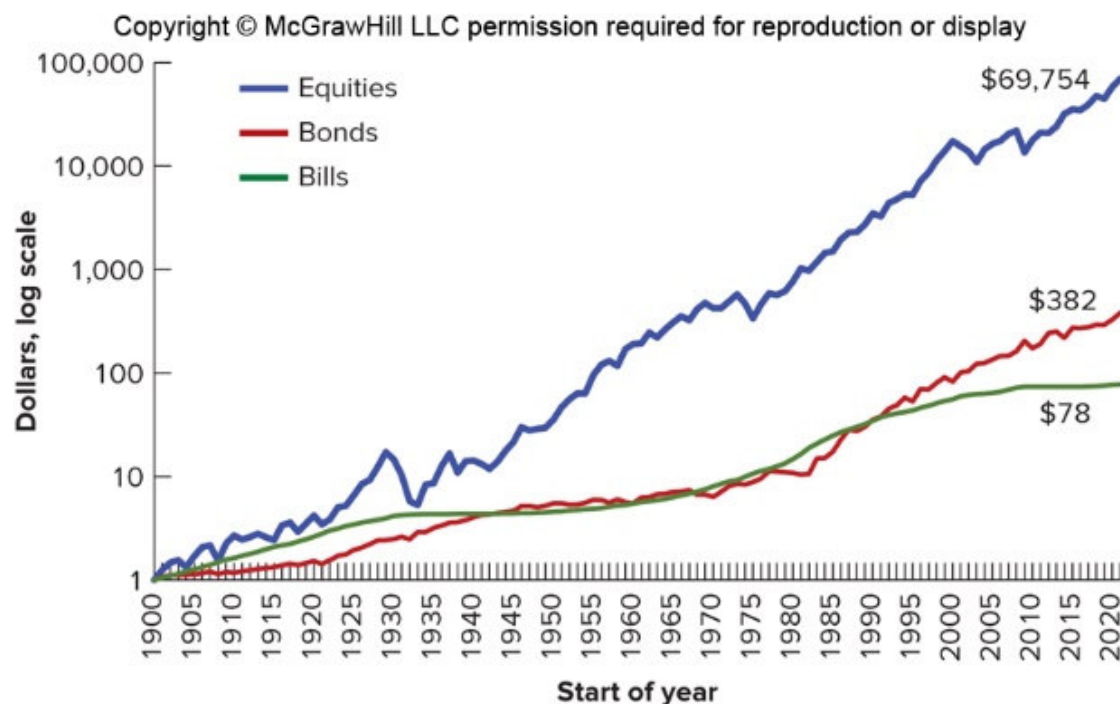
Part Three: Risk and Return 5. Capital Asset Pricing Model

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- Introductory Remarks
- Recap Returns
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Part Three: Risk and Return 5. Capital Asset Pricing Model

A Century of Capital Market History



- **Treasury bills** issued by U.S. government, 3 months time to maturity
- **Treasury bonds** issued by U.S. government, 10 years time to maturity
- Diversified portfolio of **stocks**

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Part Three: Risk and Return 5. Capital Asset Pricing Model

A Century of Capital Market History

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Portfolio	Average Annual Rate of Return, %	Average Premium, % (Extra return versus Treasury bills)
Treasury bills	3.7	
Treasury bonds	5.4	1.7
Common stocks	11.5	7.8

Source: E. Dimson, P. R. Marsh, and M. Staunton, *Triumph of the Optimists: 101 Years of Global Investment Returns* (Princeton, NJ: Princeton University Press, 2002), with updates kindly provided by *Triumph's* authors.

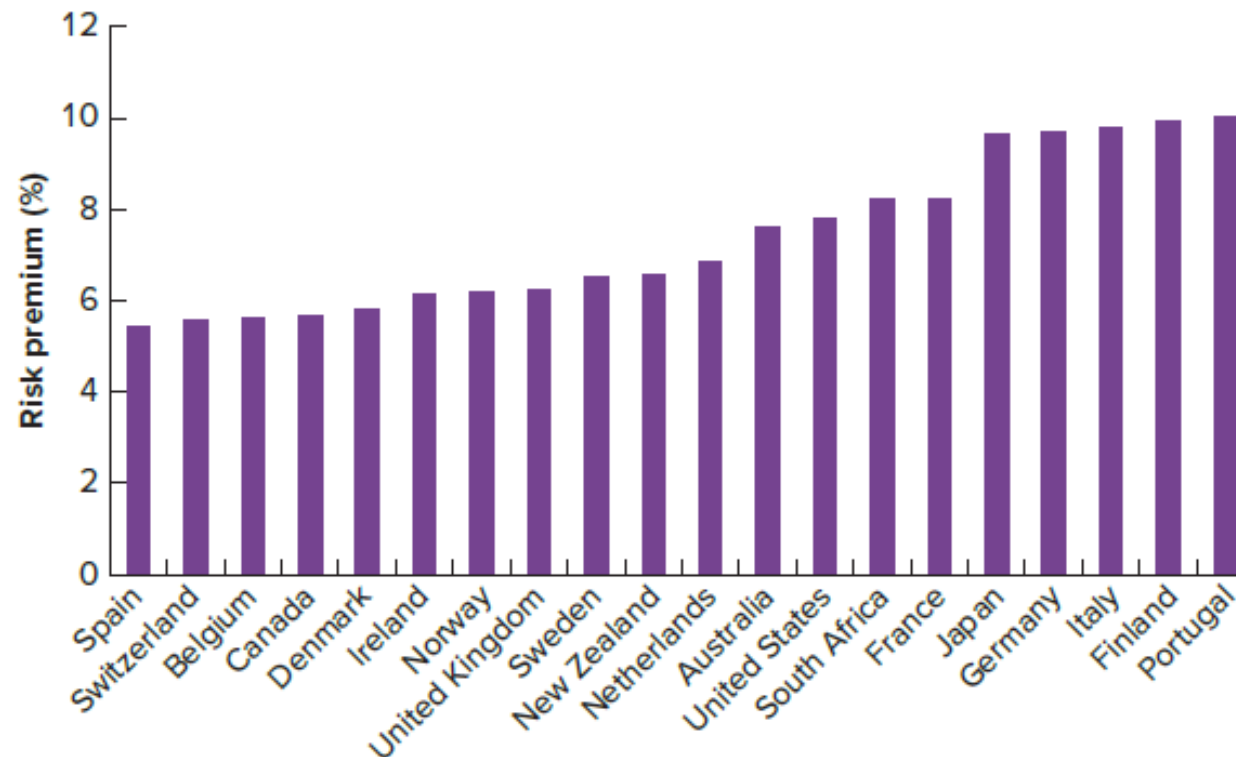
- Maturity Premium
- Risk Premium



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Part Three: Risk and Return 5. Capital Asset Pricing Model

A Century of Capital Market History: Country Risk Premiums (1900 to 2020)



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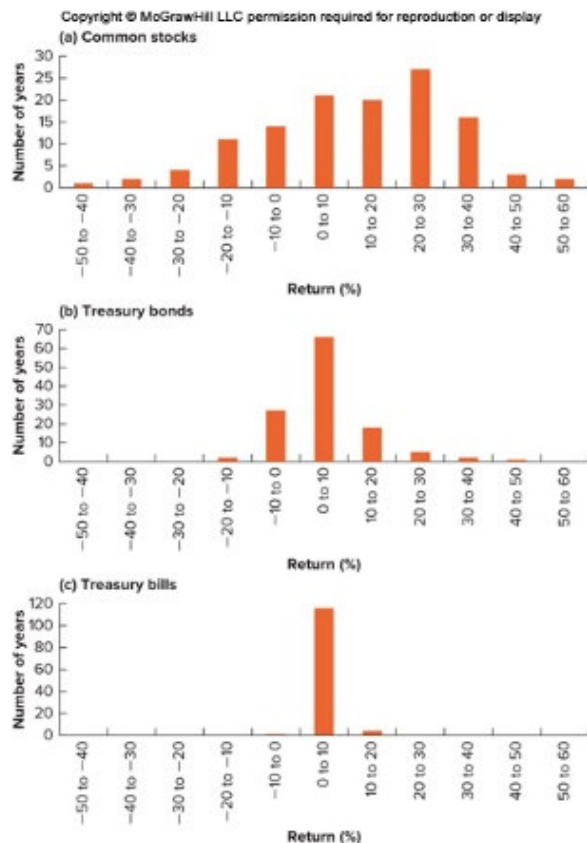
Part Three: Risk and Return 5. Capital Asset Pricing Model

Capital Asset Pricing Model (CAPM)

- Theory of the relationship between risk and return
- It states that the expected risk premium on any security equals its beta times the market risk premium
- The expected risk premium is the difference between market return and risk-free rate (e.g., Treasury bill). It is compensation for bearing market risk premium for investing
- Risk premium on the market is $r_M - r_f$
- Risk premium on the asset is $r_A - r_f$
- Expected return on any asset is $r_A = r_f + \beta(r_M - r_f)$

Small text at the bottom: "These slides are a summary of the Capital Asset Pricing Model. They are not intended to be a substitute for the textbook or other sources of information." HSBA logo.

Part Three: Risk and Return 5. Capital Asset Pricing Model



A Century of Capital Market History

- What is the relationship between the volatility of these securities and their expected returns?

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Part Three: Risk and Return 5. Capital Asset Pricing Model

A Century of Capital Market History: Historical Risk (1900-2020)

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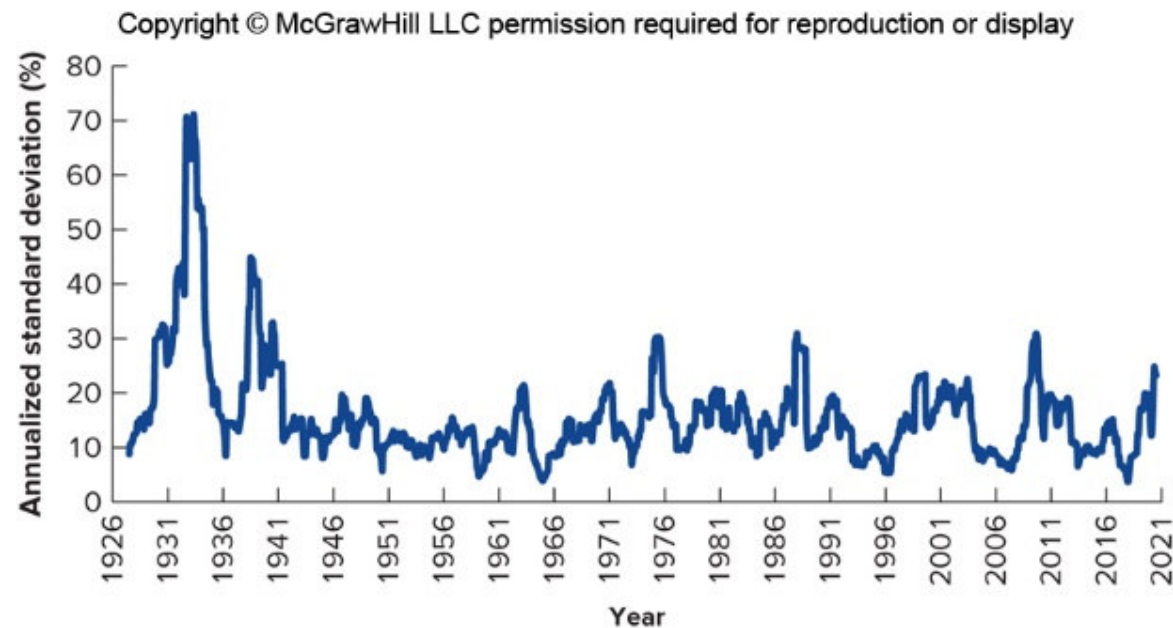
Portfolio	Standard Deviation (%)
Treasury bills	2.8
Long-term government bonds	8.9
Common stocks	19.5

Source: Authors' calculations using data from E. Dimson, P. R. Marsh, and M. Staunton, *Triumph of the Optimists: 101 Years of Global Investment Returns* (Princeton, NJ: Princeton University Press, 2002), with updates kindly provided by *Triumph's* authors.

Taken from the slides accompanying Brealey/Myers/Marcus: Fundamentals of Corporate Finance, 11th edition.

Part Three: Risk and Return 5. Capital Asset Pricing Model

A Century of Capital Market History: Historical Risk (1900-2020)



Taken from the slides accompanying Brealey/Myers/Marcus: Fundamentals of Corporate Finance, 11th edition.

Part Three: Risk and Return 5. Capital Asset Pricing Model

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- Introductory Remarks
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Part Three: Risk and Return 5. Capital Asset Pricing Model

Recap (from Statistics): Measuring Risk

- **Variance**
 - Average value of squared deviations from mean.
 - A measure of volatility.
- **Standard Deviation**
 - Square root of variance.
 - Another measure of volatility.
- **In financial markets, the standard deviation is often called volatility (or “vola”).**

Taken from the slides accompanying Brealey/Myers/Marcus: Fundamentals of Corporate Finance, 11th edition.

Part Three: Risk and Return 5. Capital Asset Pricing Model

Recap (from Statistics): Measuring Risk_Coin-Toss Game

(1) Percent Rate of Return	(2) Deviation from Expected Return	(3) Squared Deviation
+40	+30	900
+10	0	0
+10	0	0
-20	-30	900

- (1) Expected return = $(.25 \times .40) + (.5 \times .10) + (.25 \times -.20) = .10$ or 10%.
- (2) Variance = average of squared deviation = $1.800/4 = 450$.
- (3) Standard deviation = square root of variance = .212, or 21.2%.

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Part Three: Risk and Return 5. Capital Asset Pricing Model

Recap (from Statistics): Measuring Risk_Stock Market Returns 2015-2020

Year	Rate of Return (%)	Deviation from Average Return (%)	Squared Deviation
2015	-1.6	-14.3	205.75
2016	13.0	0.2	0.04
2017	21.0	8.2	66.96
2018	-6.1	-18.9	357.88
2019	29.8	17.0	289.94
2020	<u>20.6</u>	7.8	<u>60.96</u>
Total	76.6		981.32

- Average return = $76.6/6 = 12.77\%$.
- Variance = average of squared deviations = $981.32/6 = 163.55$.
- Standard deviation = square root of variance = 12.79% .

Taken from the slides accompanying Brealey/Myers/Marcus: Fundamentals of Corporate Finance, 11th edition.

Part Three: Risk and Return 5. Capital Asset Pricing Model

Interim Conclusion

- Historic data supports the intuition that a higher risk leads to a higher return in the long run. There seems to be a premium for waiting and a premium for worrying.
- **So far, risk is measured as standard deviation (total risk). Is this the correct risk measure?**

Part Three: Risk and Return 5. Capital Asset Pricing Model

Intermediate Exercise

Recall the coin-toss game from one of the previous slides:

(1) Percent Rate of Return	(2) Deviation from Expected Return	(3) Squared Deviation
+40	+30	900
+10	0	0
+10	0	0
-20	-30	900

You are offered a second game with the following profile:

(1) Percent Rate of Return
+70
+10
+10
-50

How does the expected return of the second game relate to the first one? Is it riskier, less risky or of equal risk?

Part Three: Risk and Return 5. Capital Asset Pricing Model

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Ticker	Company	Standard deviation (%)
X	U.S. Steel	76.4
PCG	Pacific Gas & Electric	55.4
MRO	Marathon Oil	48.9
NEM	Newmont Mining	36.7
AMZN	Amazon	28.3
BA	Boeing	27.4
CPB	Campbell Soup	22.6
IBM	IBM	22.5
F	Ford	22.1
INTC	Intel	22.1

Risk and Diversification

Ticker	Company	Standard deviation (%)
UNP	Union Pacific	21.4
DIS	Disney	21.2
GOOG	Google/Alphabet	20.1
SBUX	Starbucks	18.7
WMT	Walmart	17.7
XOM	ExxonMobil	19.4
PFE	Pfizer	16.6
MCD	McDonald's	14.4
KO	Coca-Cola	12.6
S&P 500		12.0

Taken from the slides accompanying Brealey/Myers/Marcus: Fundamentals of Corporate Finance, 11th edition.

Part Three: Risk and Return 5. Capital Asset Pricing Model

Risk and Diversification

$$\begin{aligned} \text{Portfolio rate of return} &= (\text{fraction of portfolio in first asset}) \\ &\quad \times (\text{rate of return on first asset}) \\ &+ \\ &\quad (\text{fraction of portfolio in second asset}) \\ &\quad \times (\text{rate of return on second asset}) \end{aligned}$$

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Part Three: Risk and Return 5. Capital Asset Pricing Model

Risk and Diversification: Two Asset Example_Gold and Auto Stocks

Scenario	▪ Auto Stock			▪ Gold Stock		
	Rate of Return (%)	Deviation from Expected Return (%)	Squared Deviation	Rate of Return (%)	Deviation from Expected Return (%)	Squared Deviation
Recession	-8	-13	169	+20	+19	361
Normal	+5	0	0	+3	+2	4
Boom	+18	+13	169	-20	-21	441
Expected return	$(-8 + 5 + 18) / 3 = 5\%$			$(-20 + 3 + 20) / 3 = 1\%$		
Variance	$(169 + 0 + 169) / 3 = 112.7$			$(361 + 4 + 441) / 3 = 268.7$		
Standard deviation	$\sqrt{112.7} = .106$ or 10.6%			$\sqrt{268.7} = .164$ or 16.4%		

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Part Three: Risk and Return 5. Capital Asset Pricing Model

Risk and Diversification: Two Asset Example_Gold and Auto Stocks

- Assume 25% in gold and 75% in auto

Rate of Return (%)				
Scenario	Probability	Auto Stock	Gold Stock	Portfolio Return (%)
Recession		-8	+20	-1.0
Normal		+5	+3	+4.5
Boom		+18	-20	+8.5
Expected return		5	1	4
Variance		112.7	268.7	15.2
Standard deviation		10.6	16.4	3.9

Taken from the slides accompanying Brealey/Myers/Marcus: Fundamentals of Corporate Finance, 11th edition.

Part Three: Risk and Return 5. Capital Asset Pricing Model

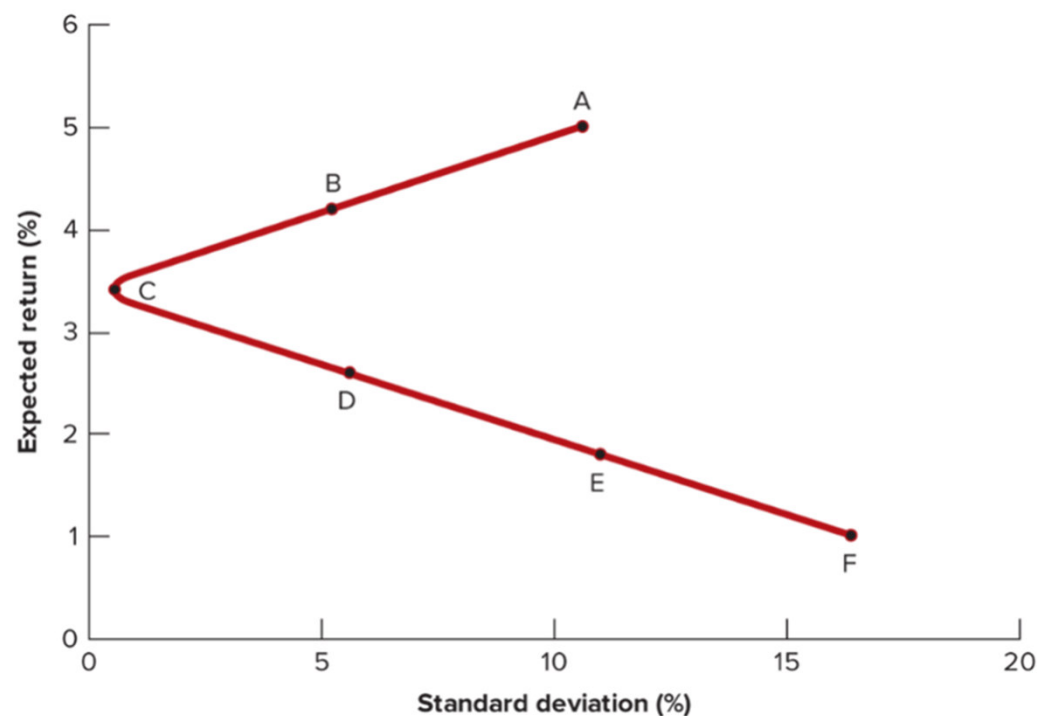
Risk and Diversification: Two Asset Example_Gold and Auto Stocks

Portfolio Weights			Portfolio Rate of Return (%)				
	Gold	Autos	Recession	Normal	Boom	Expected Return	Standard Deviation
A	0.0	1.0	−8.0	5.0	18.0	5.0	10.6
B	0.2	0.8	−2.4	4.6	10.4	4.2	5.2
C	0.4	0.6	3.2	4.2	2.8	3.4	0.6
D	0.6	0.4	8.8	3.8	−4.8	2.6	5.6
E	0.8	0.2	14.4	3.4	−12.4	1.8	11.0
F	1.0	0.0	20.0	3.0	−20.0	1.0	16.4

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Part Three: Risk and Return 5. Capital Asset Pricing Model

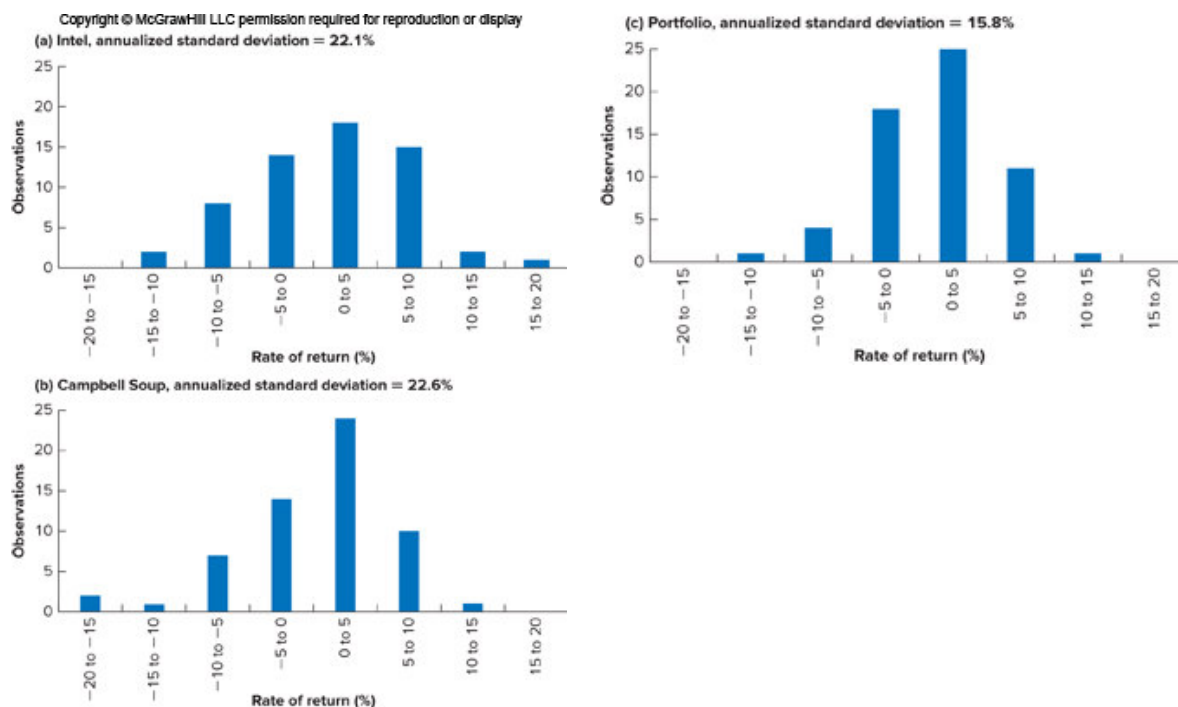
Risk and Diversification: Two Asset Example_Gold and Auto Stocks



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Part Three: Risk and Return 5. Capital Asset Pricing Model

Risk and Diversification: Intel and Campbell Soup



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Part Three: Risk and Return 5. Capital Asset Pricing Model

Ticker	Company	Standard deviation(%)
X	U.S. Steel	76.4
PCG	Pacific Gas & Electric	55.4
MRO	Marathon Oil	48.9
NEM	Newmont Mining	36.7
AMZN	Amazon	28.3
BA	Boeing	27.4
CPB	Campbell Soup	22.6
IBM	IBM	22.5
F	Ford	22.1
INTC	Intel	22.1

Risk and Diversification

Ticker	Company	Standard deviation(%)
UNP	Union Pacific	21.4
DIS	Disney	21.2
GOOG	Google/Alphabet	20.1
SBUX	Starbucks	18.7
WMT	Walmart	17.7
XOM	ExxonMobil	19.4
PFE	Pfizer	16.6
MCD	McDonald's	14.4
KO	Coca-Cola	12.6
S&P 500		12.0

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Part Three: Risk and Return 5. Capital Asset Pricing Model

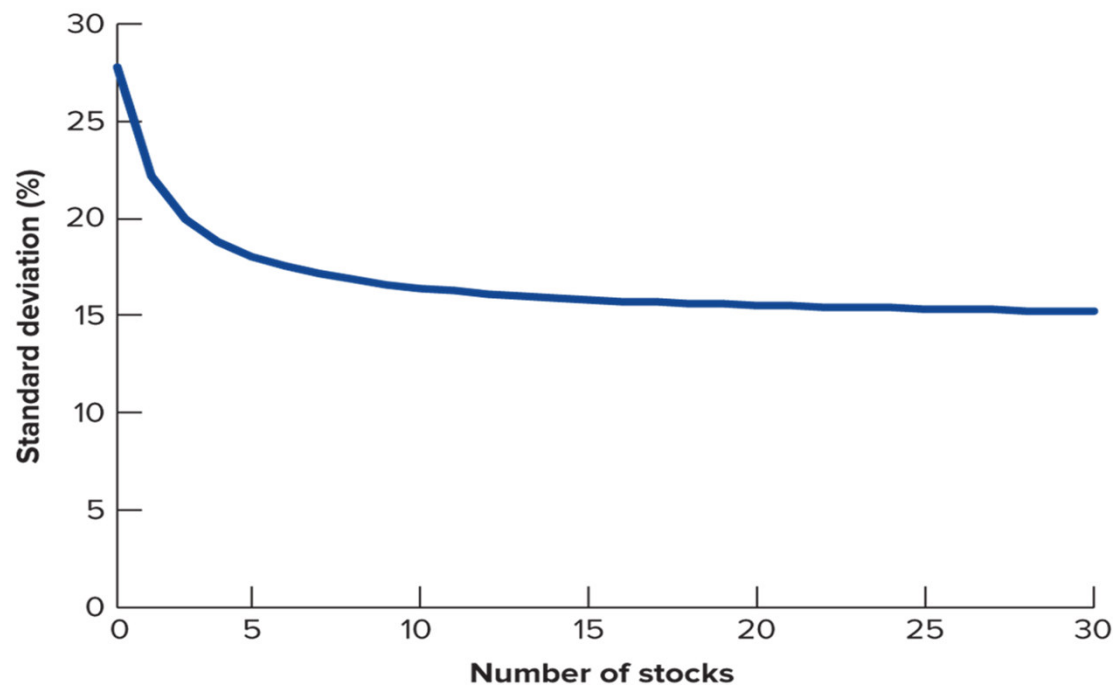
Risk and Diversification

- **Diversification**
 - Strategy designed to reduce risk by spreading the portfolio across many investments. Effect is particularly strong when correlation among portfolio assets is low.
- **Specific Risk**
 - Risk factors affecting only that firm or industry.
 - Also called **diversifiable risk** or **idiosyncratic risk**.
- **Market Risk**
 - Economy-wide sources of risk that affect the overall stock market.
 - Also called **systematic risk** or **macro risk**.

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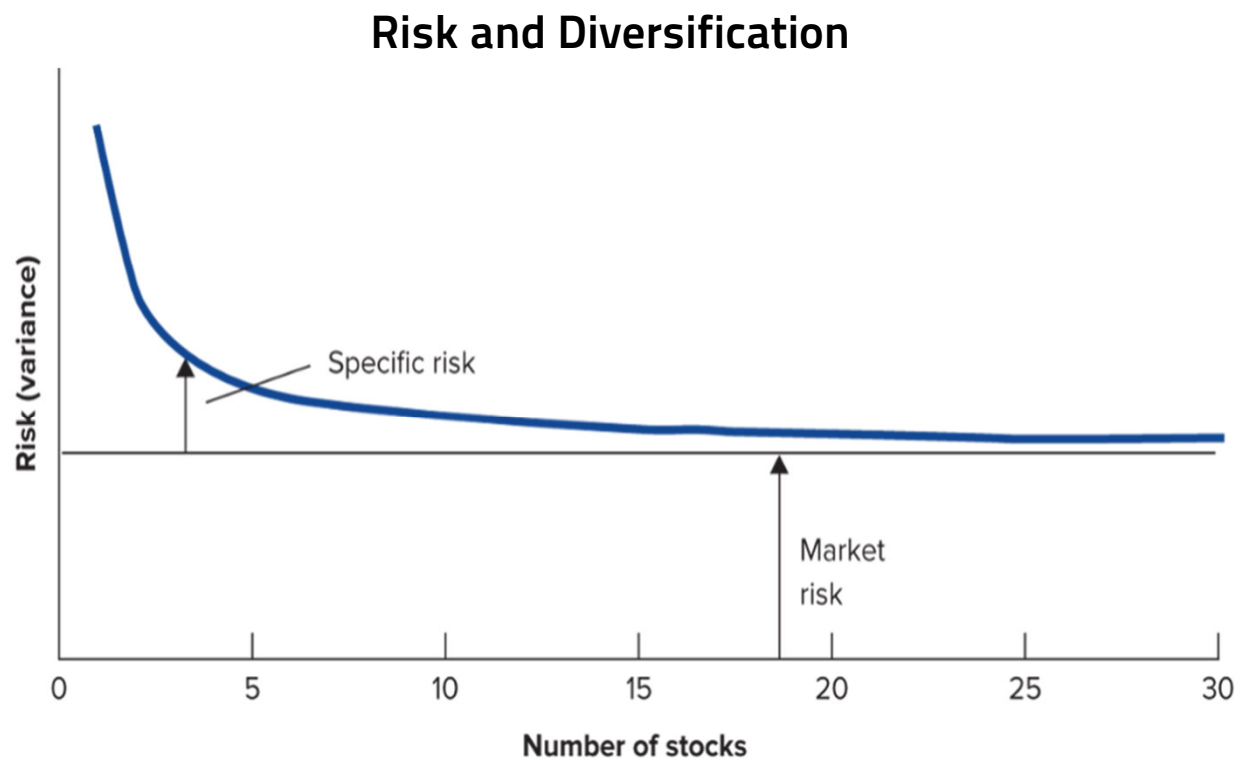
Part Three: Risk and Return 5. Capital Asset Pricing Model

Risk and Diversification



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Part Three: Risk and Return 5. Capital Asset Pricing Model

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- Beta
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Part Three: Risk and Return 5. Capital Asset Pricing Model

Interim Summary

- Historic data supports the intuition that a higher risk leads to a higher return in the long run. There seems to be a premium for waiting and a premium for worrying.
- So far, risk is measured as standard deviation (total risk). However, part of the risk can be diversified away with no/almost no cost. Looking at the whole group of capital market participants, nobody has to bear diversifiable risk, but someone has to bear systematic risk.
- Thus, you cannot expect a “reward for worrying” for bearing diversifiable risk but only for bearing systematic risk.
- Thus, we need a measure for systematic risk, because the standard deviation measures specific and systematic risk.

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Part Three: Risk and Return 5. Capital Asset Pricing Model

Interim Summary

- This measure will be beta.
- Moreover, we have to substantiate that there is also a link between systematic risk and return.
- And we need a model to estimate (future) expected returns for a given beta from capital market data → CAPM.

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Part Three: Risk and Return 5. Capital Asset Pricing Model

Intermediate Exercise

So far, your stock portfolio only consists of VW shares. You saved some additional money and want to buy shares of a second company. From a diversification point of view, would you rather buy BMW shares or SAP shares?

Part Three: Risk and Return 5. Capital Asset Pricing Model

Intermediate Exercise

Assume that you want to set up a portfolio that includes equally weighted the stocks of Disney, Starbucks, and Coca-Cola. The expected returns for the three stocks are 15% (Disney), 11% (Starbucks), and 13% (Coca-Cola). The standard deviations can be found in the table.

What is the expected return of your portfolio?

Thus far, you are not in a position to calculate the exact value of the portfolio's standard deviation, but you are able to calculate the theoretical maximum value of the standard deviation. Please calculate it. Do you think that the portfolio standard deviation will be equal to the maximum value that you calculated or below? Please explain.

Ticker	Company	Standard deviation (%)
UNP	Union Pacific	21.4
DIS	Disney	21.2
GOOG	Google/Alphabet	20.1
SBUX	Starbucks	18.7
WMT	Walmart	17.7
XOM	ExxonMobil	19.4
PFE	Pfizer	16.6
MCD	McDonald's	14.4
KO	Coca-Cola	12.6
S&P 500		12.0

Part Three: Risk and Return 5. Capital Asset Pricing Model

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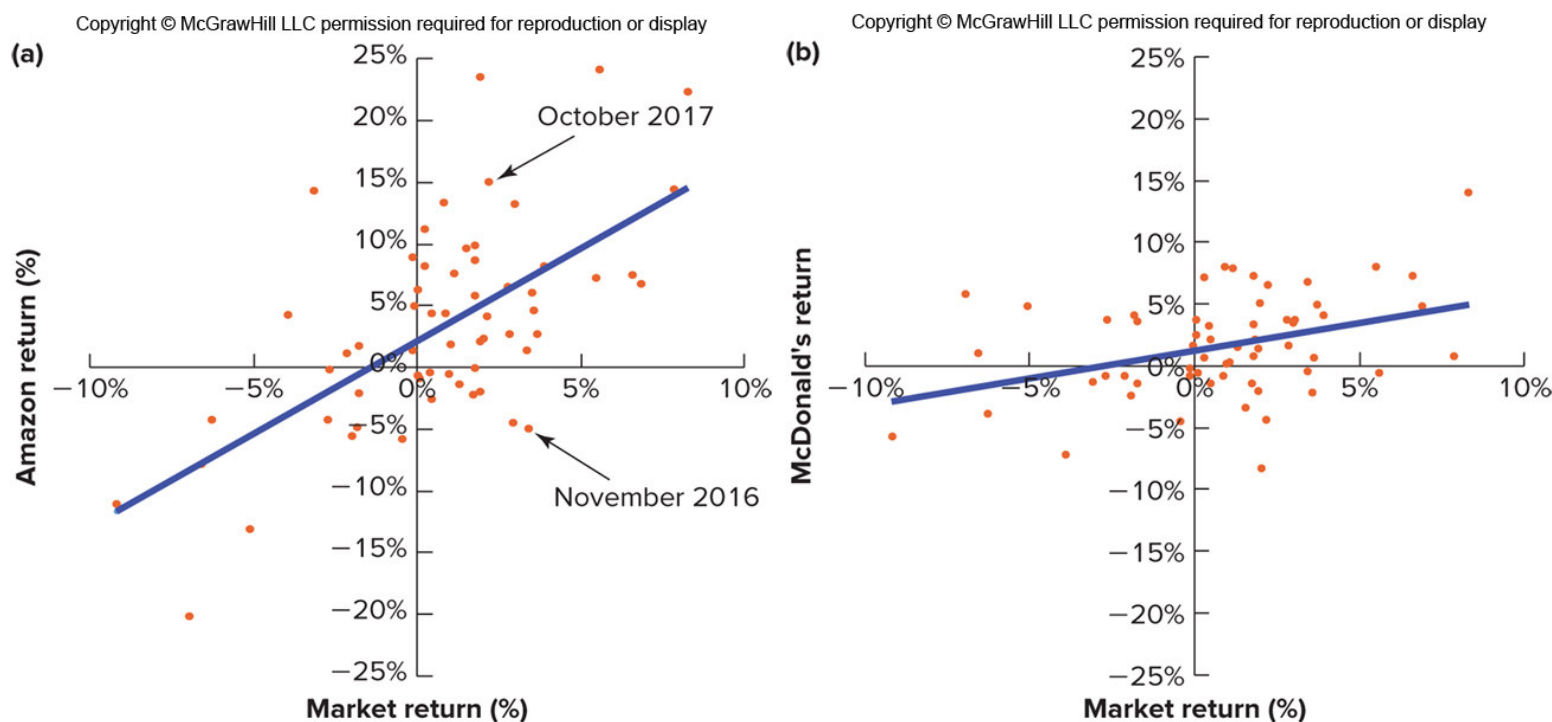
Beta

- **Beta:** Sensitivity of a stock's return to the return on the market portfolio.
- **Market Portfolio:** Portfolio of all assets in the economy. In practice a broad stock market index is used to represent the market.

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Part Three: Risk and Return 5. Capital Asset Pricing Model

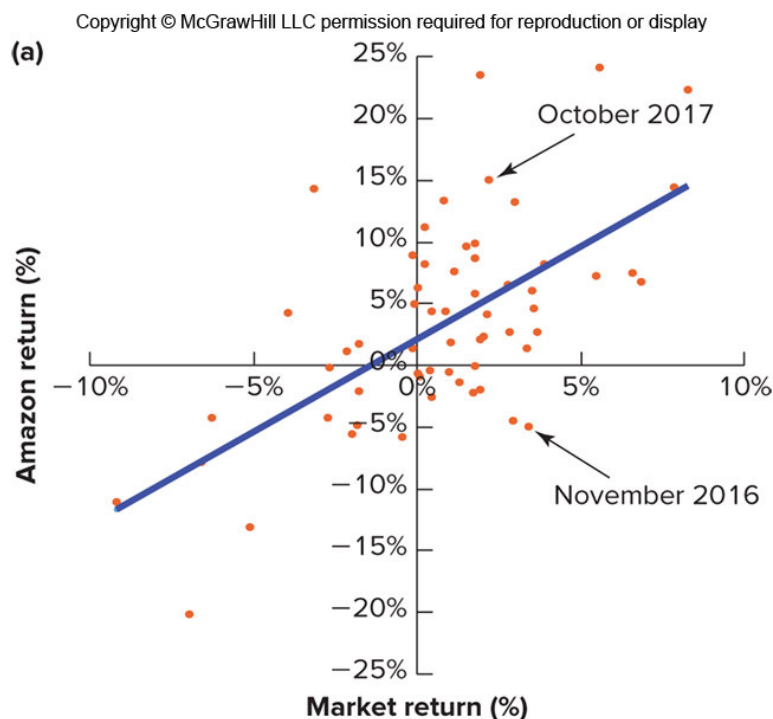
Beta: Real World Cases



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Part Three: Risk and Return 5. Capital Asset Pricing Model

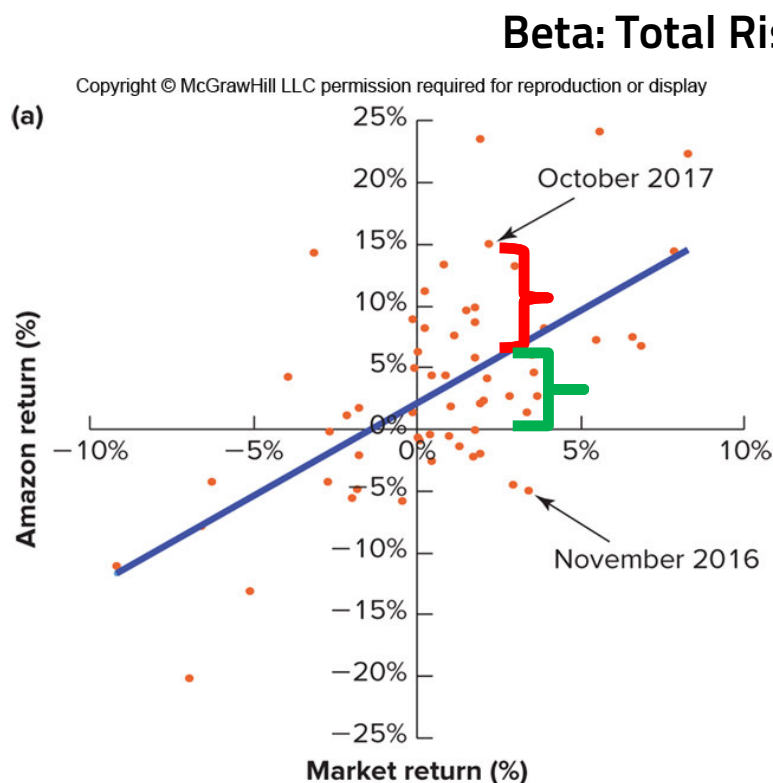
Beta: Real World Cases



- The **slope of the fitted line** represents beta.
- Technically, it is a **univariate regression** with the market return as the explanatory variable and the security return as the dependent variable.
- Stocks with a $\beta > 1$ are called **aggressive** stocks, stocks with a $\beta < 1$ are called **defensive** stocks.

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Part Three: Risk and Return 5. Capital Asset Pricing Model



- The **vertical distance** between the fitted line and the data point represents the security's return that is not explained by the general market movement → caused by specific risk factors. }
- The **vertical distance** between the fitted line and the horizontal axis represents the security's return that is explained by the general market movement → caused by systematic risk factors. }

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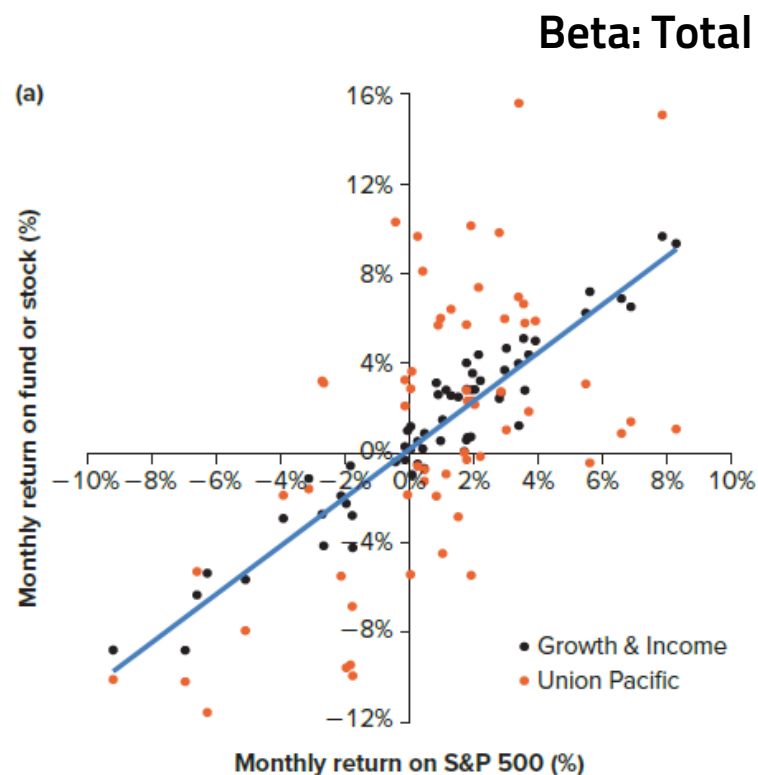
Part Three: Risk and Return 5. Capital Asset Pricing Model

Beta: Event Study of Bayer's Announcement to Acquire Monsanto on 19 May 2016

Beta (60 days)	0,8138009
Return Bayer 19.5.16	-0,0780015
Return DAX 19.5.16	-0,0148181
Normal Return Bayer	-0,012059
Abnormal Return Bayer	-0,0659425



Part Three: Risk and Return 5. Capital Asset Pricing Model

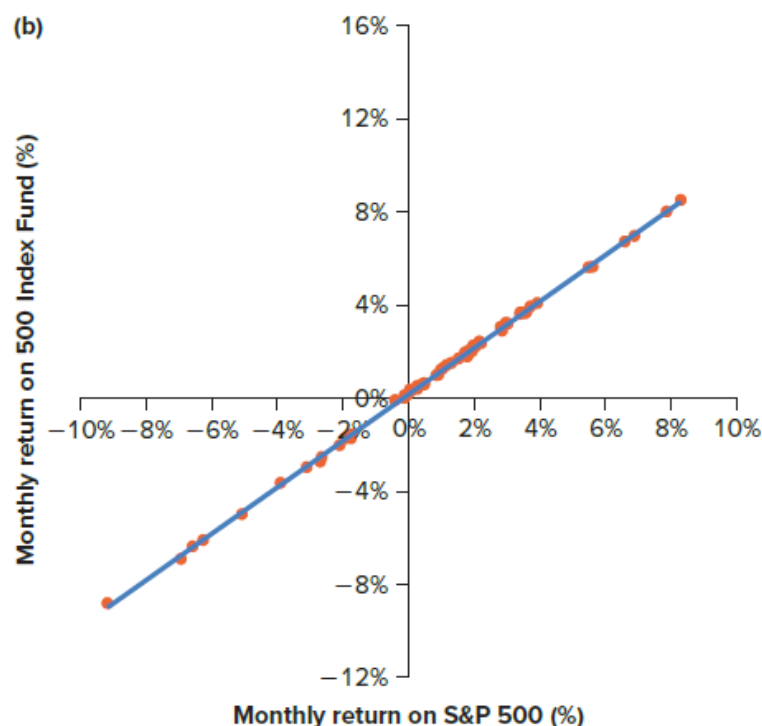


- See the data for Vanguard's Growth and Income Fund (portfolio) and Union Pacific (single stock).
- Betas are identical.
- However, not surprisingly, the specific risk of the portfolio is much lower than that of the single stock.
- Total risk (standard deviation) of Union Pacific will be much bigger.

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Part Three: Risk and Return 5. Capital Asset Pricing Model

Beta: Total Risk and Market Risk



- See the data for Beta of Vanguard's 500 Index Fund that replicates the S&P 500.

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Part Three: Risk and Return 5. Capital Asset Pricing Model

Beta: Total Risk and Market Risk (2015-2019)

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X	U.S. Steel	76.4
PCG	Pacific Gas & Electric	55.4
MRO	Marathon Oil	48.9
NEM	Newmont Mining	36.7
AMZN	Amazon	28.3
BA	Boeing	27.4
CPB	Campbell Soup	22.6
IBM	IBM	22.5
F	Ford	22.1
INTC	Intel	22.1

Ticker	Company	Beta
X	U.S. Steel	3.03
MRO	Marathon Oil	2.35
AMZN	Amazon	1.51
IBM	IBM	1.33
BA	Boeing	1.19
F	Ford	1.09
UNP	Union Pacific	1.07
GOOG	Google/Alphabet	1.01
DIS	Disney	1.00
XOM	ExxonMobil	1.00
INTC	Intel	0.91
PFE	Pfizer	0.65
PCG	Pacific Gas & Electric	0.55
SBUX	Starbucks	0.51
MCD	McDonald's	0.45
CPB	Campbell Soup	0.43
KO	Coca-Cola	0.43
WMT	Walmart	0.35
NEM	Newmont Mining	-0.01
Average		0.99

Note: Betas are calculated from five years of monthly data.

Ticker	Company	Standard deviation(%)
UNP	Union Pacific	21.4
DIS	Disney	21.2
GOOG	Google/Alphabet	20.1
SBUX	Starbucks	18.7
WMT	Walmart	17.7
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S&P 500		12.0

Taken from the slides accompanying Brealey/Myers/Marcus: Fundamentals of Corporate Finance, 11th edition.

Part Three: Risk and Return 5. Capital Asset Pricing Model

Beta: Portfolio Betas

- Diversification decreases variability from unique risk, but not from market risk.
- Portfolio beta equals the weighted average of the betas of the securities in the portfolio.
- The S&P Composite Index 500 stocks have an average beta of 1.0.
- The market portfolio is a portfolio of all assets in the economy. In practice a broad stock market index is used to represent the market. The beta of the market portfolio is by definition 1.0.

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Part Three: Risk and Return 5. Capital Asset Pricing Model

Beta: Portfolio Betas

- Portfolio beta
 - = (fraction of portfolio in stock 1 $\times$ β of stock 1)
 - + (fraction of portfolio in stock 2 $\times$ β of stock 2)
 - ...
 - + (fraction of portfolio in stock t $\times$ β of stock t)

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Part Three: Risk and Return 5. Capital Asset Pricing Model

Intermediate Exercise

The beta of IBM is 1.33. You learn that the general market, e.g., measured with the S&P 500, increased today by 1%. What is your best estimate for the return of the IBM share on that day?

In fact, the IBM share increased by 2% on that day. Decompose to what extent the return was caused by macro factors and to which extent by specific factors. Give examples for specific risk factors.

Part Three: Risk and Return 5. Capital Asset Pricing Model

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Part Three: Risk and Return 5. Capital Asset Pricing Model

Capital Asset Pricing Model (CAPM)

- Theory of the relationship between risk and return.
- It states that the **expected risk premium on any security** equals its beta times the market risk premium.
- The **market risk premium** is the difference between market return and risk-free return, e.g., Treasury bills. It is compensation for bearing market risk (premium for worrying).

Risk premium on the market = $r_m - r_f$

Risk premium on any asset i = $r_i - r_f$

Expected return on any asset i = $r_f + \beta \times (r_m - r_f)$

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Part Three: Risk and Return 5. Capital Asset Pricing Model

Capital Asset Pricing Model (CAPM): Example

- The return on the stock market is 10%; the risk-free rate of return is 3%. What is the risk premium for Stock X that has a beta of 0.5? What is the expected return of that stock?

Market Risk Premium: $r_m - r_f = .10 - .03 = .07$

Risk Premium for Stock X: $r_X - r_f = \beta \times (r_m - r_f) = .5 \times .07 = .035$

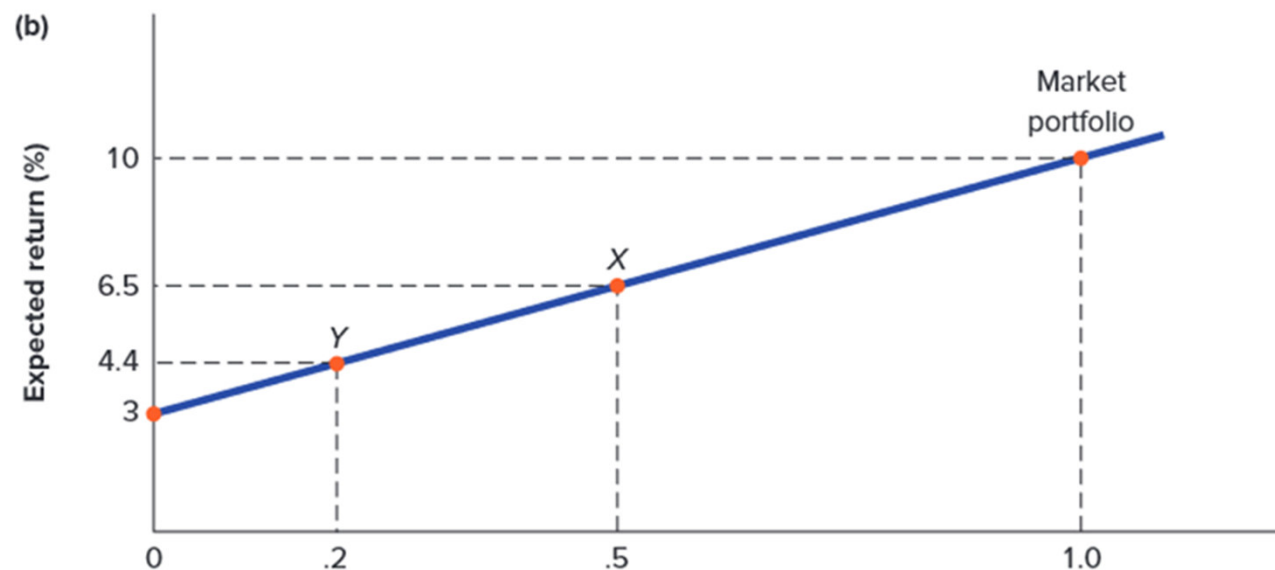
Expected Return for Stock X: $r_f + \beta \times (r_m - r_f) = .03 + .035 = .065$, or 6.5%

Taken from the slides accompanying Brealey/Myers/Marcus: Fundamentals of Corporate Finance, 11th edition.

Part Three: Risk and Return 5. Capital Asset Pricing Model

Capital Asset Pricing Model (CAPM): Example

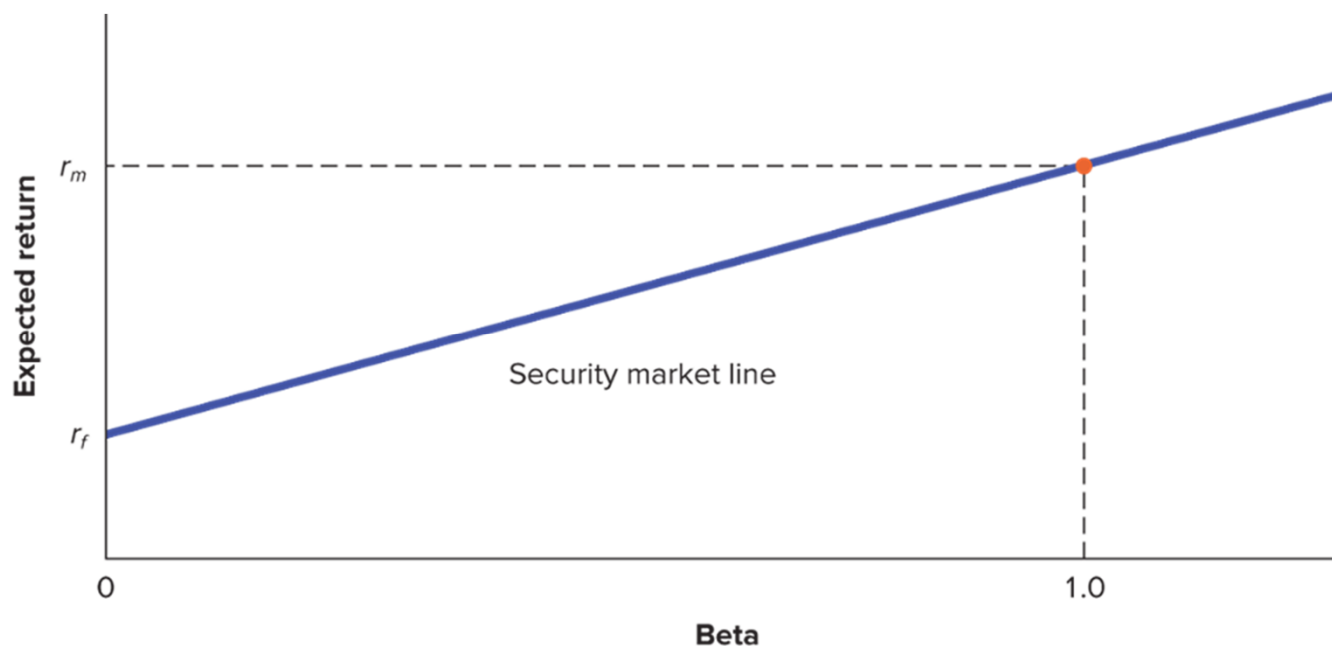
- Graph shows stock X as well as a stock Y ($\beta = .2$) and the market portfolio. (Note: horizontal axis = β , vertical axis = expected return.)



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Part Three: Risk and Return 5. Capital Asset Pricing Model

Capital Asset Pricing Model (CAPM): Security Market Line (SML)



$$\text{SML equation: } r = r_f + \beta(r_m - r_f)$$

Taken from the slides accompanying Brealey/Myers/Marcus: Fundamentals of Corporate Finance, 11th edition.

Part Three: Risk and Return 6: Capital Market Based Cost of Capital	
BASIC Required Rate of Return	
• Data	
• $r_f = 4\%$	
• Equity	
• $\beta = 1.2$ (assumed)	
• SML eq. assuming constant growth	$r = r_f + \beta(r_m - r_f)$
	$r = 4\% + 1.2(10\% - 4\%) = 11.2\%$

Part Three: Risk and Return 5. Capital Asset Pricing Model

Capital Asset Pricing Model (CAPM): Calculating Expected Returns

Ticker	Company	Beta	Expected Return (%)
X	U.S. Steel	3.03	22.2
MRO	Marathon Oil	2.35	17.4
AMZN	Amazon	1.51	11.5
IBM	IBM	1.33	10.3
BA	Boeing	1.19	9.3
F	Ford	1.09	8.6
UNP	Union Pacific	1.07	8.5
GOOG	Google/Alphabet	1.01	8.1
DIS	Disney	1.00	8.0
XOM	ExxonMobil	1.00	8.0

Ticker	Company	Beta	Expected Return (%)
INTC	Intel	0.91	7.3
PFE	Pfizer	0.65	5.6
PCG	Pacific Gas & Electric	0.55	4.9
SBUX	Starbucks	0.51	4.5
MCD	McDonald's	0.45	4.1
CPB	Campbell Soup	0.43	4.0
KO	Coca-Cola	0.43	4.0
WMT	Walmart	0.35	3.5
NEM	Newmont Mining	-0.01	0.9
Average		0.99	7.94

Expected return
= 1% + beta × 7%,
i.e., r_f = 1% and
market risk
premium = 7%

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Part Three: Risk and Return 5. Capital Asset Pricing Model

Capital Asset Pricing Model (CAPM): Testing the CAPM

- There is empirical evidence not in line with the CAPM, e.g., that portfolios with smaller firms and portfolios with firms with lower growth rates ("value stocks") earn higher returns than can be explained by the CAPM.
- The CAPM is a single-factor model. In reaction to the evidence, multi-factor models have been developed. However, they are lacking the theoretical foundation of the CAPM.
- Empirical evidence also questions that the relation between risk (beta) and return is linear.
- Nevertheless, the CAPM is still widely used in business.

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Part Three: Risk and Return 5. Capital Asset Pricing Model

Intermediate Exercise

Why is the expected return according to the CAPM for Newmont Mining the lowest (0.9%) although its standard deviation is the second highest in the respective group of shares? Why is that consistent with the view that expected return increases with risk?

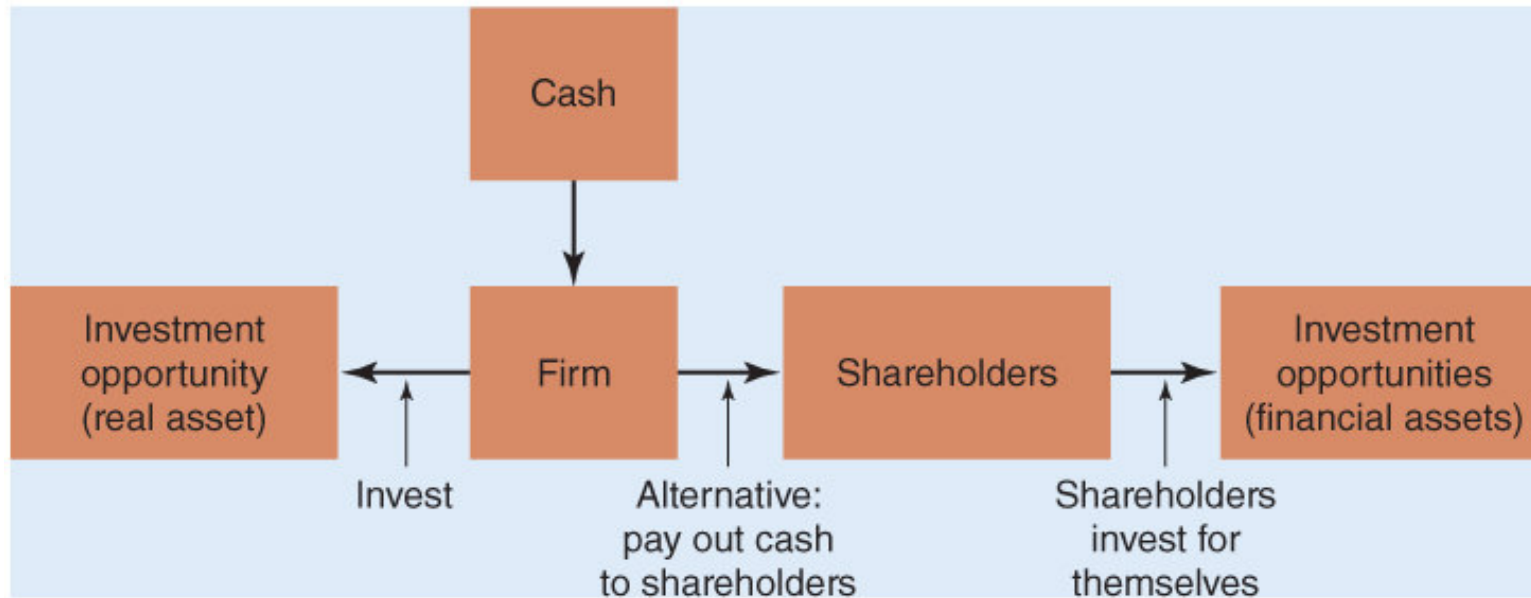
The expected return for the market portfolio is at 10%, the risk-free rate of return at 3. The beta of stock A is at 1.2.

How high is the market risk premium?

How high is the risk premium of stock A?

What is the expected return of stock A according to the CAPM?

The Course Journey



Taken from the slides accompanying Brealey/Myers/Marcus: Fundamentals of Corporate Finance, 11th edition.

capital
budgeting rules
(chapter 2)

- benchmark debt (ch. 3)
- benchmark equity (ch. 4)
- risk/return (ch. 5/6)
- WACC (ch. 6)

Part Three: Risk and Return 5. Capital Asset Pricing Model

Terms and concepts you should be familiar with after chapter 5

- Maturity premium, risk premium
- Variance, standard deviation, correlation
- Total risk, systematic risk, specific risk, diversification
- Beta
- Capital Asset Pricing Model (CAPM), Security Market Line (SML)
- Market risk premium

Part Three: Risk and Return 5. Capital Asset Pricing Model

Formulas from chapter 5

- Expected (Mean) Return: Calculated as a weighted average of the possible returns, where the weights correspond to the probabilities.

$$\text{Expected Return} = E[R] = \sum_R P_R \times R$$

- Variance: The expected squared deviation from the mean.

$$\text{Var}(R) = E[(R - E[R])^2] = \sum_R P_R \times (R - E[R])^2$$

- Variance Estimate Using Realized Returns: $\text{Var}(R) = \frac{1}{T-1} \sum_{t=1}^T (R_t - \bar{R})^2$

- Standard Deviation: The square root of the variance.

$$SD(R) = \sqrt{\text{Var}(R)}$$

Taken from the slides accompanying Berk/DeMarzo: Corporate Finance, 6th edition.

Part Three: Risk and Return 5. Capital Asset Pricing Model

Formulas from chapter 5

- Portfolio rate of return
$$= \begin{aligned} &(\text{fraction of portfolio in asset 1} \times \text{rate of return on asset 1}) \\ &+ (\text{fraction of portfolio in asset 2} \times \text{rate of return on asset 2}) \\ &\dots \\ &+ (\text{fraction of portfolio in asset } t \times \text{rate of return on asset } t) \end{aligned}$$
- Portfolio beta
$$\begin{aligned} &= (\text{fraction of portfolio in stock 1} \times \beta \text{ of stock 1}) \\ &+ (\text{fraction of portfolio in stock 2} \times \beta \text{ of stock 2}) \\ &\dots \\ &+ (\text{fraction of portfolio in stock } t \times \beta \text{ of stock } t) \end{aligned}$$

Taken from the slides accompanying Brealey/Myers/Marcus: Fundamentals of Corporate Finance, 11th edition.

Part Three: Risk and Return 5. Capital Asset Pricing Model

Formulas from chapter 5

- Risk premium on the market = $r_m - r_f$
- Risk premium on any asset i = $r_i - r_f$
- SML equation: $r = r_f + \beta(r_m - r_f)$

Taken from the slides accompanying Brealey/Myers/Marcus: Fundamentals of Corporate Finance, 11th edition.

Agenda

Part Three: Risk and Return

5. Capital Asset Pricing Model 18./26.5.26

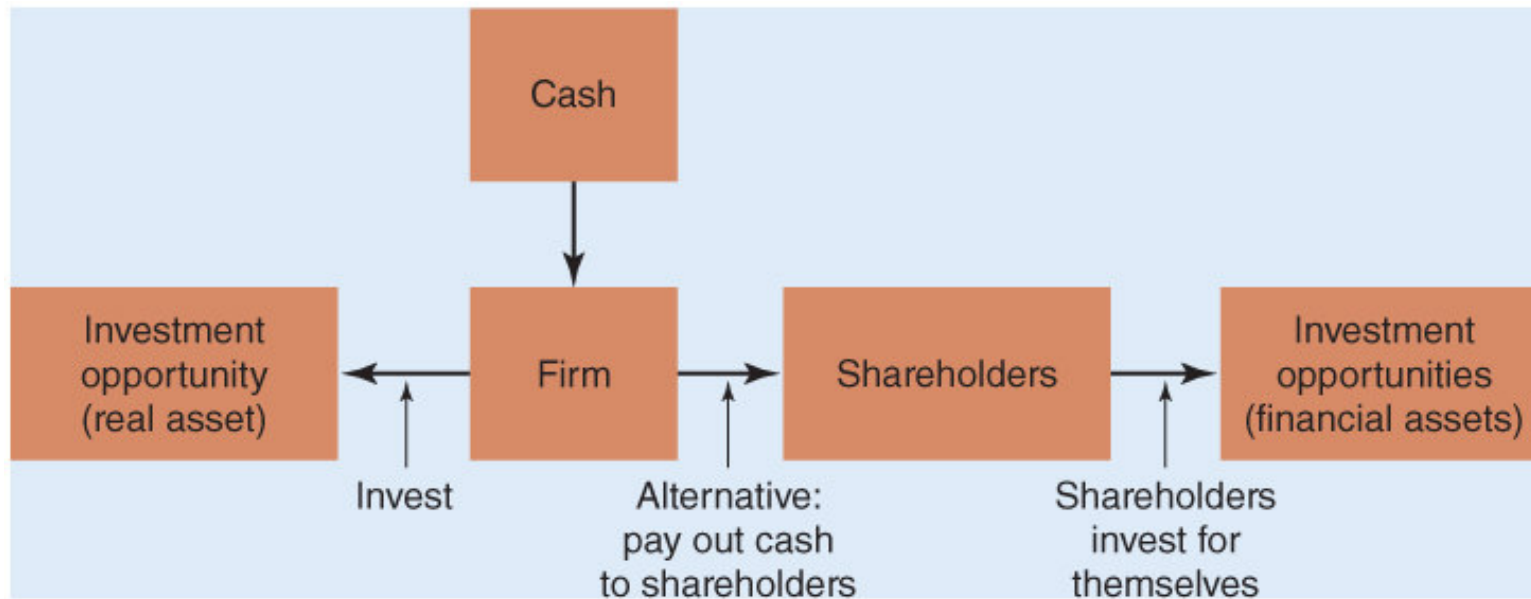
6. Capital Market Based Cost of Capital 26.5./1.6.26



Part Three: Risk and Return 6. Capital Market Based Cost of Capital: Literature

- BMM, chapter 12.4-13.5
- BDM, chapter 12

The Course Journey



Taken from the slides accompanying Brealey/Myers/Marcus: Fundamentals of Corporate Finance, 11th edition.

capital
budgeting rules
(chapter 2)

- benchmark debt (ch. 3)
- benchmark equity (ch. 4)
- risk/return (ch. 5/6)
- WACC (ch. 6)

Part Three: Risk and Return 6. Capital Market Based Cost of Capital

Terms and concepts you should be familiar with after chapter 6

- Applying the CAPM in capital budgeting
- Project-specific opportunity cost of capital
- Capital structure
- Tax advantage of debt
- Weighted-average cost of capital (WACC)

Part Three: Risk and Return 6. Capital Market Based Cost of Capital

Structure of Chapter 6

- The CAPM and the Opportunity Cost of Capital
- Introducing the Weighted-Average Cost of Capital (WACC)
- Tax Advantage of Debt
- WACC
- Interpreting WACC
- WACC for Selected Firms

Part Three: Risk and Return 6. Capital Market Based Cost of Capital

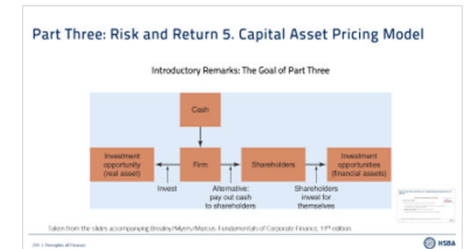
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Part Three: Risk and Return 6. Capital Market Based Cost of Capital

The CAPM and the Opportunity Cost of Capital

- **Company cost of capital.**
 - Opportunity cost of capital for investment in the entire firm.
 - The company cost of capital is the appropriate discount rate for an average-risk investment project undertaken by the firm.
- The **project cost of capital** depends on the use of the capital.
 - It depends on the risk of the project and not the risk of the company.



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Part Three: Risk and Return 6. Capital Market Based Cost of Capital

The CAPM and the Opportunity Cost of Capital: Example_Coca-Cola Project

- **Problem**
 - Suppose that Coca-Cola is contemplating an investment of \$10 million to expand a warehouse. It has forecast the cash flows and calculated an IRR of 3%. We assume the risk-free rate is 1% and the market risk premium is 7%. Should Coca-Cola invest?

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Part Three: Risk and Return 6. Capital Market Based Cost of Capital

The CAPM and the Opportunity Cost of Capital: Example_Coca-Cola Project

- **Solution**
 - We know for Coca-Cola as the whole company ($\beta=0.43$) that, according to the CAPM, the expected return on equity is 4% [$1\% + 0.43 * (8\% - 1\%)$].
 - This would be the correct benchmark (discount rate) for the warehouse project only **if the project had the same risk profile as the whole company**. In this case, the project would be rejected as its expected return (IRR = 3%) is below what investors could expect to earn in the stock market from investments with the same risk (4%).

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Part Three: Risk and Return 6. Capital Market Based Cost of Capital

The CAPM and the Opportunity Cost of Capital: Example_Coca-Cola Project

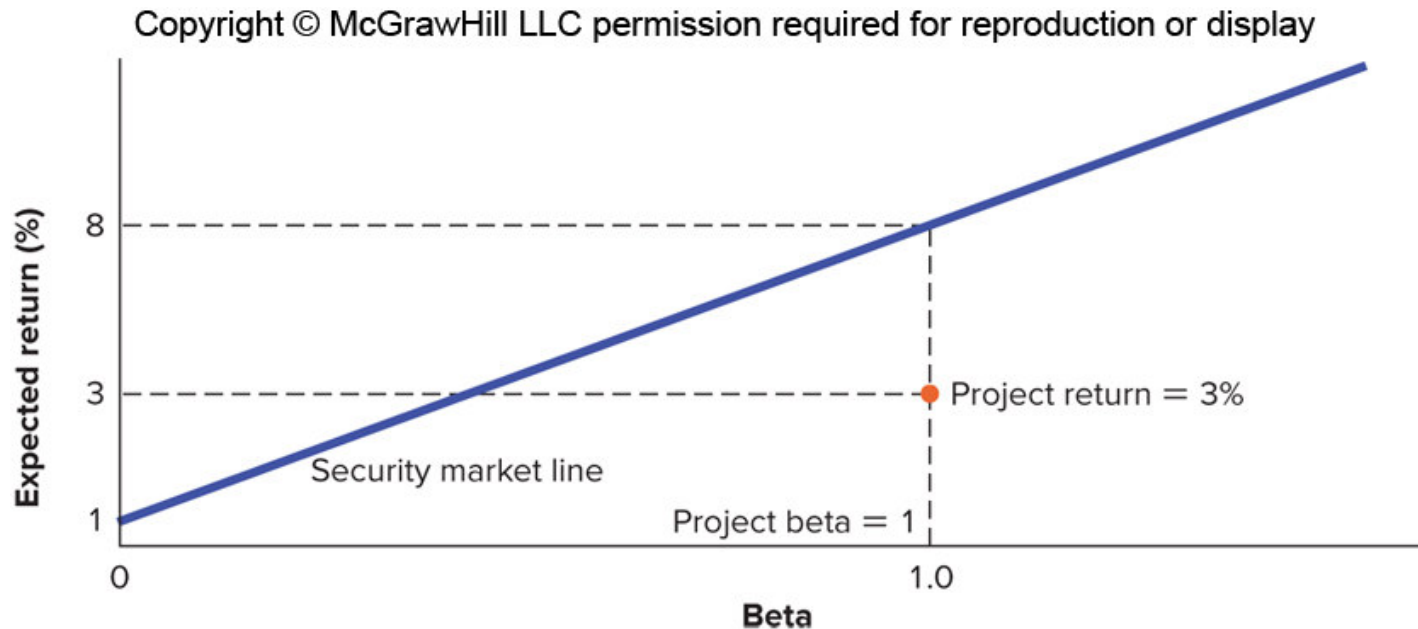
- **Solution**
 - **If the project were risk-free** ($\rightarrow \beta=0$), the benchmark (discount rate) would be 1% [$1\% + 0 * (8\% - 1\%)$]. The project would be realized.
 - **If the project had a beta of one**, the benchmark (discount rate) would be 8% [$1\% + 1 * (8\% - 1\%)$]. In this case, the project would be rejected as its expected return (IRR = 3%) is below what investors could expect to earn in the stock market from investments with the same risk (8%).

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Part Three: Risk and Return 6. Capital Market Based Cost of Capital

The CAPM and the Opportunity Cost of Capital: Example_Coca-Cola Project

- Solution



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Part Three: Risk and Return 6. Capital Market Based Cost of Capital

The CAPM and the Opportunity Cost of Capital: Project Beta

- A project's beta can be estimated by forecasting the project's cash flows and the return of the market portfolio for different scenarios, e.g., boom, normal, and recession.
- Determinants of a project's beta are:
 - How cyclical is the project? (Remember, e.g., that US Steel had a much higher beta than Coca-Cola.)
 - Operating leverage, i.e., a large percentage of the costs is fixed, increases beta.
 - Financial leverage, i.e., a bigger share of debt financing, increases beta.

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Part Three: Risk and Return 6. Capital Market Based Cost of Capital

The CAPM and the Opportunity Cost of Capital: Interim Conclusion

- The CAPM helps us to estimate the cost of **equity** and to make capital budgeting decisions.
- However: What do we do when a company also uses debt financing?
- **Thus: We have to widen the scope of our cost of capital calculations.**

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Part Three: Risk and Return 6. Capital Market Based Cost of Capital

Structure of Chapter 6

- The CAPM and the Opportunity Cost of Capital
- **Introducing the Weighted-Average Cost of Capital (WACC)**
- Tax Advantage of Debt
- WACC
- Interpreting WACC
- WACC for Selected Firms

Part Three: Risk and Return 6. Capital Market Based Cost of Capital

Introducing the Weighted-Average Cost of Capital (WACC)

- **Cost of Capital**
 - The return the firm's investors could expect to earn if they invested in securities with comparable degrees of risk.
- **Capital Structure**
 - The mix of long-term debt and equity financing.

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Part Three: Risk and Return 6. Capital Market Based Cost of Capital

Introducing the WACC: Example

- Geothermal Inc. has the following structure.

Market value of assets	\$647	100%
Market value of equity	\$453	70%
Market value of debt	\$194	30%

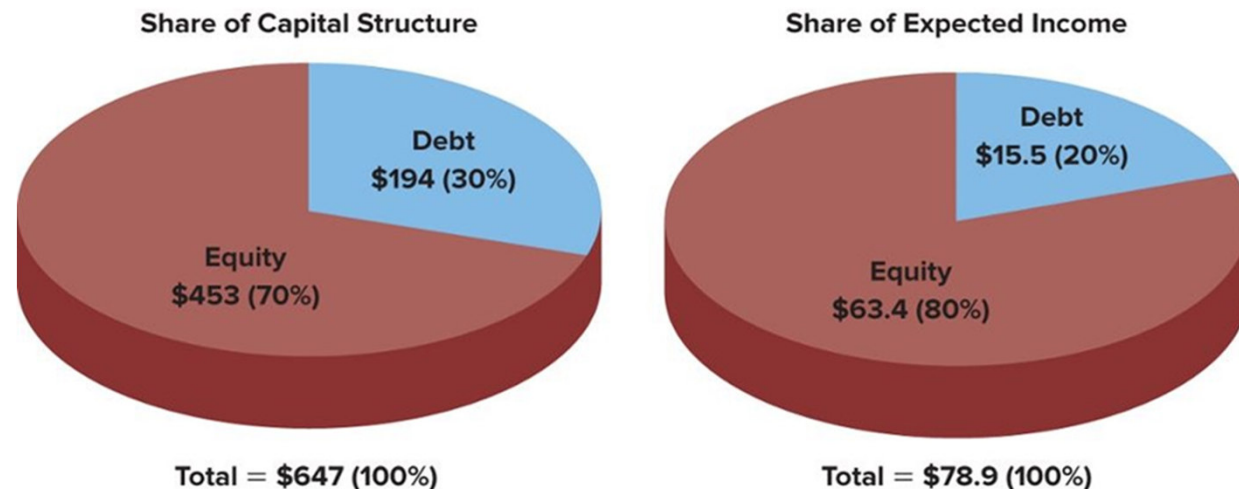
- Given that Geothermal Inc. pays 8% for debt and 14% for equity, what is the company cost of capital?
- Portfolio return = $(.3 \times 8\%) + (.7 \times 14\%) = 12.2\%$.

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Part Three: Risk and Return 6. Capital Market Based Cost of Capital

Introducing the WACC: Example

- Geothermal's **debtholders** account for **30%** of the capital structure, but they **get a smaller share of income** because their 8% return is **guaranteed** by the company. The **stockholders bear more risk** and **receive, on average, greater return**. If you buy all the debt and all the equity, you get all the income.



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Part Three: Risk and Return 6. Capital Market Based Cost of Capital

Introducing the WACC

- **Generalization**

$$r_{\text{assets}} = \frac{\text{total income}}{\text{value of investments}} = \frac{(D \times r_{\text{debt}}) + (E \times r_{\text{equity}})}{V}$$

$$r_{\text{assets}} = \frac{D}{V} \times r_{\text{debt}} + \frac{E}{V} \times r_{\text{equity}}$$

- D = market value of debt.
- E = market value of equity = # shares $\times$ price per share.
- $V = D + E$.

r_{debt} = YTM on bonds or loans.

r_{equity} = **CAPM** = $r_f + \beta(r_m - r_f)$ or from the **DDM**.

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Part Three: Risk and Return 6. Capital Market Based Cost of Capital

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Part Three: Risk and Return 6. Capital Market Based Cost of Capital

Tax Advantage of Debt

- Taxes are an important consideration because interest payments are deducted from income before tax is calculated.

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Part Three: Risk and Return 6. Capital Market Based Cost of Capital

Tax Advantage of Debt: Example

- Criss-Cross Industries has earnings before interest and taxes (EBIT) of \$10 million. Interest payments are \$2 million, and the corporate tax rate is 21%. How much more tax would Criss-Cross pay if it were financed solely by equity?

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	Criss-Cross	Firm with No Debt
EBIT	\$10.0	\$10.0
Interest expense	2.0	0.0
Taxable income	8.0	10.0
Taxes owed	1.7	2.1
Net income	6.3	7.9
Total income accruing to debt- plus equityholders	8.3	7.9

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Part Three: Risk and Return 6. Capital Market Based Cost of Capital

Tax Advantage of Debt

- Taxes are an important consideration because interest payments are deducted from income before tax is calculated.
- After-tax cost of debt = pretax cost $\times$ (1 – tax rate)

$$= r_{\text{debt}} \times (1 - T_c).$$

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Part Three: Risk and Return 6. Capital Market Based Cost of Capital

Intermediate Exercise

In your own words: What is the tax advantage of debt compared to equity?

The Better World AG issues a bond at par with the following terms:

time to maturity: 5 years

coupon rate: 5%

face value: EUR100

What is the return for bondholders who buy the bond on issue day and hold it until maturity?
(Think before you start calculating!)

What is the cost of debt for this bond for Better World AG before and after taxes? Assume a corporate tax rate of 30%

Part Three: Risk and Return 6. Capital Market Based Cost of Capital

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Part Three: Risk and Return 6. Capital Market Based Cost of Capital

WACC

- The expected rate of return on a portfolio of all the firm's securities, adjusted for tax savings due to interest payments.

$$\text{WACC} = \left[\frac{D}{V} \times (1 - T_c) r_{\text{debt}} \right] + \left[\frac{E}{V} \times r_{\text{equity}} \right]$$

- D = market value of debt.
- E = market value of equity = # shares × price per share.
- V = D + E.

r_{debt} = YTM on bonds or loans.

r_{equity} = **CAPM** = $r_f + \beta(r_m - r_f)$ or from the **DDM**.

T_c = firm's average tax rate.

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Part Three: Risk and Return 6. Capital Market Based Cost of Capital

WACC: Example

- Given that Geothermal Inc. pays 8% for debt and 14% for equity, what is the WACC?
- Interest is tax deductible.
- At 21% tax rate, debt only costs 6.32% ($8.00\% \times .79$).
- $WACC = (.3 \times 6.32\%) + (.7 \times 14\%) = 11.7\%$.

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Part Three: Risk and Return 6. Capital Market Based Cost of Capital

WACC

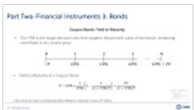
- **Three Steps to Calculating Cost of Capital**
 - Calculate the value of each security/funding source as a proportion of the firm's market value (**weights**).
 - Determine the required **rate of return** on each security.
 - Calculate a **weighted average** of the after-tax return on the debt and the return on the equity.

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Part Three: Risk and Return 6. Capital Market Based Cost of Capital

WACC: Weights

- **Measuring Capital Structure**
 - In estimating WACC, **do not use the book value** of securities.
 - In estimating WACC, **use the market value** of the securities.
 - Book values often do not represent the true market value of a firm's securities.
 - **Market Value of Bonds** – PV of all coupons and par value discounted at the current yield to maturity (YTM). Accordingly for loans if necessary.
 - **Market Value of Equity** – Market price per share multiplied by the number of outstanding shares.



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Part Three: Risk and Return 6. Capital Market Based Cost of Capital

WACC: Weights_Example

- **Big Oil Book Value Liabilities and Equity (mil)**

Bank debt	\$200	25.0%
LT bonds	\$200	25.0%
Common stock	\$100	12.5%
Retained earnings	\$300	37.5%
Total	\$800	100.0%

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Part Three: Risk and Return 6. Capital Market Based Cost of Capital

WACC: Weights_Example

- If the **long-term bonds** pay an 8% coupon and mature in 12 years, what is their market value (mil) assuming a 9% YTM?

$$PV = \frac{\$16}{1.09} + \frac{\$16}{1.09^2} + \frac{\$16}{1.09^3} + \dots + \frac{\$216}{1.09^{12}} = \$185.70$$

- If the **common stock** of \$100 million represent the aggregated face value of all outstanding shares, each stock has a face value of \$1, and the stock currently trades at \$12, total market value of equity amounts to \$1,200 million (100 million shares * \$12 per share).

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Part Three: Risk and Return 6. Capital Market Based Cost of Capital

WACC: Weights_Example

- **Big Oil Market Value Liabilities and Equity (mil)**

Bank debt	\$200.0	12.6%
LT bonds	\$185.7	11.7%
Total debt	\$385.7	24.3%
Common stock (\$12 per share)	\$1,200.0	75.7%
Total	\$1,585.7	100.0%

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Part Three: Risk and Return 6. Capital Market Based Cost of Capital

WACC: Required Rates of Return

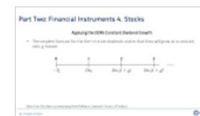
- Debt

- $r_d = \text{YTM.}$

- Equity

- $r_e = \text{CAPM} = r_f + \beta(r_m - r_f).$

- DDM, e.g., assuming constant growth: $P_0 = \frac{\text{Div}_1}{r_e - g}$ $r_e = \frac{\text{Div}_1}{P_0} + g$



Taken from the slides accompanying Brealey/Myers/Marcus: Fundamentals of Corporate Finance, 11th edition.

Part Three: Risk and Return 6. Capital Market Based Cost of Capital

WACC: Concluding Example

- Executive Fruit has issued debt and common stock. The market value of these securities are \$4 million and \$6 million, respectively. The required returns are 6% and 18%, respectively.
- Determine the WACC for Executive Fruit, Inc.

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Part Three: Risk and Return 6. Capital Market Based Cost of Capital

WACC: Concluding Example

- **Step 1**

- Firm Value = \$4 + 6 = \$10 (millions).

- **Step 2**

- Required returns are given.

- **Step 3**

$$\text{WACC} = \left[\frac{\$4}{\$10} \times (1 - .21) \times .06 \right] + \left(\frac{\$6}{\$10} \times .18 \right) = .1270, \text{ or } 12.70\%$$

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Part Three: Risk and Return 6. Capital Market Based Cost of Capital

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Part Three: Risk and Return 6. Capital Market Based Cost of Capital

Interpreting WACC

- The WACC is an appropriate discount rate only for a project that is a carbon copy of the firm's existing business.
- **Changing Capital structure**
 - There are two costs of increased debt financing.
 - The **explicit cost of debt** is the rate of interest bondholders demand; increasing debt → increased interest demanded.
 - The **implicit cost** is the required increase in return from equity due to increased risk.
 - As a result, there are two counteracting effects: tax advantage of debt (→ more debt) and the risk effect of debt (→ less debt). Discussion of capital structure in Major Finance.

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Part Three: Risk and Return 6. Capital Market Based Cost of Capital

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Part Three: Risk and Return 6. Capital Market Based Cost of Capital

WACC for Selected Firms

- **Notes on the data:**
 - Expected return on equity = 1% + Beta × 7%.
 - Interest rate on debt is estimated as yield to maturity in September 2020 on comparably rated corporate bonds.
 - D is the book value of the firm's debt, and E is the market value of equity.
 - $$WACC = \left(\frac{D}{V}\right) \times (1 - .21) \times r_{\text{debt}} + \left(\frac{E}{V}\right) \times r_{\text{equity}}.$$

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Part Three: Risk and Return 6. Capital Market Based Cost of Capital

WACC for Selected Firms

Ticker	Company	Beta	Expected Return on equity (%)	Interest Rate on debt (%)	Proportion of Equity (E/V)	Proportion of Debt (D/V)	WACC (%)
X	U.S. Steel	3.03	22.2	5.7	0.34	0.66	10.5
MRO	Marathon Oil	2.35	17.4	2.3	0.40	0.60	8.1
AMZN	Amazon	1.51	11.5	1.5	0.96	0.04	11.1
IBM	IBM	1.33	10.3	1.6	0.65	0.35	7.2
BA	Boeing	1.19	9.3	2.9	0.81	0.19	8.0
F	Ford	1.09	8.6	3.5	0.22	0.78	4.1
UNP	Union Pacific	1.07	8.5	1.9	0.84	0.16	7.4
GOOG	Google/Alphabet	1.01	8.1	1.5	0.99	0.01	8.0
DIS	Disney	1.00	8.0	1.9	0.86	0.14	7.1

Taken from the slides accompanying Brealey/Myers/Marcus: Fundamentals of Corporate Finance, 11th edition.

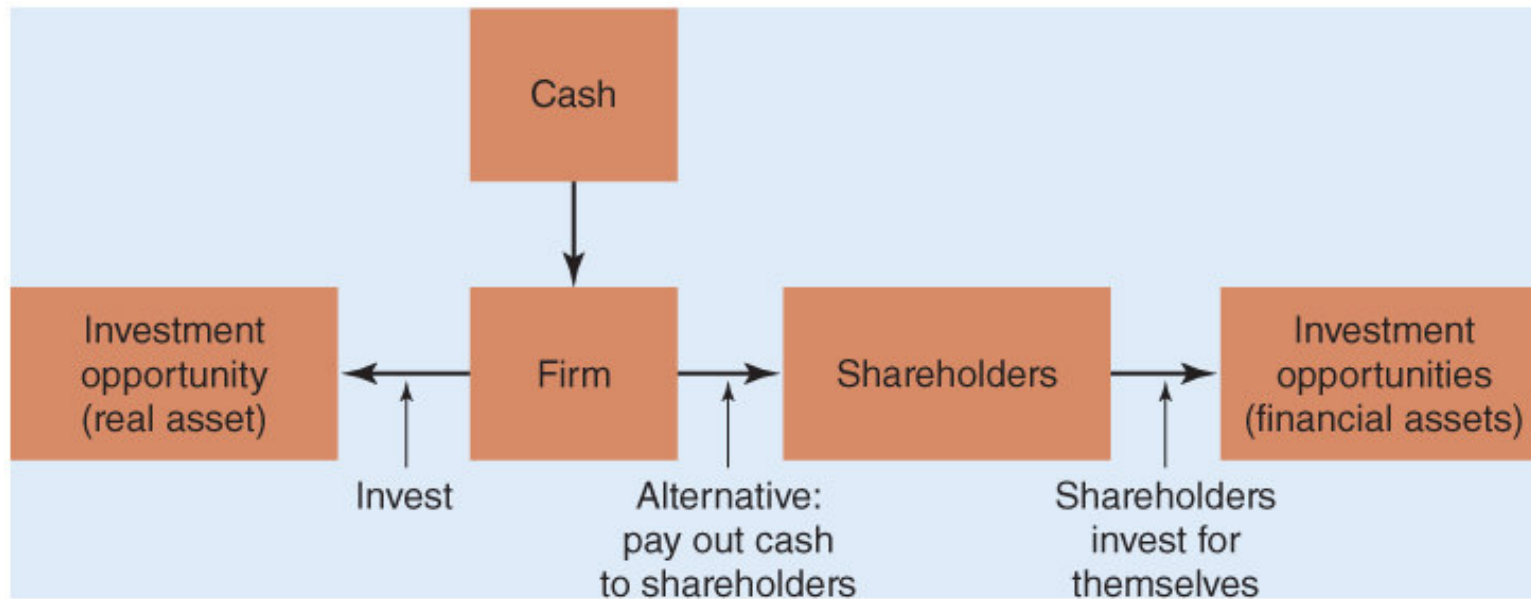
Part Three: Risk and Return 6. Capital Market Based Cost of Capital

WACC for Selected Firms

Ticker	Company	Beta	Expected Return on equity (%)	Interest Rate on debt (%)	Proportion of Equity (E/V)	Proportion of Debt (D/V)	WACC (%)
XOM	ExxonMobil	1.00	8.0	1.5	0.84	0.16	6.9
INTC	Intel	0.91	7.3	1.6	0.89	0.11	6.7
PFE	Pfizer	0.65	5.6	1.5	0.85	0.15	4.9
PCG	Pacific Gas & Electric	0.55	4.9	2.9	0.45	0.55	3.5
SBUX	Starbucks	0.51	4.5	2.1	0.90	0.10	4.3
MCD	McDonald's	0.45	4.1	2.1	0.78	0.22	3.6
CPB	Campbell Soup	0.43	4.0	2.9	0.73	0.27	3.5
KO	Coca-Cola	0.43	4.0	1.6	0.88	0.12	3.7
WMT	Walmart	0.35	3.5	1.5	0.86	0.14	3.1
NEM	Newmont Mining	-0.01	0.9	2.3	0.88	0.12	1.0

Taken from the slides accompanying Brealey/Myers/Marcus: Fundamentals of Corporate Finance, 11th edition.

The Course Journey



Taken from the slides accompanying Brealey/Myers/Marcus: Fundamentals of Corporate Finance, 11th edition.

capital
budgeting rules
(chapter 2)

- benchmark debt (ch. 3)
- benchmark equity (ch. 4)
- risk/return (ch. 5/6)
- WACC (ch. 6)

Part Three: Risk and Return 6. Capital Market Based Cost of Capital

A Final, Concluding Example

- Assume Amazon considers developing a new, AI-based foundation for its online store. The project requires an investment of USD100 billion. The financial department at Amazon estimated the expected cash flows of the project and calculated that the project has an IRR of 15%.
- Assuming the project has the same risk profile as the current business of Amazon: How would you calculate the NPV of this project, how would you derive a capital budgeting decision using the IRR? What would be the result?
- How about a situation where the project is assumed to dispose of a higher risk than the current business of Amazon?

Part Three: Risk and Return 6. Capital Market Based Cost of Capital

Terms and concepts you should be familiar with after chapter 6

- Applying the CAPM in capital budgeting
- Project-specific opportunity cost of capital
- Capital structure
- Tax advantage of debt
- Weighted-average cost of capital (WACC)

Part Three: Risk and Return 6. Capital Market Based Cost of Capital

Formulas from chapter 6

- $r_{\text{assets}} = \frac{\text{total income}}{\text{value of investments}} = \frac{(D \times r_{\text{debt}}) + (E \times r_{\text{equity}})}{V}$
- $r_{\text{assets}} = \frac{D}{V} \times r_{\text{debt}} + \frac{E}{V} \times r_{\text{equity}}$
- After-tax cost of debt $= r_{\text{debt}} \times (1 - T_c)$.
- $\text{WACC} = \left[\frac{D}{V} \times (1 - T_c) r_{\text{debt}} \right] + \left[\frac{E}{V} \times r_{\text{equity}} \right]$

Taken from the slides accompanying Brealey/Myers/Marcus: Fundamentals of Corporate Finance, 11th edition.

Agenda

Part One: Time and Interest Rates

1. Time Value of Money 27.7.26

2. Investment Decision Rules 3./10.8.26

Part Two: Financial Instruments

3. Bonds 10./17.8.26

4. Stocks 17./24.8.26



Agenda

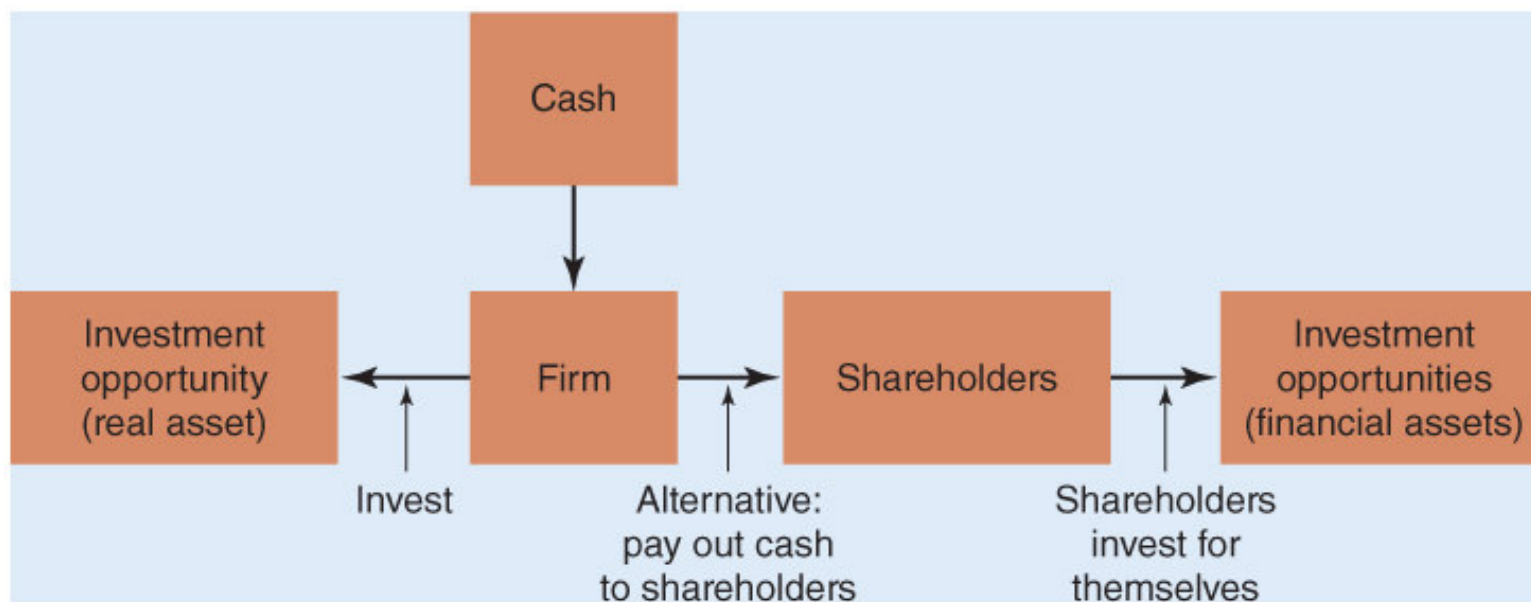
Part Three: Risk and Return

5. Capital Asset Pricing Model 31.8./7.9.26

6. Capital Market Based Cost of Capital 7./14.9.26



Reviewing the Course Journey



Taken from the slides accompanying Brealey/Myers/Marcus: Fundamentals of Corporate Finance, 11th edition.

capital
budgeting rules
(chapter 2)

- benchmark debt (ch. 3)
- benchmark equity (ch. 4)
- risk/return (ch. 5/6)
- WACC (ch. 6)

Course Review

- **Chapter 1**

- recap of tools already known
- time value of money

- **Chapter 2**

- place of value creation: mainly capital budgeting, but not in sale of financial instruments (4)
- main tool NPV; IRR also popular → important to know weak spots
- high-quality benchmark needed → chapters 3-6
- opportunity approach; market-based approach

Course Review

- **Chapter 3**
 - market-based r_{debt}
 - investor perspective: YTM; company perspective: cost of debt
 - making familiar with the major issues and mechanics
- **Chapter 4**
 - DDM → market-based r_{equity}
 - market efficiency
 - no positive NPV for financing to be expected
 - use market data (wisdom of the crowd) and models to derive data

Course Review

- **Chapter 5**
 - CAPM/SML → market-based r_{equity}
 - risk and return
 - standard deviation (total risk); diversification; specific and systematic risk; beta
- **Chapter 6**
 - market-based WACC → link to chapter 2: improve quality of capital budgeting
 - market-based rates and (often) weights
 - tax advantage of debt

Part Two: Financial Instruments 4. Stocks

Market Efficiency: Semi-Strong Form

- Testing the semi-strong form: Do the prices react immediately when new information becomes known?
- **Aurubis Ad Hoc Profit Warning 26.4.19**

- **Aurubis AG: Aurubis AG generates preliminary quarterly earnings of EUR 63 million and reduces full-year forecast**

Aurubis AG / Key word(s): Profit Warning/Change in Forecast
Aurubis AG: Aurubis AG generates preliminary quarterly earnings of EUR 63 million and reduces full-year forecast

26-Apr-2019 / 17:00 CET/CEST

Disclosure of an inside information acc. to Article 17 MAR of the Regulation (EU) No 596/2014, transmitted by DGAP - a service of EQS Group AG.

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<https://www.dgap.de/dgap/News/adhoc/aurubis-aurubis-generates-preliminary-quarterly-earnings-eur-million-and-reduces-fullyear-forecast/?newsID=1144891>

Part Two: Financial Instruments 4. Stocks

Market Efficiency: Semi-Strong Form



**Aurubis Ad Hoc Profit
Warning 26.4.19**

Taken from Comdirect.

Part Three: Risk and Return 5. Capital Asset Pricing Model

Beta Event Study of Bayer's Announcement to Acquire Monsanto on 19 May 2016

Beta (60 days)	0.8138009
Return Bayer 19.5.16	-0.0780015
Return DAX 19.5.16	-0.0148181
Normal Return Bayer	-0.012059
Abnormal Return Bayer	-0.0659425